

GMTSAR installation and processing Guide - Practical InSAR handbook series

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By

Siavash Shami

Preface

Interferometric Synthetic Aperture Radar (InSAR) uses the pairwise phase difference of SAR images to estimate the amount of deformation and displacement of the earth's surface. InSAR technique is used to estimate the displacement caused by earthquakes, volcanoes, subsidence, and settlement. InSAR has good spatial accuracy and density compared to other methods of estimating surface changes, such as the global positioning system (GPS). It is also economically acceptable due to its time and cost savings.

Interpreting the results of InSAR requires the processing of SAR images by SAR image processing software. [GMTSAR](#) software is written the C Programming Language by Sandwell, D. et al. This software has high speed in processing SAR images and can process SAR products such as Envisat, ALOS-1, TerraSAR-X, COSMOS-SkyMed, Sentinel-1, and ALOS-2. This software is open-source (GNU General Public License) that can be used to process Small Baseline Subset (SBAS) time-series

This book is written in two major general chapters. In the *first chapter*, we will show step by step how to install GMTSAR software and other software on [Ubuntu 20.04](#) is taught. Also, in this chapter, download the necessary data such as Sentinel-1 images, and DEM is taught. In the *second* part, which is the main chapter of the book, the processing of the Sentinel-1 images with GMTSAR software is explained. First of all, the process of processing two earthquake-related images using the DInSAR technique is explained so that the user can obtain the amount of displacement caused by the earthquake with this technique. The SBAS time-series is then processed using GMTSAR software to estimate time series ground deformation determine the rate of subsidence of the earth's crust.

It should be noted that this document does not deal with the theory and mathematics of InSAR and algorithms. Dear all readers are requested to send their comments, suggestions, and questions about this book to the following email address: (siavashshami@email.kntu.ac.ir)

Sincerely
Siavash Shami

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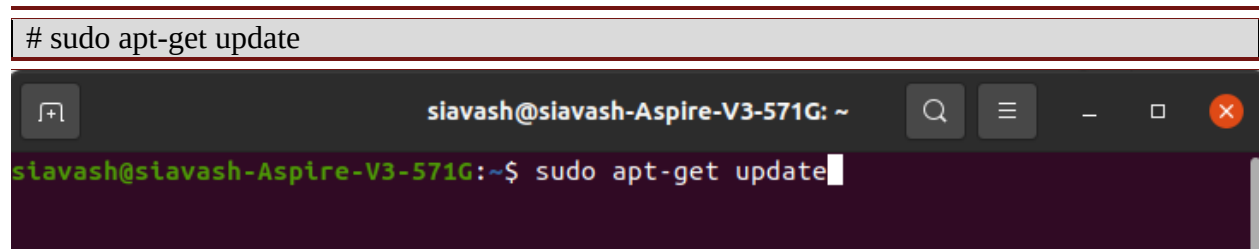
Chapter 1: Install Software and Data Download

This chapter will teach you how to install StaMPS/MTI, GMT, GMTSAR, and other software which is needed for processing steps. Sentinel-1 Radar images are free and Accessible in most parts of the world, and users can download their images to do their research. For this reason, these images have been used in the process of teaching how to process in this book. Thus, in the first chapter of the book, an explanation will be given about the structure of these images and how to download them. You will also learn how to download a DEM file and the orbital file of Sentinel-1 images.

1.1. Install Synaptic

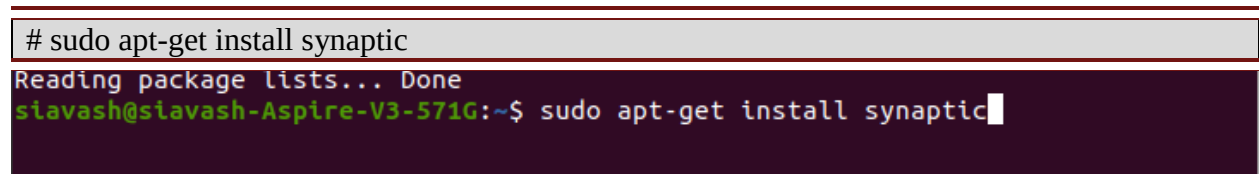
First of all, update your system by running the following command in the Terminal.

```
# sudo apt-get update
```



Then run the following command to install synaptic software in the Terminal.

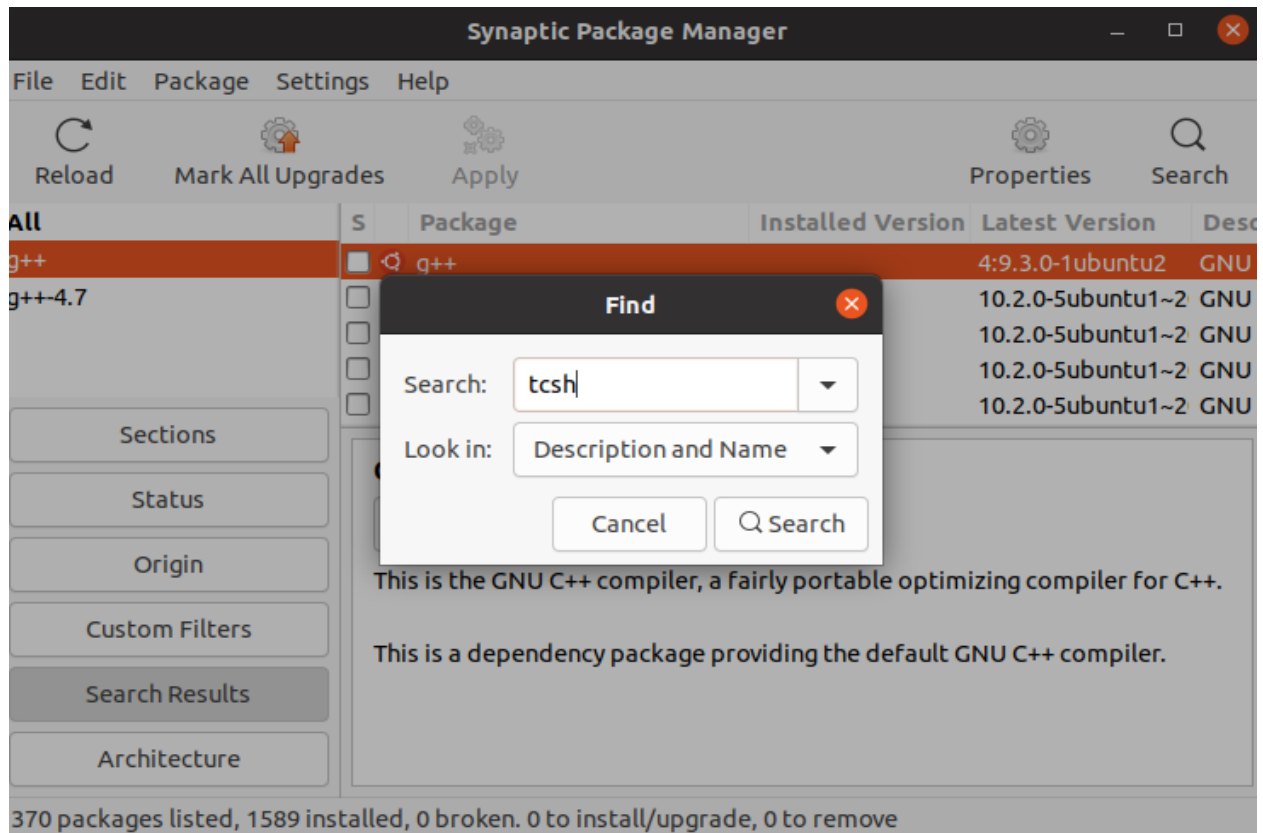
```
# sudo apt-get install synaptic
```



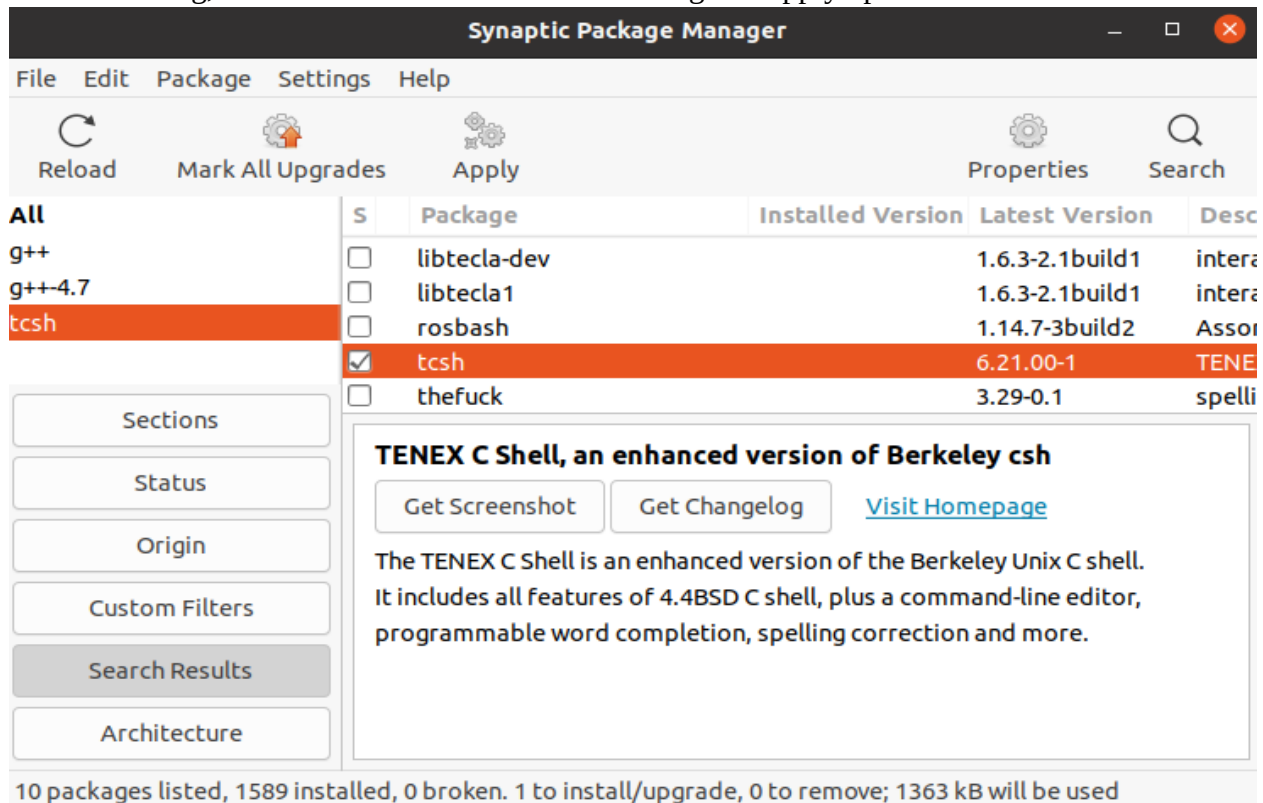
After installing the synaptic software, you must use it to install the following software.

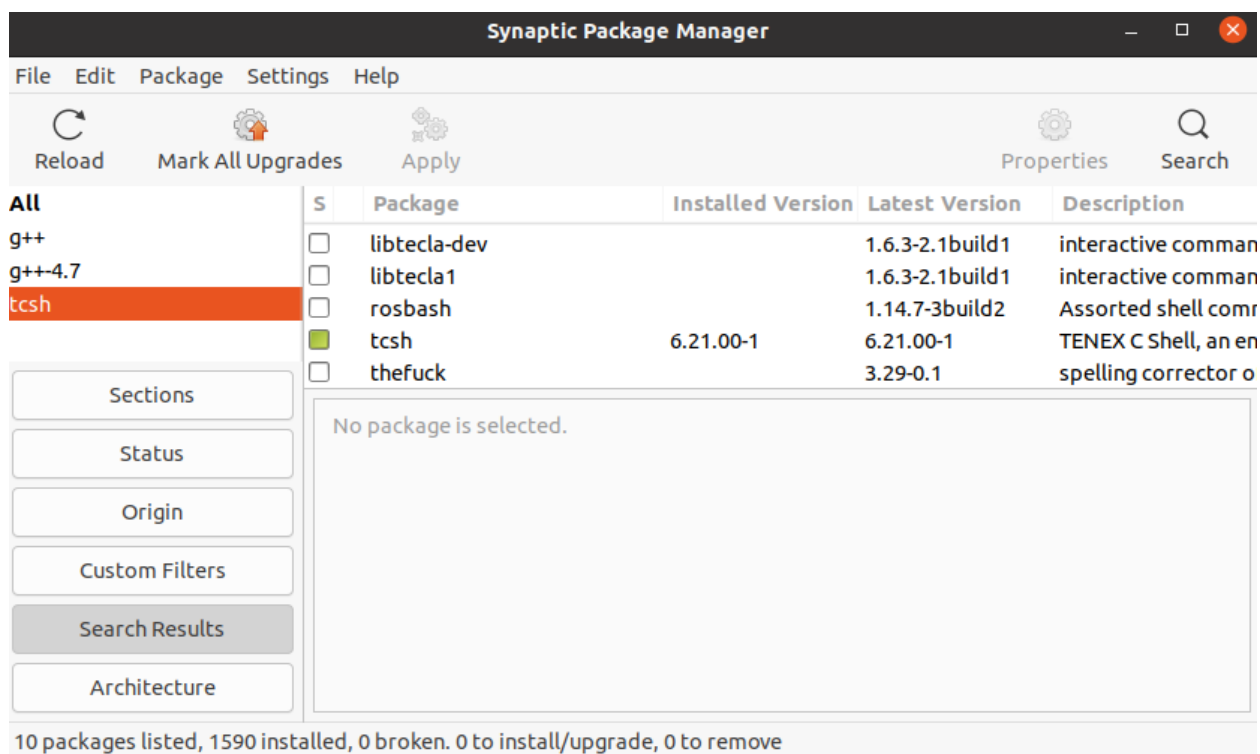
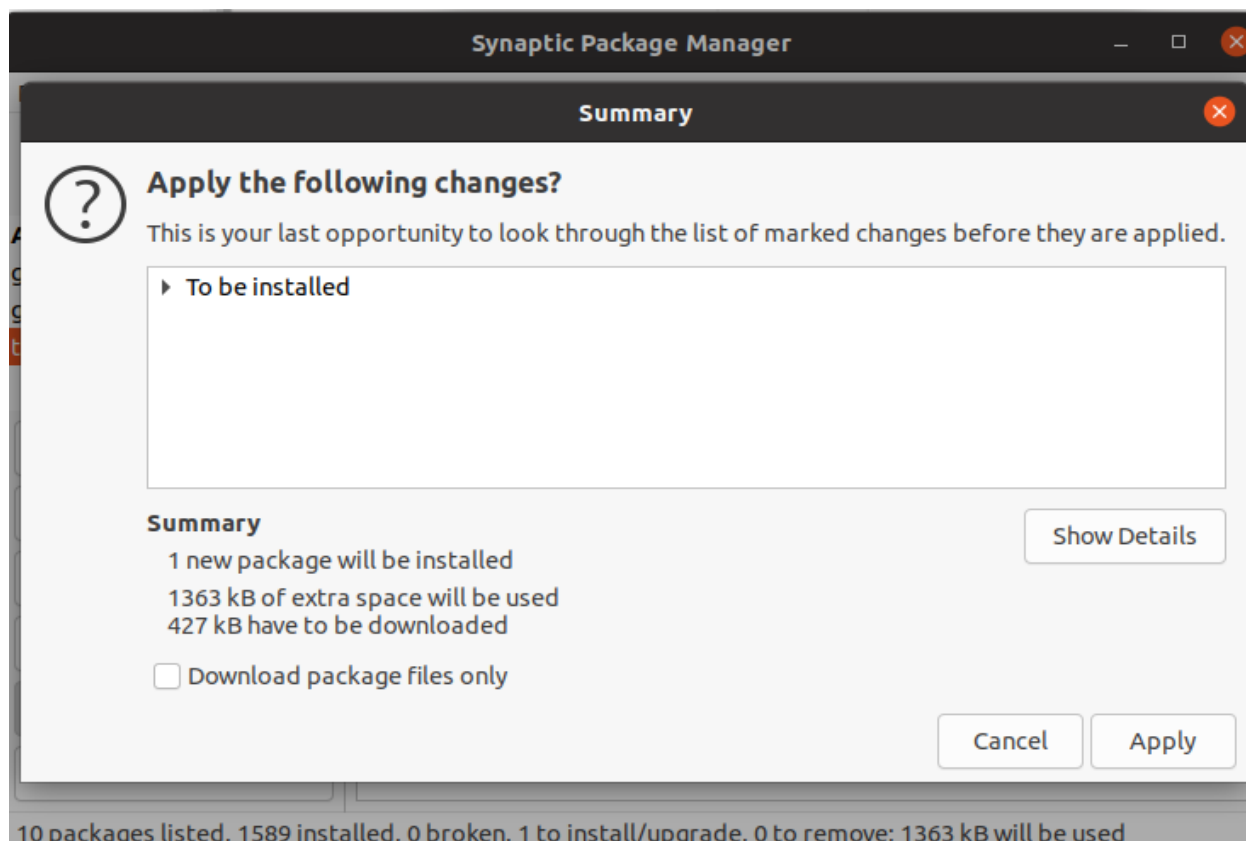
autoconf	g++-4.7	g++	gcc-4.7	gcc	curl
libboost-all-dev	cmake	subversion	gawk	make	gmt
libcurl4-gnutls-dev	gfortran-4.7	gfortran	libgdal1-dev	build-essential	csh
libfftw3-dev	libtiff5-dev	libhdf5-dev	libpcre3-dev	libnetcdf-dev	tcsh
liblapack-dev	libx11-dev				

For example; Search and how to install tcsh software is taught in the pictures below. Installing the other software above also has similar steps to installing this software.



After searching, select it and install the software using the Apply option.





- If the software is successfully installed, the software version will be written in front of it.

If version 4.7 of gcc, g ++, gfortran software is not available, first run the following commands in the Terminal and then install version 4.7 of the software through synaptic software.

```
# sudo add-apt-repository 'deb http://archive.ubuntu.com/ubuntu/ trusty main'
# sudo add-apt-repository 'deb http://archive.ubuntu.com/ubuntu/ trusty universe'
```

```
slavash@slavash-Aspire-V3-571G:~$ sudo add-apt-repository 'deb http://archive.ubuntu.com/ubuntu/ trusty main'
[sudo] password for slavash:
Hit:1 http://ir.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://ir.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://ir.archive.ubuntu.com/ubuntu focal-backports InRelease
Ign:4 http://archive.ubuntu.com/ubuntu trusty InRelease
Get:5 http://archive.ubuntu.com/ubuntu trusty Release [58.5 kB]
Hit:6 http://security.ubuntu.com/ubuntu focal-security InRelease
Get:7 http://archive.ubuntu.com/ubuntu trusty Release.gpg [933 B]
Get:8 http://archive.ubuntu.com/ubuntu trusty/main i386 Packages [1,348 kB]
Get:9 http://archive.ubuntu.com/ubuntu trusty/main amd64 Packages [1,350 kB]
Fetched 2,758 kB in 52s (52.9 kB/s)
Reading package lists... Done
slavash@slavash-Aspire-V3-571G:~$ sudo add-apt-repository 'deb http://archive.ubuntu.com/ubuntu/ trusty universe'
Hit:1 http://ir.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://ir.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://ir.archive.ubuntu.com/ubuntu focal-backports InRelease
Ign:4 http://archive.ubuntu.com/ubuntu trusty InRelease
Hit:5 http://archive.ubuntu.com/ubuntu trusty Release
Get:7 http://archive.ubuntu.com/ubuntu trusty/universe amd64 Packages [5,859 kB]
Hit:8 http://security.ubuntu.com/ubuntu focal-security InRelease
Get:9 http://archive.ubuntu.com/ubuntu trusty/universe i386 Packages [5,866 kB]
```

The g ++, gcc, and gfortran compilers are installed by default on the Ubuntu operating system. But Doris software works with version 4.7 of this software. To do this, change the default version of the above software to version 4.7 by executing the following commands in the Terminal.

```
# cd /usr/bin/
# sudo rm -f gfortran
# sudo ln -s gfortran-4.7 gfortran
# ls -ltr gfortran
# sudo rm -f g++
# sudo ln -s g++-4.7 g++
# ls -ltr g++
# sudo rm -f gcc
# sudo ln -s gcc-4.7 gcc
# ls -ltr gcc
```

```

mesg                                zmore
mformat                             znew
siavash@siavash-Aspire-V3-571G:/usr/bin$ sudo rm -f gfortran
[sudo] password for siavash:
siavash@siavash-Aspire-V3-571G:/usr/bin$ sudo ln -s gfortran-4.7 gfortran
siavash@siavash-Aspire-V3-571G:/usr/bin$ ls -ltr gfortran
lrwxrwxrwx 1 root root 12 19:25 17   مليس gfortran -> gfortran-4.7
siavash@siavash-Aspire-V3-571G:/usr/bin$ sudo rm -f g++
siavash@siavash-Aspire-V3-571G:/usr/bin$ sudo ln -s g++-4.7 g++
siavash@siavash-Aspire-V3-571G:/usr/bin$ ls -ltr g++
lrwxrwxrwx 1 root root 7 19:26 17   مليس g++ -> g++-4.7
siavash@siavash-Aspire-V3-571G:/usr/bin$ sudo rm -f gcc
siavash@siavash-Aspire-V3-571G:/usr/bin$ sudo ln -s gcc-4.7 gcc
siavash@siavash-Aspire-V3-571G:/usr/bin$ ls -ltr gcc
lrwxrwxrwx 1 root root 7 19:26 17   مليس gcc -> gcc-4.7
siavash@siavash-Aspire-V3-571G:/usr/bin$

```

- If the software version is converted correctly, the software will turn green, as shown above.

Then download the StaMPS/MTI software package from the following address and decompress it.

https://drive.google.com/file/d/1Z_tgc65EcGbXRN5SKaMmNHxyWjWE81Fl/view

Then enter the following commands in the Terminal and copy the unzipped software file to /home/usr/local/

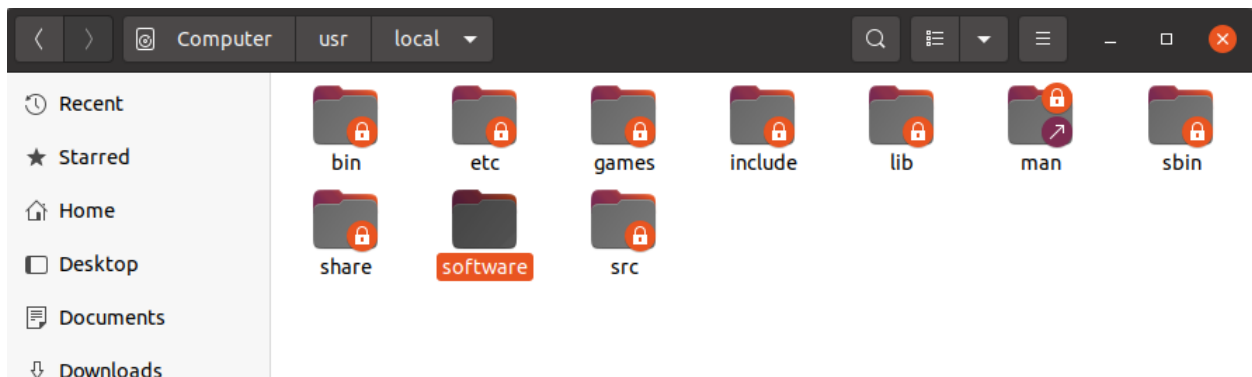
```

# cd /usr/
# sudo chown username local

siavash@siavash-Aspire-V3-571G:~$ cd /usr/
siavash@siavash-Aspire-V3-571G:/usr$ sudo chown siavash local
[sudo] password for siavash:
siavash@siavash-Aspire-V3-571G:/usr$

```

- Replace your system username instead of the username in the command above.



1.2. Install fftw

Open the Terminal and run the following commands.

```
# cd /usr/local/
# mkdir fftw
# mkdir doris
# cd software
# tar -xvf fftw-3.3.4.tar.gz
# cd fftw-3.3.4
# ./configure --prefix=/usr/local/fftw --enable-float
# make
# make install
# cd ../
```

```
siavash@siavash-Aspire-V3-571G:~$ cd /usr/local/
siavash@siavash-Aspire-V3-571G:/usr/local$ mkdir fftw
siavash@siavash-Aspire-V3-571G:/usr/local$ mkdir doris
siavash@siavash-Aspire-V3-571G:/usr/local$ cd software
siavash@siavash-Aspire-V3-571G:/usr/local/software$ tar -xvf fftw-3.3.4.tar.gz
fftw-3.3.4/
fftw-3.3.4/m4/
fftw-3.3.4/m4/f03api.sh
fftw-3.3.4/m4/Makefile.am
```

```
siavash@siavash-Aspire-V3-571G:/usr/local/software$ cd fftw-3.3.4
siavash@siavash-Aspire-V3-571G:/usr/local/software/fftw-3.3.4$ ./configure --pre
fix=/usr/local/fftw --enable-float
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a thread-safe mkdir -p... /usr/bin/mkdir -p
```

```
siavash@siavash-Aspire-V3-571G:/usr/local/software/fftw-3.3.4$ make
```

```
make[1]: Leaving directory '/usr/local/software/fftw-3.3.4'
```

```
siavash@siavash-Aspire-V3-571G:/usr/local/software/fftw-3.3.4$ make install
```

```
make[1]: Leaving directory '/usr/local/software/fftw-3.3.4/m4'
```

```
siavash@siavash-Aspire-V3-571G:/usr/local/software/fftw-3.3.4$ cd ..
```

1.3. Install Doris

Doris software is for making interferograms. Following the previous step, run the following commands in the Terminal to install version v4.06beta2 of Doris software.

```
# tar -xzf doris_v4.06beta2.tar.gz
# cd doris_v406beta2/src/
# ./configure
```

```
siavash@siavash-Aspire-V3-571G:/usr/local/software$ tar -xzf doris_v4.06beta2.tar.gz
siavash@siavash-Aspire-V3-571G:/usr/local/software$ cd doris_v406beta2/src/
siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/src$ ./configure
```

Then make the settings for the compilers as shown below.

```

compiler: g++
FFTW library: y
FFTW LIB DIR: /usr/local/fftw/lib
FFTW INCLUDE DIR: /usr/local/fftw/include
VECLIB library: n
LAPACK library: n
Little Endian: y
DEBUG version: n
Doris directory /usr/local/bin: n
Installation directory: /usr/local/doris

```

```

==> What is your C++ compiler? [g++]

==> Do you have the FFTW library (y/n)? [n]
y
==> What is the path to the FFTW library (libfftw3f.a or libfftw3f.so)? []
/usr/local/fftw/lib
==> What is the path to the FFTW include file (fftw3.h)? []
/usr/local/fftw/include
==> Do you have the VECLIB library (y/n)? [n]
n
==> Do you have the LAPACK library (y/n)? [n]
n
==> Are you working on a Little Endian (X86 PC, Intel) machine (y/n)? [y]
y
==> Do you want to compile a more verbose DEBUG version (y/n)? [n]
      (You can first compile an optimal version, then perform a make clean,
      and then compile a debug version, which you can use in case doris
      does not produce expected results)
n
==> Installation of Doris in directory: /usr/local/bin (y/n)? [y]
n
==> Enter installation directory (use absolute path):
/usr/local/doris

Creating Makefile for:
compiler:      g++
fftw:          y
FFTW LIB DIR:  /usr/local/fftw/lib
FFTW INCLUDE DIR: /usr/local/fftw/include
veclib:        n
lapack:        n
Little endian: y
DEBUG version: n
Install in dir: /usr/local/doris

```

Finally, after applying the settings and confirming them, install the software by executing the following commands.

```

# make
# make install
# cd ../

```



```

siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/src$ make
*****
* ...Compilation finished... *
*****

siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/src$ make install
* Installing doris in: /usr/local/doris
( cd ../bin; cp -f helpdoris baseline.doris.sh baseline.doris.csh construct_dem.sh coregpm.do
ris doris* heightamb phasefilt.doris plotcpm* plotoffsets* run viewanddel cpx2ps lonlathei2as
cii ascii2ascii_UTM ascii2ps* tsx* rs2* csk* gammaReadfiles.csh hhmmss2sec.py sec2hhmmss.py /
usr/local/doris/. )

* Check that /usr/local/doris is in your path search: echo $PATH .

siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/src$ cd ..

```

1.4. Install SARtools

Then run the following commands.

```

# cd SARtools
# gedit Makefile

siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2$ cd SARtools
siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/SARtools$ gedit Makefile

```

Then in the opened text file, change the software storage location to the following address according to the image and save it.

```

INSTALL_DIR = /usr/local/doris

18 #%%// TUDelft      16-Apr-2009                                     #
19 #####
20
21 ### The shell we're using ###
22 SHELL = /bin/sh
23
24 ### Installdirdef should exist!
25 INSTALL_DIR = /usr/local/doris|
26
27 ### GCC compiler
28 CC = g++
29 CFLAGS = -O3 -D_FILE_OFFSET_BITS=64 -D_LARGEFILE_SOURCE -D_LARGEFILE
30

```

Then run the following commands to complete the installation process at this stage.

```

# make
# make install

```



```
# cd ../../

siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/SARtools$ make
g++ -O3 -D_FILE_OFFSET_BITS=64 -D_LARGEFILE_SOURCE -D_LARGEFILE64_SOURCE dateconv.o -o dateconv
siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/SARtools$ make install
make[1]: Leaving directory '/usr/local/software/doris_v406beta2/SARtools'
siavash@siavash-Aspire-V3-571G:/usr/local/software/doris_v406beta2/SARtools$ cd ../../
```

1.5. Install Envisattools

Run the following commands in the Terminal.

```
# tar -xzf ENVISAT_TOOLS_v2.3.tar.gz
# cd ENVISAT_TOOLS_v2.3
# gedit Makefile

siavash@siavash-Aspire-V3-571G:/usr/local/software$ tar -xzf ENVISAT_TOOLS_v2.3.tar.gz
siavash@siavash-Aspire-V3-571G:/usr/local/software$ cd ENVISAT_TOOLS_v2.3
siavash@siavash-Aspire-V3-571G:/usr/local/software/ENVISAT_TOOLS_v2.3$ gedit Makefile
```

Then in the opened text file, change the software storage location to the following address according to the image and save it.

```
INSTALL_DIR = /usr/local/doris

18 #####
19
20
21 ### Check if you agree with the options below (install_dir OK?)
22 ### then simply type: "make install"
23 SHELL = /bin/sh
24 CC = gcc
25 #CFLAGS = -O3
26 #CFLAGS = -m32 # for 64-bit systems, it requires compatibility
27 LFLAGS = -lm
28 INSTALL_DIR = /usr/local/doris|
29
30
31
```

Then, run the following commands to complete the installation process at this stage.

```
# make
# make install
# cd ../../

siavash@siavash-Aspire-V3-571G:/usr/local/software/ENVISAT_TOOLS_v2.3$ make
```

```
type [enabled by default]
siavash@siavash-Aspire-V3-571G:/usr/local/software/ENVISAT_TOOLS_v2.3$ make install
```

```
-----
FINISHED INSTALLING ENVISAT TOOLS...
executables in: /usr/local/doris
-----
```

```
siavash@siavash-Aspire-V3-571G:/usr/local/software/ENVISAT_TOOLS_v2.3$ cd ../
```

1.6. Install getorb

The getorb software is for reading orbital information. Run the following commands to install it in the Terminal.

```
# tar -xzf getorb_2.3.2.tar.gz
# cd getorb
# gedit Makefile
```

```
siavash@siavash-Aspire-V3-571G:/usr/local/software$ tar -xzf getorb_2.3.2.tar.gz
siavash@siavash-Aspire-V3-571G:/usr/local/software$ cd getorb
siavash@siavash-Aspire-V3-571G:/usr/local/software/getorb$ gedit Makefile
```

Then in the opened text file, change the software storage location to the following address according to the image and remove the # sign next to gfortran to identify the system as a compiler and finally save it.

```
BIN_DIR = /usr/local/doris
```

```
21 # 1. After compiling and testing of the software, you may like to put the
22 #     executable in a 'bin' directory that is in your path.
23 #
24 #BIN_DIR           = /home/altim/bin
25 BIN_DIR           = /usr/local/doris
26 #
27 # 2. 'make' is usually able to determine which is your Fortran compiler and
28 #     C compiler. However, if 'make' crashes, because it can't find the
29 #     compilers, specify the proper compiler names.
30 #     The C-preprocessor usually has to be defined.
31 #
32 #FC                = fc          # Generic
33 #FC                = g77         # Linux, older versions of gcc
34 FC                = gfortran    # Linux, gcc version 4 or higher
35 #FC                = ifort       # Intel fortran compiler
36 #
37 #CC                = cc          # Generic
38 #CC                = gcc         # Linux
```

Then run the following commands to complete the installation process at this stage.

```
# make
# make install
```

```
# cd ../
siavash@siavash-Aspire-V3-571G:/usr/local/software/getorb$ gedit Makefile
siavash@siavash-Aspire-V3-571G:/usr/local/software/getorb$ make
gfortran      mdate_main.o getorb.a -o mdate
siavash@siavash-Aspire-V3-571G:/usr/local/software/getorb$ make install
chmod 755 getorb lodr mdate
mv getorb lodr mdate /usr/local/doris
siavash@siavash-Aspire-V3-571G:/usr/local/software/getorb$ cd ../
```

1.7. Install snaphu

Snaphu software is used for unwrapping. Run the following commands to install this software in the Terminal.

```
# tar -xzf snaphu-v1.4.2.tar.gz
# cd snaphu-v1.4.2/src/
# gedit Makefile
siavash@siavash-Aspire-V3-571G:/usr/local/software$ tar -xzf snaphu-v1.4.2.tar.gz
siavash@siavash-Aspire-V3-571G:/usr/local/software$ cd snaphu-v1.4.2/src
siavash@siavash-Aspire-V3-571G:/usr/local/software/snaphu-v1.4.2/src$ gedit Makefile
```

Then in the opened text file, change the software storage location to the following address according to the image and save it.

```
INSTALLDIR = /usr/local/doris
14 # CS2 MCF solver module.
15
16 CC          =      cc
17 CFLAGS      =      -O3    # -D NO_CS2
18 LIBS        =      -lm
19 BINDIR      =      ../bin
20 INSTALLDIR  =      /usr/local/doris
21 MANDIR      =      /usr/local/man
22 SNAPHU      =      $(BINDIR)/snaphu
23 SNAPHUMAN   =      ../man/man1/snaphu.1
24 OBJS        =      snaphu_tile.o \
25 snaphu_solver.o \
```

Then run the following commands to complete the installation process at this stage.

```
# sudo mkdir /usr/local/man/man1
# make
# sudo make install
```

```
# cd ../../

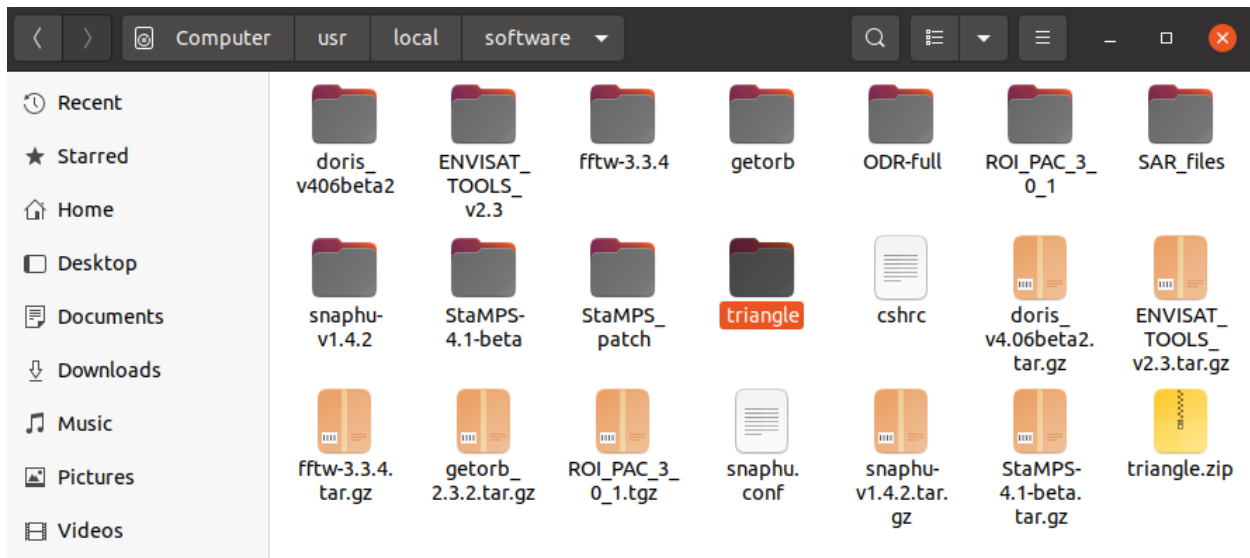
siavash@siavash-Aspire-V3-571G:/usr/local/software/snaphu-v1.4.2/src$ sudo mkdir /usr/local/man/man1
[sudo] password for siavash:
siavash@siavash-Aspire-V3-571G:/usr/local/software/snaphu-v1.4.2/src$ make

snaphu -o ./bin/snaphu \
-lm
siavash@siavash-Aspire-V3-571G:/usr/local/software/snaphu-v1.4.2/src$ sudo make install

cp ../man/snaphu.1 /usr/local/man/man1/
siavash@siavash-Aspire-V3-571G:/usr/local/software/snaphu-v1.4.2/src$ cd ../../
```

1.8. Install triangle

Triangle software is used for triangulation. To install this software, first, manually extract the triangle zip file from the software folder in /home/usr/local/



Then run the following commands in the Terminal.

```
# cd triangle
# make
# cd ../

siavash@siavash-Aspire-V3-571G:/usr/local/software$ cd triangle
siavash@siavash-Aspire-V3-571G:/usr/local/software/triangle$ make

cc -O -DLINUX -I/usr/X11R6/include -L/usr/X11R6/lib -o ./showme ./showme.c -lX11
siavash@siavash-Aspire-V3-571G:/usr/local/software/triangle$ cd ../
```

1.9. StaMPS/MTI installation

Install StaMPS software v4.1b by running the following commands in the Terminal. This software is used for PS and SBAS time-series.

```
# tar -xzf StaMPS-4.1-beta.tar.gz
# cd StaMPS-4.1-beta/src
# make
# make install
# cp ../StaMPS_CONFIG.tcsh ~
# cd ../../

slavash@slavash-Aspire-V3-571G:/usr/local/software$ tar -xzf StaMPS-4.1-beta.tar.gz
slavash@slavash-Aspire-V3-571G:/usr/local/software$ cd StaMPS-4.1-beta/src
slavash@slavash-Aspire-V3-571G:/usr/local/software/StaMPS-4.1-beta/src$ make

g++ -O3 pscdem.o -o pscdem
slavash@slavash-Aspire-V3-571G:/usr/local/software/StaMPS-4.1-beta/src$ make install

* Removed executables in source dir: /usr/local/software/StaMPS-4.1-beta/src.
slavash@slavash-Aspire-V3-571G:/usr/local/software/StaMPS-4.1-beta/src$ cp ../StaMPS_CONFIG.tcsh ~
slavash@slavash-Aspire-V3-571G:/usr/local/software/StaMPS-4.1-beta/src$ cd ../../
```

1.10. Install ROI_PAC

ROI_PAC software is used to convert or create Level 1 or SLC images. Run the following commands to install this software.

```
# tar -xzvf ROI_PAC_3_0_1.tgz
# cd ROI_PAC_3_0_1/ROI_PAC
# which ifort g95 f90 pgf95 f95 xlf gfortran cc gcc icc
# touch aclocal.m4 Makefile.in configure
# ./contrib/install-fftw.sh CC=g++

slavash@slavash-Aspire-V3-571G:/usr/local/software$ tar -xzvf ROI_PAC_3_0_1.tgz
ROI_PAC_3_0_1/
ROI_PAC_3_0_1/.DS_Store
ROI_PAC_3_0_1/Makefile

slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ which ifort g95 f90 pgf95 f95 xlf gfortran cc gcc icc
/usr/bin/gfortran
/usr/bin/cc
/usr/bin/gcc
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ touch aclocal.m4 Makefile.in
configure
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ ./contrib/install-fftw.sh CC=g++
```

Then 4 files with export names FFTW_INC_DIR... and setenv FFTW_LIB_DIR... are created in the Terminal.

```

100 3413k 100 3413k 0 0 437k 0 0:00:07 0:00:07 --:--:-- 714k
Doing: tar xzf /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetDist/fftw-3.2.2.tar.gz
Doing: configure --enable-float --prefix=/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-21031
0-1331 CC=g++
make install
(This may take a few minutes.)
Looks like build was successful.

To setup your environment for multibuild.sh

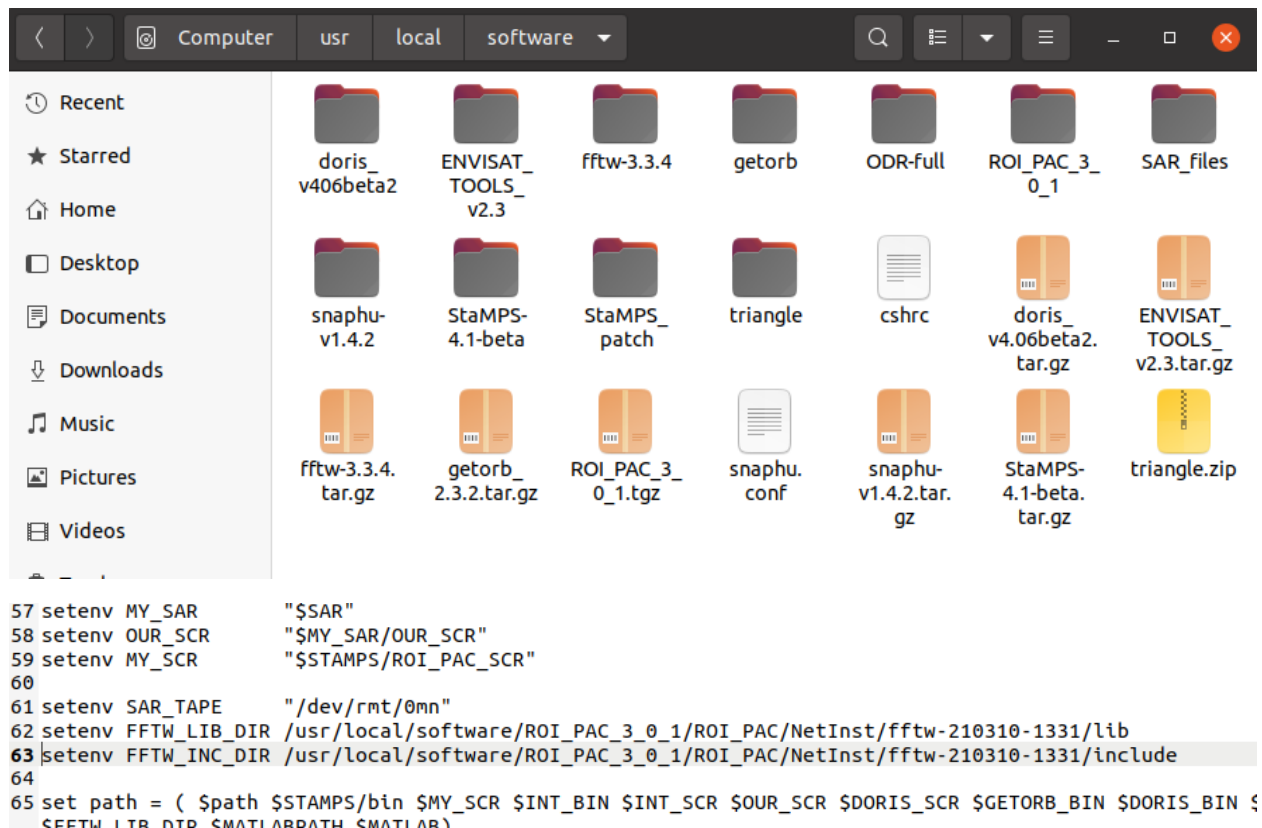
export FFTW_LIB_DIR=/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/lib
export FFTW_INC_DIR=/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/include

or

setenv FFTW_LIB_DIR /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/lib
setenv FFTW_INC_DIR /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/include
siavash@siavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$

```

Then open the cshrc file in /home/usr/local/software/ and copy the two setenv FFTW_LIB_DIR lines created in the Terminal instead of the previous two setenv FFTW_LIB_DIR lines and save the file.



- setenv code FFTW_LIB_DIR... in cshrc file

Also run the two export lines FFTW_INC_DIR... built in the Terminal, respectively.

```
export FFTW_LIB_DIR=/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/lib
export FFTW_INC_DIR=/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/include

or

setenv FFTW_LIB_DIR /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/lib
setenv FFTW_INC_DIR /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/include
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ export FFTW_LIB_DIR=/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/lib
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ export FFTW_INC_DIR=/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/include
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$
```

After, run the following commands according to the images below in the Terminal.

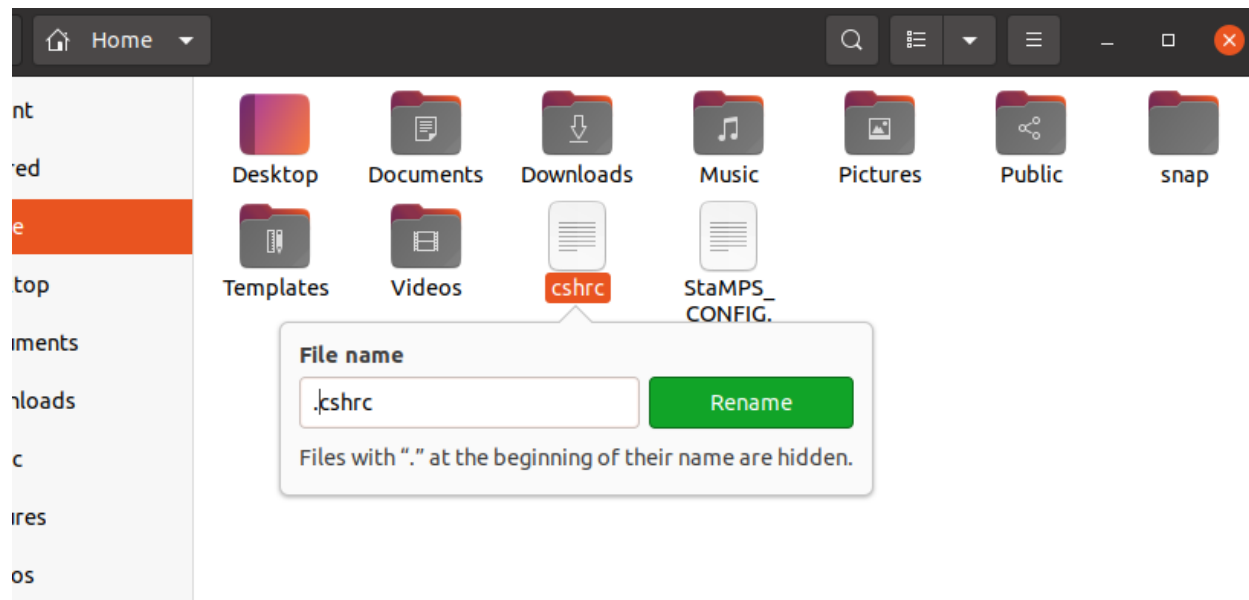
```
# ./contrib/multibuild.sh
# mkdir INT_BIN
# cp multibuild-*/installs/defaults/bin/* INT_BIN
# cp DOC/SAR_CONFIG .
# source SAR_CONFIG

slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ ./contrib/multibuild.sh
Will configure using /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/configure
Doing builds in multibuild-210310-1337
common_options are >LDFLAGS=-L/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/lib#C
PPFLAGS=-I/usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-210310-1331/include<
installation directory is /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/multibuild-210310-1337/installs

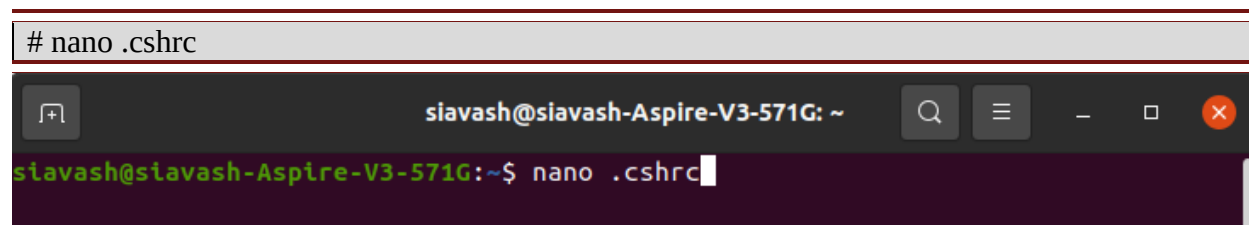
See README_ROI_BUILD for information on how to run
  contrib/multitest.sh
to do test runs using data from TEST_DIR.tar
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ mkdir INT_BIN
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ cp multibuild-*/installs/defaults/bin/* INT_BIN
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ cp DOC/SAR_CONFIG .
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$ source SAR_CONFIG
slavash@slavash-Aspire-V3-571G:/usr/local/software/ROI_PAC_3_0_1/ROI_PAC$
```

In the following:

- Copy the SAR_files file from /home/usr/local/software/ to /home/usr/local/
- Copy the cshrc file from /home/usr/local/software/ to /home/ and rename it to .cshrc



Then run the following command in the Terminal.



After change the existing addresses as shown in the image and save it.

The screenshot shows a terminal window with the title 'slavash@slavash-Aspire-V3-571G: ~'. The window displays the contents of the '.cshrc' file being edited with 'GNU nano 4.8'. The file contains environment variable settings for various software tools and directories. At the bottom, a status bar indicates 'Read 67 lines' and provides keyboard shortcuts for common editor actions.

```

GNU nano 4.8                                .cshrc
setenv EDITOR xemacs
setenv PAGER less
setenv STAMPS      /usr/local/software/StaMPS-4.1-beta
setenv SAR          /usr/local/software/ROI_PAC_3_0_1
setenv GETORB_BIN   /usr/local/doris
setenv SAR_ODR_DIR  /usr/local/SAR_files/ODR
#setenv SAR_PRC_DIR  "/home/ahopper/software/SAR_FILES/PRC"
setenv VOR_DIR      /usr/local/SAR_files/DOR_VOR
setenv INS_DIR      /usr/local/SAR_files/INS
setenv DORIS_BIN    /usr/local/doris
setenv TRIANGLE_BIN /usr/local/software/triangle
setenv SNAPHU_BIN   /usr/local/doris

setenv ROI_PAC      "$SAR/ROI_PAC"
#####
# ROI_PAC VERSION 3
#####
setenv INT_BIN      "$ROI_PAC/INT_BIN"
setenv INT_SCR      "$ROI_PAC/INT_SCR"
#####
[ Read 67 lines ]
^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify  ^C Cur Pos
^X Exit      ^R Read File ^\ Replace   ^U Paste Text ^T To Spell ^_ Go To Line

```

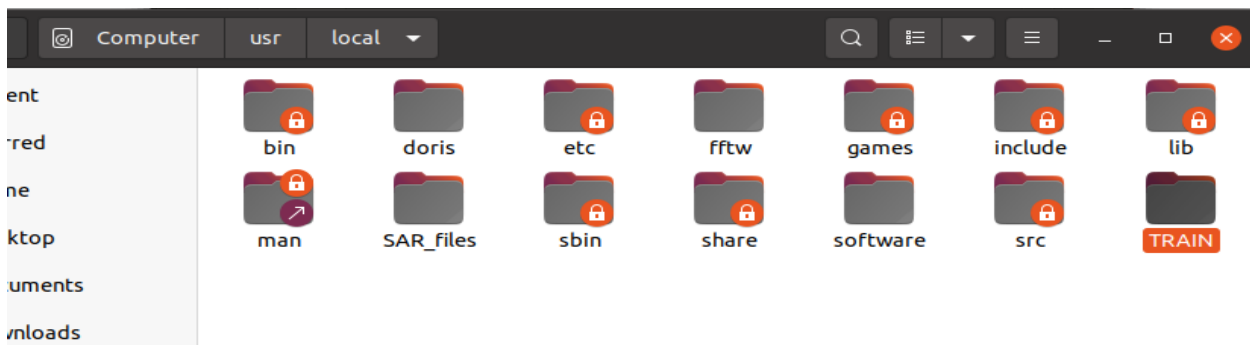
1.11. Install TRAIN

TRAIN software is used to correct the atmospheric phase over time series.

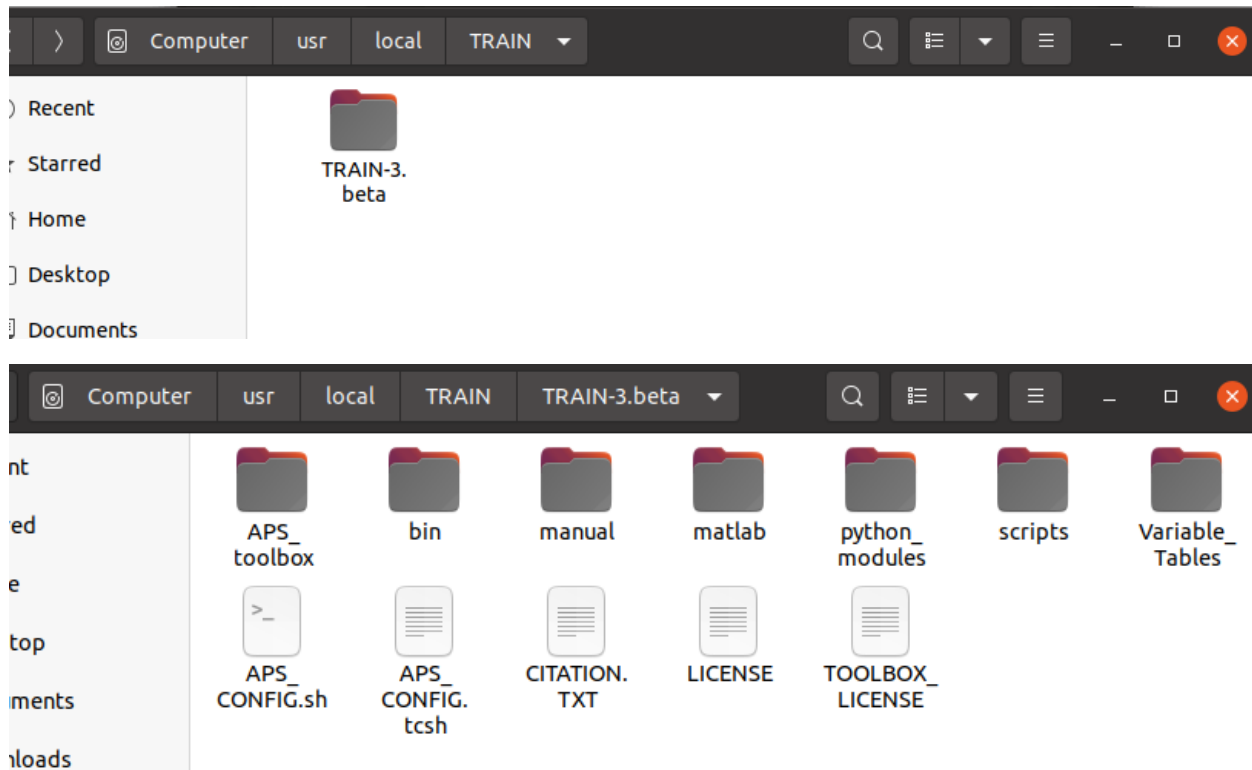
Download the TRAIN software from the following address and unzip it

https://drive.google.com/file/d/1vgJ26zMuHf33Joan1uhaDwTx8lUf0JI_/view

Then create a folder named TRAIN in /usr/local/



Then copy the TRAIN-3.beta folder into the TRAIN folder and create a folder called APS_toolbox.



Open the APS_CONFIG.tcsh file and edit the APS_toolbox folder location address and save the file.

The screenshot shows a text editor window titled 'APS_CONFIG.tcsh' with the path '/usr/local/TRAIN/TRAIN-3.beta'. The file contains the following text:

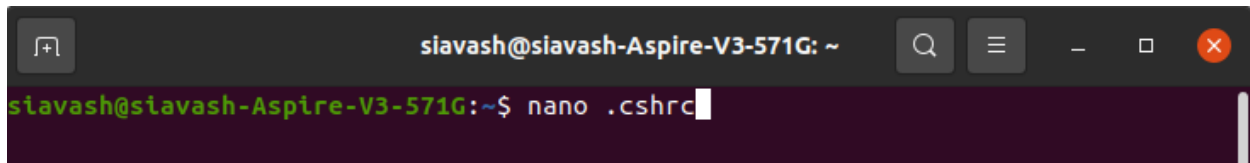
```

9 # This program is distributed in the hope that it will be useful,
10 # but WITHOUT ANY WARRANTY; without even the implied warranty of
11 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
12 # GNU General Public License for more details.
13 #
14 # You should have received a copy of the GNU General Public License along
15 # with this program; if not, write to the Free Software Foundation, Inc.,
16 # 51 Franklin Street, Fifth Floor, Boston, MA 02110-1301 USA.
17 #
18
19
20 # Give the correct path to the APS toolbox
21 setenv APS_toolbox "/usr/local/TRAIN/TRAIN-3.beta"
22 setenv PYTHONPATH "/Users/dbekaert/Software/svn_aps/trunk/python_modules/"
23 # full path to the get_modis.py file
24 setenv get_modis_filepath "/Users/dbekaert/Software/oscar/get_modis.py"
25 setenv get_ecmwf_filepath "/Users/dbekaert/Software/ecmwfapi/api.py"

```

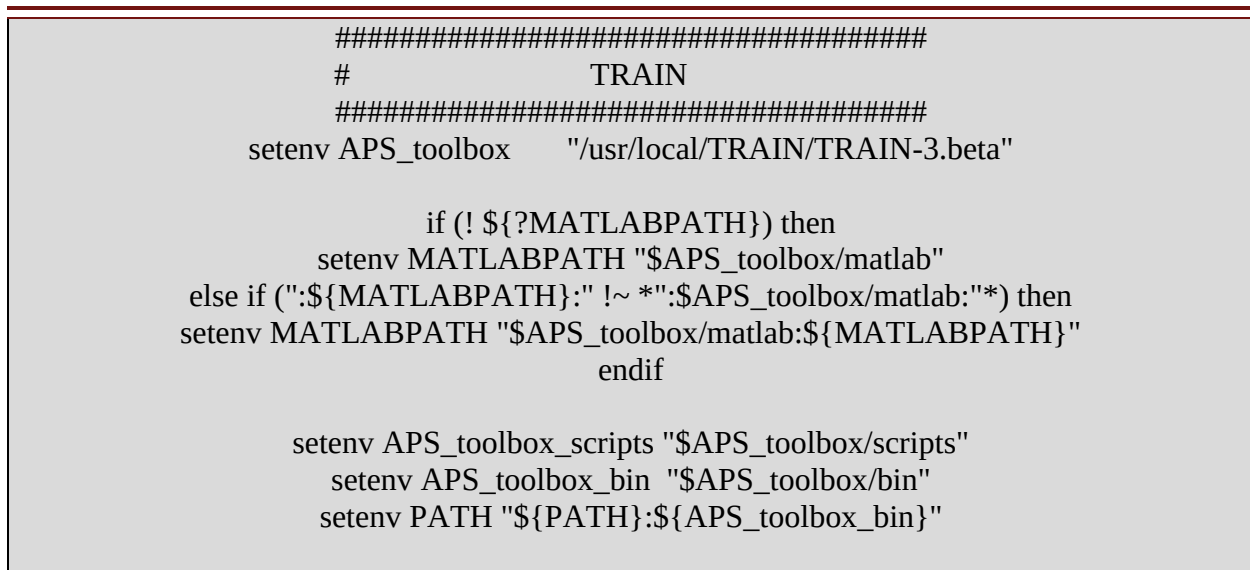
Then run the following command.

```
# nano .cshrc
```



```
siavash@siavash-Aspire-V3-571G: ~$ nano .cshrc
```

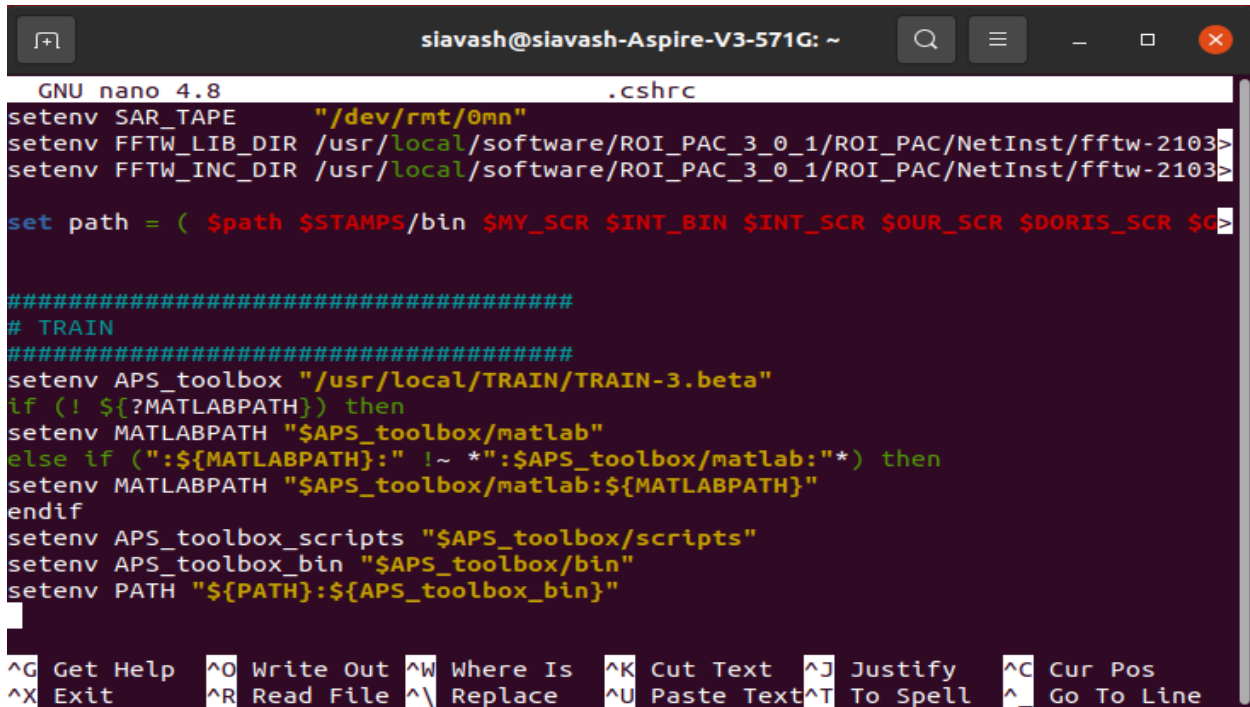
Add the following code to the .cshrc file and save it.



```
#####
# TRAIN
#####
setenv APS_toolbox "/usr/local/TRAIN/TRAIN-3.beta"

if (! ${?MATLABPATH}) then
    setenv MATLABPATH "$APS_toolbox/matlab"
else if (":${MATLABPATH}:" !~ *":$APS_toolbox/matlab:") then
    setenv MATLABPATH "$APS_toolbox/matlab:${MATLABPATH}"
endif

setenv APS_toolbox_scripts "$APS_toolbox/scripts"
setenv APS_toolbox_bin "$APS_toolbox/bin"
setenv PATH "${PATH}:${APS_toolbox_bin}"
```



```
GNU nano 4.8 .cshrc
setenv SAR_TAPE "/dev/rmt/0mn"
setenv FFTW_LIB_DIR /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-2103>
setenv FFTW_INC_DIR /usr/local/software/ROI_PAC_3_0_1/ROI_PAC/NetInst/fftw-2103>

set path = ( $path $STAMPS/bin $MY_SCR $INT_BIN $INT_SCR $OUR_SCR $DORIS_SCR $G>

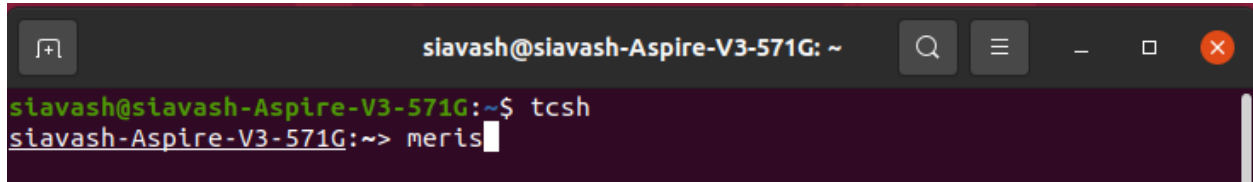
#####
# TRAIN
#####
setenv APS_toolbox "/usr/local/TRAIN/TRAIN-3.beta"
if (! ${?MATLABPATH}) then
    setenv MATLABPATH "$APS_toolbox/matlab"
else if (":${MATLABPATH}:" !~ *":$APS_toolbox/matlab:") then
    setenv MATLABPATH "$APS_toolbox/matlab:${MATLABPATH}"
endif
setenv APS_toolbox_scripts "$APS_toolbox/scripts"
setenv APS_toolbox_bin "$APS_toolbox/bin"
setenv PATH "${PATH}:${APS_toolbox_bin}"

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line
```

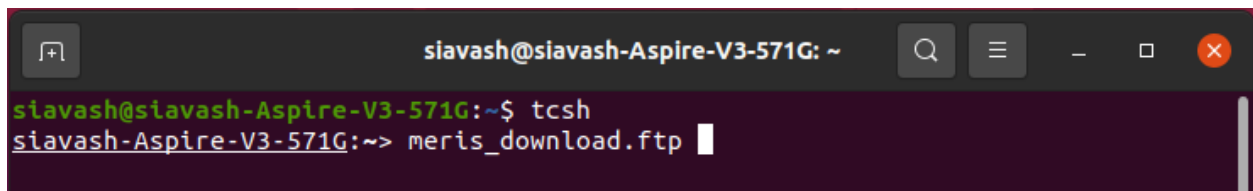
Finally, by opening the Terminal and executing the following commands, you can make sure the installation is complete.

```
# tcsh  
# meris
```

After the meris command, by pressing the Tab key on the keyboard, you will see the following image, which shows the correct installation of the software.

A terminal window titled 'siavash@siavash-Aspire-V3-571G: ~' with search, menu, and window control icons. The prompt is 'siavash@siavash-Aspire-V3-571G:~\$'. The user enters 'tcsh', and the prompt changes to 'siavash-Aspire-V3-571G:~>'. The user then enters 'meris' followed by a cursor.

```
siavash@siavash-Aspire-V3-571G:~$ tcsh  
siavash-Aspire-V3-571G:~> meris
```

A terminal window titled 'siavash@siavash-Aspire-V3-571G: ~' with search, menu, and window control icons. The prompt is 'siavash@siavash-Aspire-V3-571G:~\$'. The user enters 'tcsh', and the prompt changes to 'siavash-Aspire-V3-571G:~>'. The user then enters 'meris_download.ftp' followed by a cursor.

```
siavash@siavash-Aspire-V3-571G:~$ tcsh  
siavash-Aspire-V3-571G:~> meris_download.ftp
```

1.12. Install GMT

Run the following commands in the Terminal.

```
# sudo apt-get install tcsh  
# sudo apt install graphicsmagick  
# sudo apt install python-sphin  
# sudo apt-get install libgmt-dev  
# sudo apt-get install gmt gmt-dcw gmt-gshhg  
# sudo apt-get install gmt  
# sudo apt-get install git
```

```

siavash@siavash-Aspire-V3-571G: ~
siavash@siavash-Aspire-V3-571G:~$ sudo apt-get install tcsh
[sudo] password for siavash:
Reading package lists... Done
Building dependency tree
Reading state information... Done
tcsh is already the newest version (6.21.00-1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
siavash@siavash-Aspire-V3-571G:~$ sudo apt-get install graphicsmagick
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  gsfontr libgraphicsmagick-q16-3
Suggested packages:
  graphicsmagick-dbg
The following NEW packages will be installed:
  graphicsmagick gsfontr libgraphicsmagick-q16-3
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 4,914 kB of archives.
After this operation, 13.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ir.archive.ubuntu.com/ubuntu focal/universe amd64 libgraphicsmagick
-q16-3 amd64 1.4+really1.3.35-1 [1,136 kB]
Get:2 http://ir.archive.ubuntu.com/ubuntu focal/universe amd64 graphicsmagick am

```

Then from the following address, files gshhg-gmt-2.3.7.tar.gz, gmt-5.4.5-src.tar.gz and dcw-gmt-1.1.4.tar.gz Download.



GMT is available on Windows, macOS, Linux and FreeBSD. Source and binary packages are provided from [GitHub](#):

- [GMT 6.1.1 \(recommended\)](#)
- [GMT 6.1.0](#)
- [GMT 6.0.0](#)
- [GMT 5.4.5](#)
- [GMT 4.5.18](#)

Support Data

- [GSHHG 2.3.7](#): The Global Self-consistent, Hierarchical, High-resolution Geography Database
- [DCW 1.1.4](#): The Digital Chart of the World Data

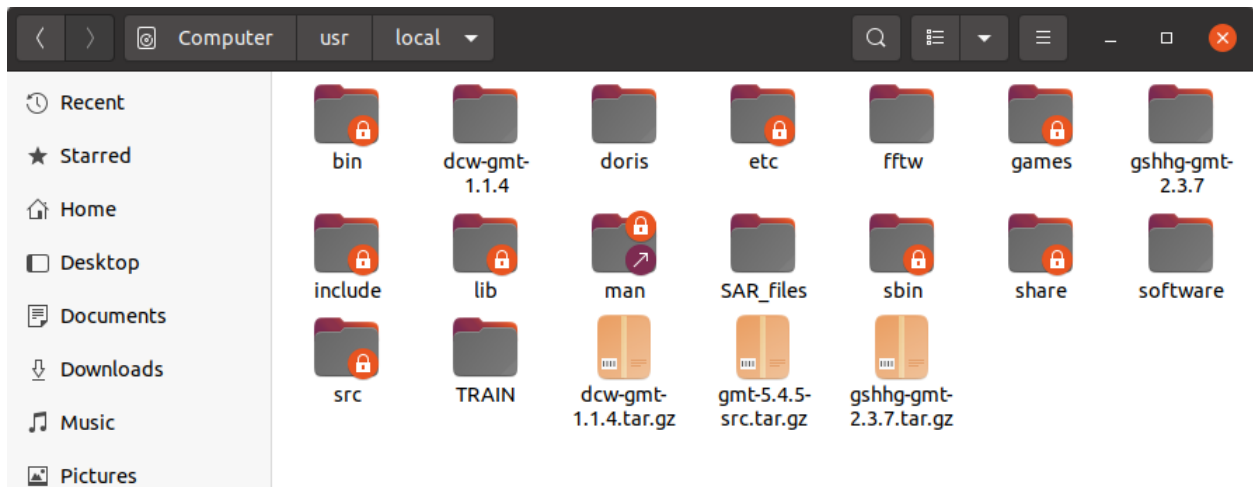
Note: You may also get GMT and its support data from any of [the GMT FTP mirrors](#). Try the site that is closest to you to minimize transmission times.

gmt-5.4.5-src.tar.gz	Source code
gmt-5.4.5-src.tar.xz	Source code
gmt-5.4.5-win32.exe	Windows installer (32bit)
gmt-5.4.5-win64.exe	Windows installer (64bit)

Assets 7

gmt-5.4.5-darwin-x86_64.dmg	212 MB
gmt-5.4.5-src.tar.gz	67.5 MB
gmt-5.4.5-src.tar.xz	56.4 MB
gmt-5.4.5-win32.exe	154 MB
gmt-5.4.5-win64.exe	158 MB
Source code (zip)	
Source code (tar.gz)	

Copy all three downloaded files to /usr/local/ and extract the two files gshhg-gmt-2.3.7.tar.gz and dcw-gmt-1.1.4.tar.gz manually.



Next, go to /usr/local/ terminal and run the following commands.

```
# tar -xvf gmt-5.4.5-src.tar.gz
# cd gmt-5.4.5
# cp cmake/ConfigUserTemplate.cmake cmake/ConfigUser.cmake
# mkdir build
# cd build
# cmake -DCMAKE_INSTALL_PREFIX=/usr/local -
DCMAKE_BUILD_TYPE=RelWithDebInfo ..
```

```
# cmake ..
# make
# sudo make install
```

```
siavash@siavash-Aspire-V3-571G:/usr/local$ tar -xvf gmt-5.4.5-src.tar.gz
gmt-5.4.5/
gmt-5.4.5/CMakeLists.txt
gmt-5.4.5/COPYING.LESSERV3
gmt-5.4.5/COPYINGv3
gmt-5.4.5/LICENSE.TXT
```

```
gmt-5.4.5/src/x2sys/x2sys_solve.c
gmt-5.4.5/src/xyz2grd.c
siavash@siavash-Aspire-V3-571G:/usr/local$ cd gmt-5.4.5
siavash@siavash-Aspire-V3-571G:/usr/local/gmt-5.4.5$ cp cmake/ConfigUserTemplate
.cmake cmake/ConfigUser.cmake
siavash@siavash-Aspire-V3-571G:/usr/local/gmt-5.4.5$ mkdir build
siavash@siavash-Aspire-V3-571G:/usr/local/gmt-5.4.5$ cd build
siavash@siavash-Aspire-V3-571G:/usr/local/gmt-5.4.5/build$ cmake -DCMAKE_INSTALL
_PREFIX=/usr/local -DCMAKE_BUILD_TYPE=RelWithDebInfo ..
CMake Deprecation Warning at CMakeLists.txt:63 (cmake_policy):
  The OLD behavior for policy CMP0026 will be removed from a future version
  of CMake.

The cmake-policies(7) manual explains that the OLD behaviors of all
```

```
-- Generating done
-- Build files have been written to: /usr/local/gmt-5.4.5/build
siavash@siavash-Aspire-V3-571G:/usr/local/gmt-5.4.5/build$ cmake ..
CMake Deprecation Warning at CMakeLists.txt:63 (cmake_policy):
  The OLD behavior for policy CMP0026 will be removed from a future version
  of CMake.
```

```
-- Generating done
-- Build files have been written to: /usr/local/gmt-5.4.5/build
siavash@siavash-Aspire-V3-571G:/usr/local/gmt-5.4.5/build$ make
```

```
[100%] Building C object src/CMakeFiles/gmt.dir/gmt.c.o
[100%] Linking C executable gmt
[100%] Built target gmt
siavash@siavash-Aspire-V3-571G:/usr/local/gmt-5.4.5/build$ sudo make install
```

Run the `gmt` or `grdinfo` command in the Terminal to sure proper installation .

```

siavash@siavash-Aspire-V3-571G: ~
siavash@siavash-Aspire-V3-571G:~$ gmt

GMT - The Generic Mapping Tools, Version 5.4.5 [64-bit] [8 cores]
(c) 1991-2019 Paul Wessel, Walter H. F. Smith, Remko Scharroo, Joaquim Luis, and
  Florian Wobbe

Supported in part by the US National Science Foundation (http://www.nsf.gov/)
and volunteers from around the world (see http://gmt.soest.hawaii.edu/).

This program comes with NO WARRANTY, to the extent permitted by law.
You may redistribute copies of this program under the terms of the

siavash@siavash-Aspire-V3-571G:~$ grdinfo
grdinfo(core) 5.4.5 [64-bit] - Extract information from grids

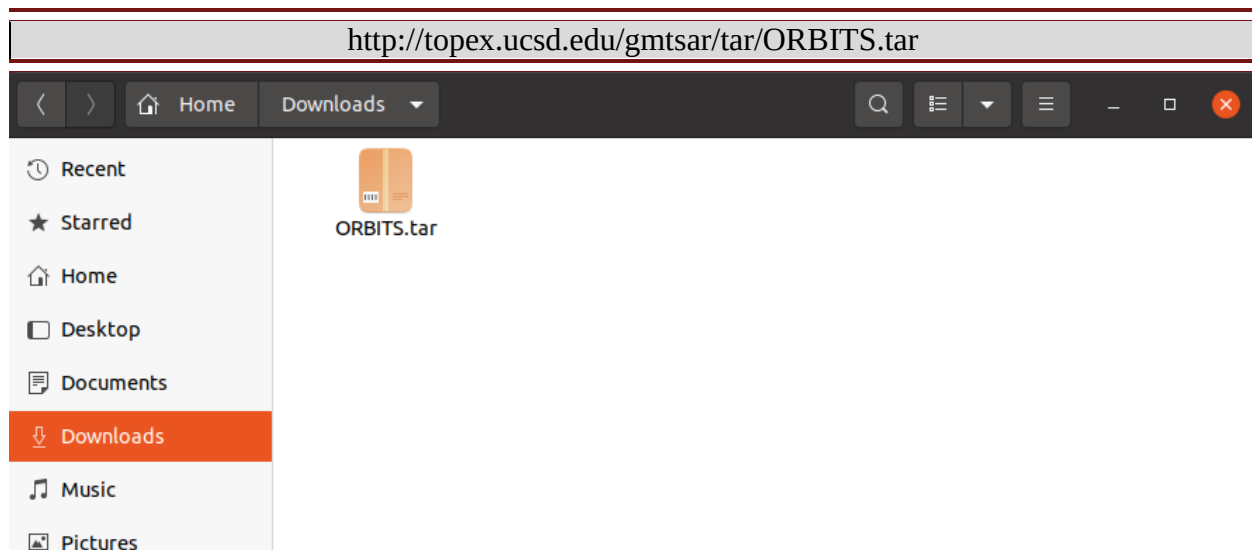
usage: grdinfo <grid> [-C] [-D[<offx>[/<offy>]][+i]] [-F] [-I[<dx>[/<dy>]]|b|i|r]]
  [-L[a|0|1|2|p]] [-M]
    [-R<west>/<east>/<south>/<north>[+r]] [-T[<dz>][+a[<alpha>]][+s]] [-V[<l
level>]] [-f[i|o]<info>]
    [-ho[<nrecs>][+c][+d][+r<remark>][+t<title>]]

<grid> may be one or more grid files

```

1.13. Install GMTSAR

Download the ORBITS.tar file from the following address and copy it to `/home/Download/`



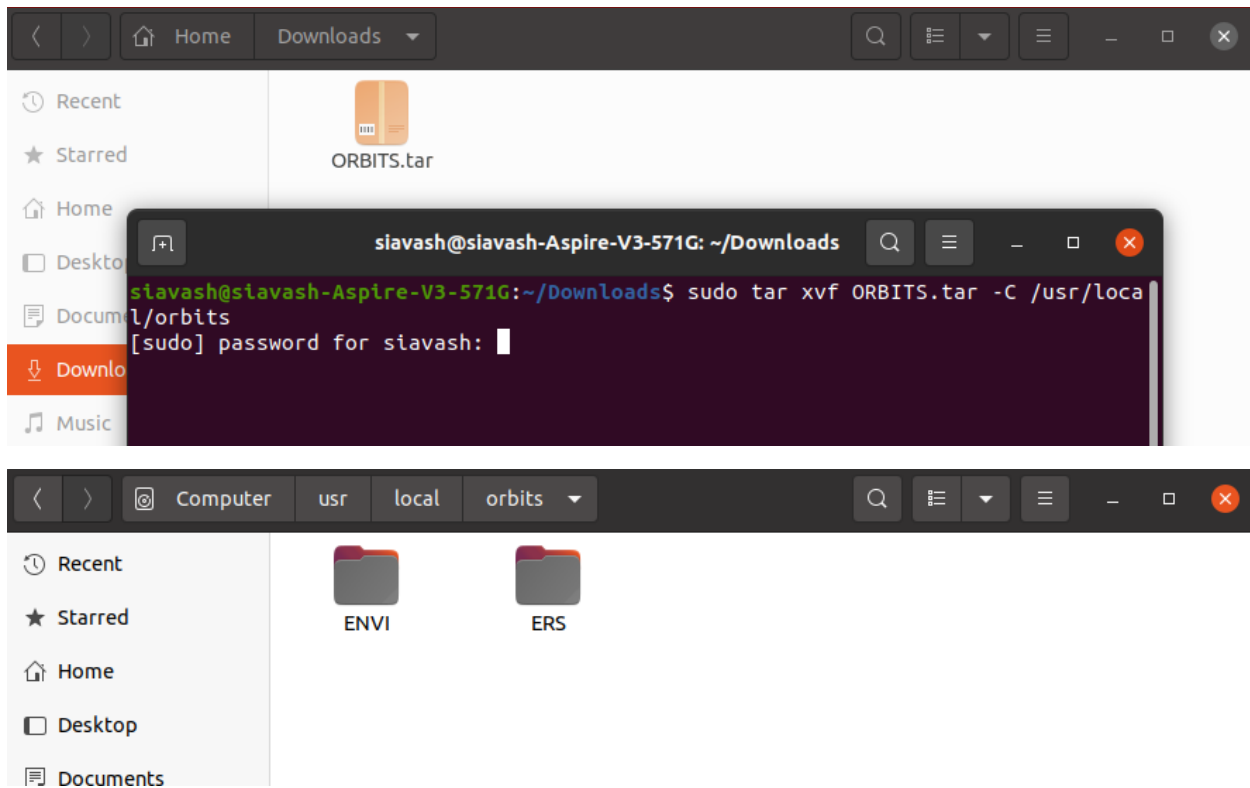
Then, run the following commands in Terminal.

```
# cd /usr/local
# sudo -i
# cd /usr/local
# mkdir orbits
# cd orbits
```

```
siavash@siavash-Aspire-V3-571G:~$ cd /usr/local
siavash@siavash-Aspire-V3-571G:/usr/local$ sudo -i
[sudo] password for siavash:
root@siavash-Aspire-V3-571G:~# cd /usr/local
root@siavash-Aspire-V3-571G:/usr/local# mkdir orbits
root@siavash-Aspire-V3-571G:/usr/local# cd orbits
root@siavash-Aspire-V3-571G:/usr/local/orbits#
```

Next, open the Terminal in the Download folder and run the following command.

```
# sudo tar xvf ORBITS.tar -C /usr/local/orbits
```

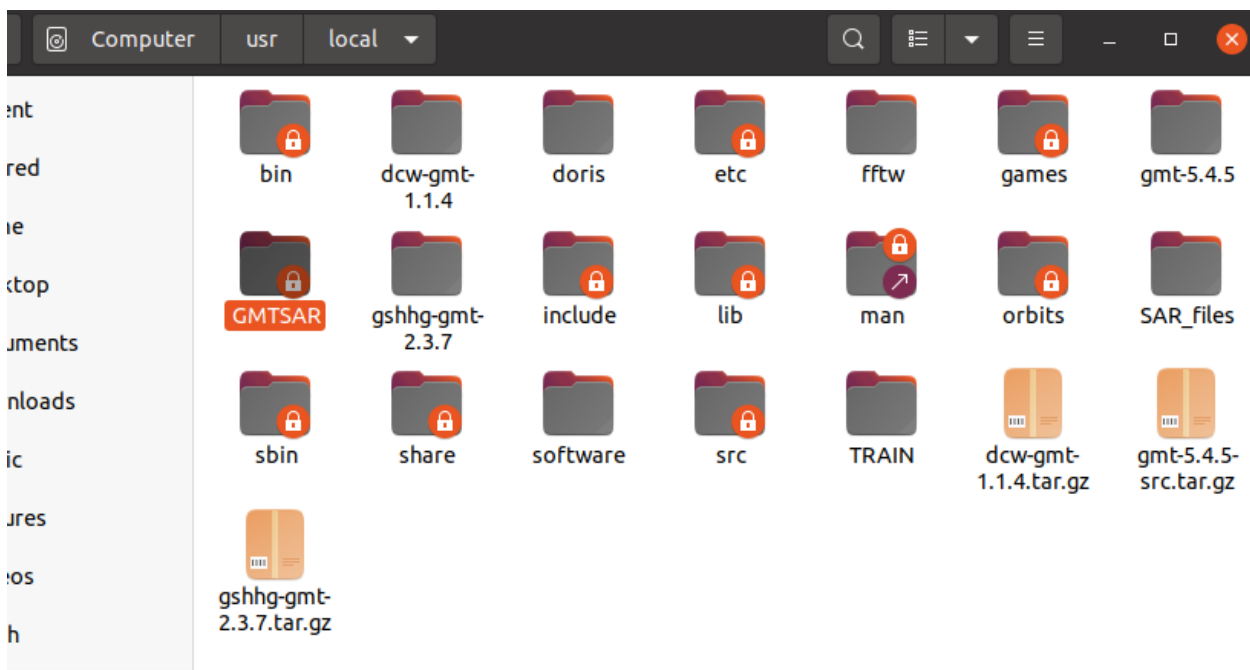


- Running the above command creates two ENVI and ERS files in the orbits folder.

Next, run the following commands.

```
# sudo -i
# cd /usr/local
# git clone --branch 6.0 https://github.com/gmtsar/gmtsar GMTSAR
```

```
siavash@siavash-Aspire-V3-571G:~$ sudo -i
[sudo] password for siavash:
root@siavash-Aspire-V3-571G:~# cd /usr/local
root@siavash-Aspire-V3-571G:/usr/local# git clone --branch 6.0 https://github.com/gmtsar/gmtsar GMTSAR
Cloning into 'GMTSAR'...
remote: Enumerating objects: 63, done.
remote: Counting objects: 100% (63/63), done.
remote: Compressing objects: 100% (48/48), done.
remote: Total 5018 (delta 30), reused 30 (delta 15), pack-reused 4955
Receiving objects: 100% (5018/5018), 7.53 MiB | 1.14 MiB/s, done.
Resolving deltas: 100% (3388/3388), done.
root@siavash-Aspire-V3-571G:/usr/local#
```



- By running the above commands, GMTSAR software will be downloaded.

Then run the following commands to install GMTSAR.

```
# cd GMTSAR
# autoconf
# ./configure --with-orbits-dir=/usr/local/orbits
```

```
# make
# make install
# cd bin
# ls
# cd ~
# pico .tcshrc
```

```
Receiving objects: 100% (5018/5018), 7.53 MiB | 1.14 MiB/s, done.
Resolving deltas: 100% (3388/3388), done.
root@siavash-Aspire-V3-571G:/usr/local# cd GMTSAR
root@siavash-Aspire-V3-571G:/usr/local/GMTSAR# autoconf
root@siavash-Aspire-V3-571G:/usr/local/GMTSAR# ./configure --with-orbits-dir=/usr/local/orbits
checking build system type... x86_64-pc-linux-gnu
```

```
root@siavash-Aspire-V3-571G:/usr/local/GMTSAR# make
```

```
root@siavash-Aspire-V3-571G:/usr/local/GMTSAR# make install
```

```
GMTSAR/share/gmtsar
root@siavash-Aspire-V3-571G:/usr/local/GMTSAR# cd bin
root@siavash-Aspire-V3-571G:/usr/local/GMTSAR/bin# ls
align_ALOS2_SCAN.csh      make_los_ascii.csh
align_ALOS_SLC.csh        make_profile.csh
align_batch_ALOS2_SCAN.csh make_raw_csk
align_batch_ALOS_SLC.csh  make_sta_tops
```

```
make_gaussian_filter
root@siavash-Aspire-V3-571G:/usr/local/GMTSAR/bin# cd ~
root@siavash-Aspire-V3-571G:~# pico .tcshrc
```

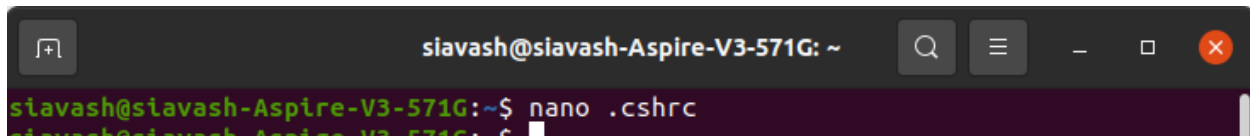
Then type the following code in the .tcshrc file and save it. (Do not copy the code, type it)

```
setenv GMTSAR /usr/local/GMTSAR
setenv PATH $GMTSAR/bin:"$PATH"
```

```
GNU nano 4.8      .tcshrc      Modified
setenv GMTSAR /usr/local/GMTSAR
setenv PATH $GMTSAR/bin:"$PATH"
```

Then run the following command.

```
# nano .cshrc
```



```
siavash@siavash-Aspire-V3-571G: ~
siavash@siavash-Aspire-V3-571G:~$ nano .cshrc
```

Then type the following code in the .cshrc file and save it.

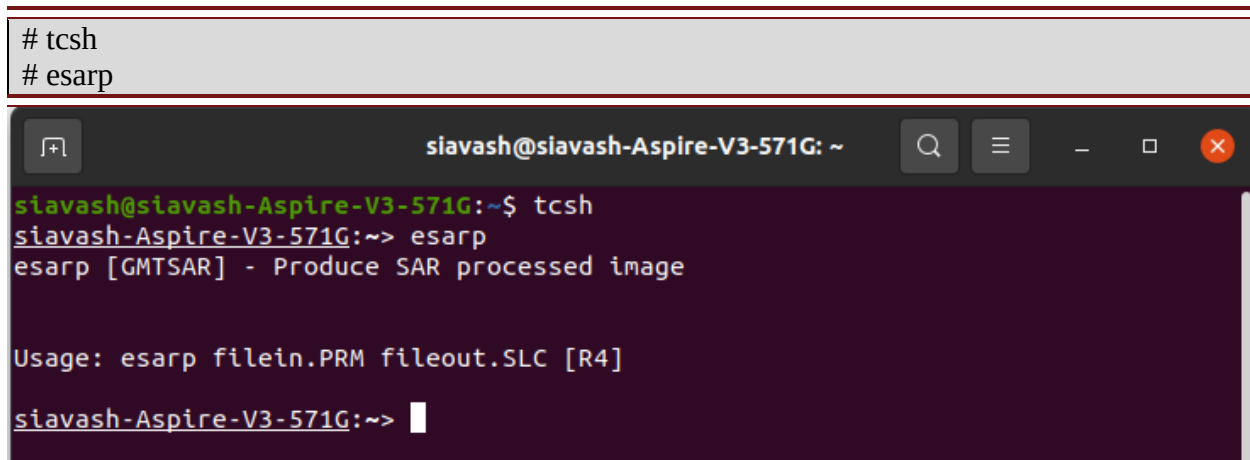


```
setenv GMTSAR /usr/local/GMTSAR
setenv PATH $GMTSAR/bin:$PATH

setenv APS_toolbox_bin "$APS_toolbox/bin"
setenv PATH "${PATH}:${APS_toolbox_bin}"

setenv GMTSAR /usr/local/GMTSAR
setenv PATH $GMTSAR/bin:$PATH
```

Running the following commands to ensure the software is installed correctly.



```
# tcsh
# esarp

siavash@siavash-Aspire-V3-571G:~$ tcsh
siavash-Aspire-V3-571G:~> esarp
esarp [GMTSAR] - Produce SAR processed image

Usage: esarp filein.PRM fileout.SLC [R4]

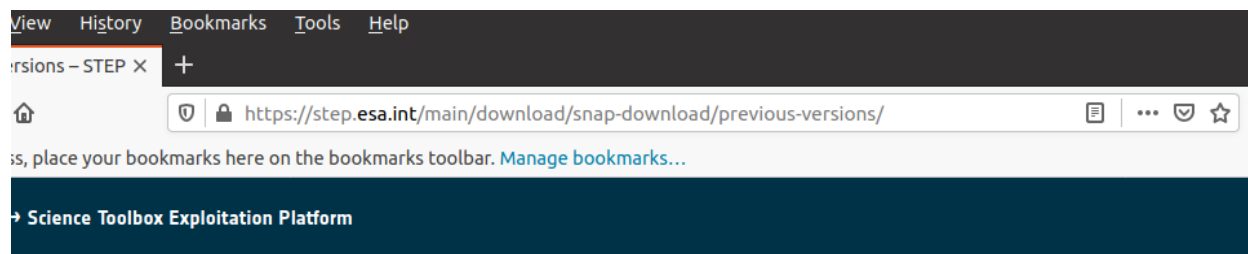
siavash-Aspire-V3-571G:~>
```

1.14. Install SNAP

Download version 6 of SNAP software for ubuntu by logging in to the site below.



```
https://step.esa.int/main/download/snap-download/previous-versions/
```



Then put the software on the Desktop and run the following commands.

```
# cd Desktop
# sudo apt-get install default-jre
# sudo apt-get install default-jdk
# sudo apt-get update
```

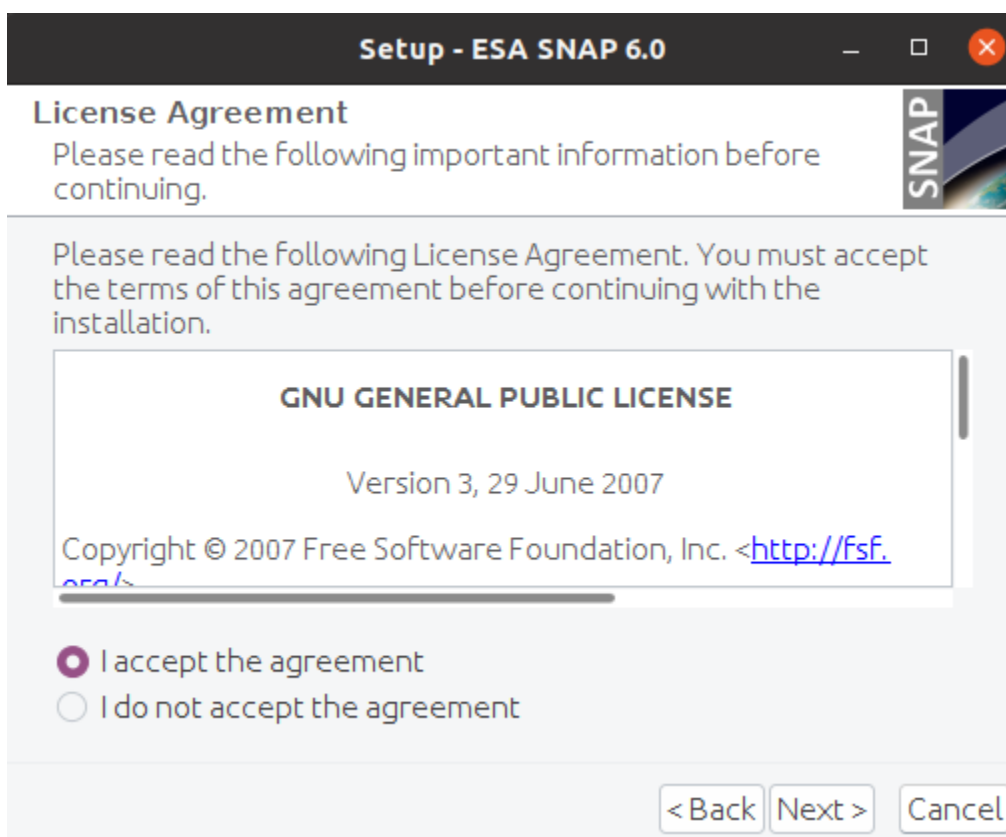
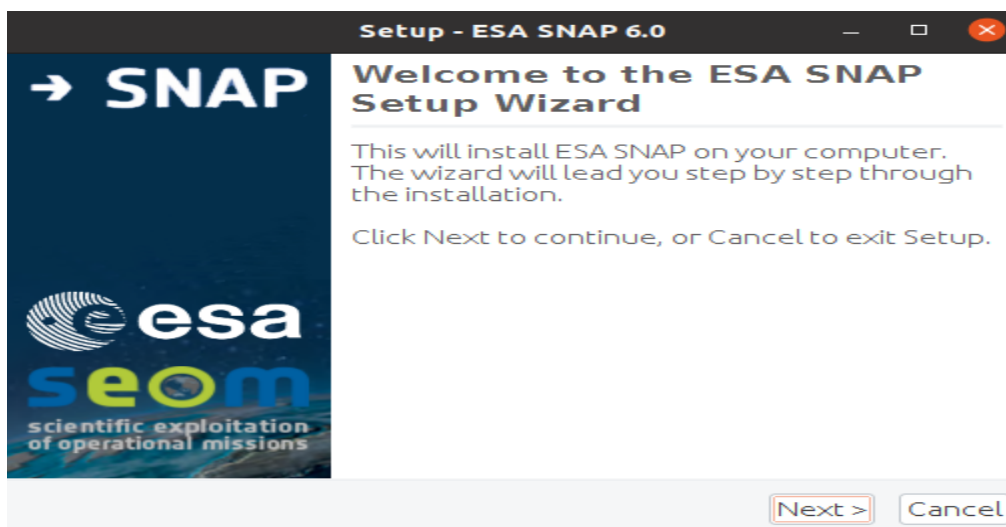
```
siavash@siavash-Aspire-V3-571G: ~/Desktop
siavash@siavash-Aspire-V3-571G:~$ cd Desktop
siavash@siavash-Aspire-V3-571G:~/Desktop$ sudo apt-get install default-jre
[sudo] password for siavash:
Reading package lists... Done
Building dependency tree
Reading state information... Done
done.
siavash@siavash-Aspire-V3-571G:~/Desktop$ sudo apt-get install default-jdk
Processing triggers for man-db (2.9.1-1) ...
siavash@siavash-Aspire-V3-571G:~/Desktop$ sudo apt-get update
[sudo] password for siavash:
Hit:1 http://ir.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://ir.archive.ubuntu.com/ubuntu focal-updates InRelease
```

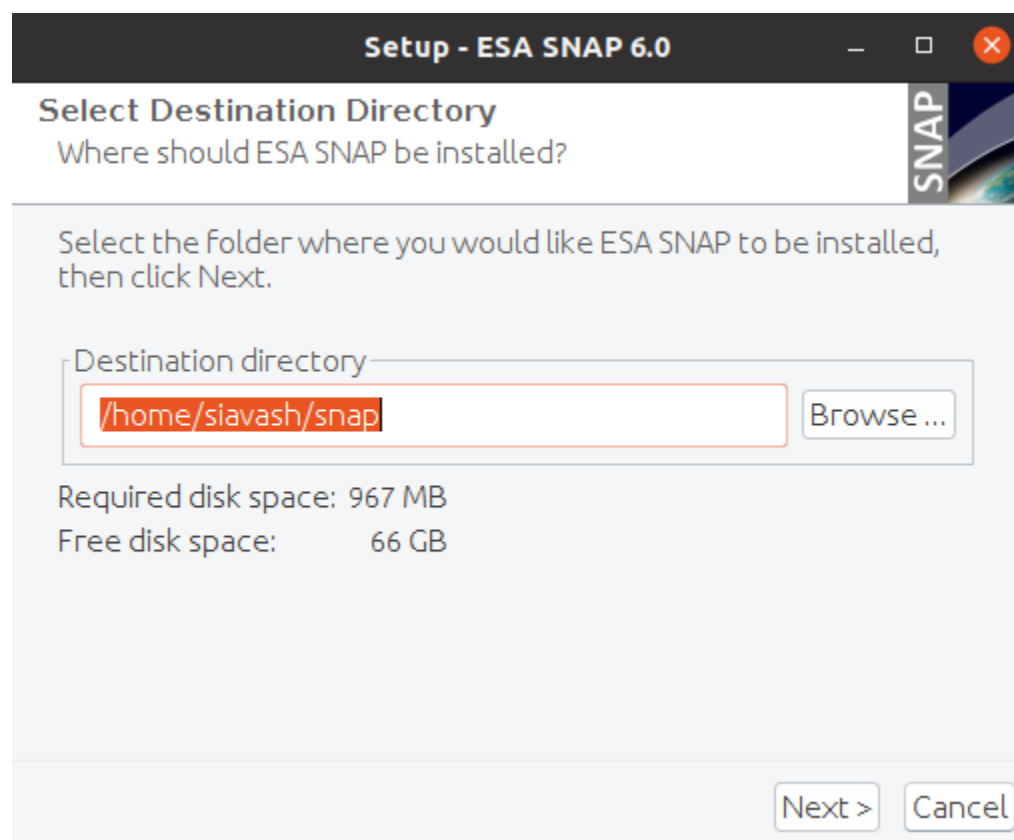
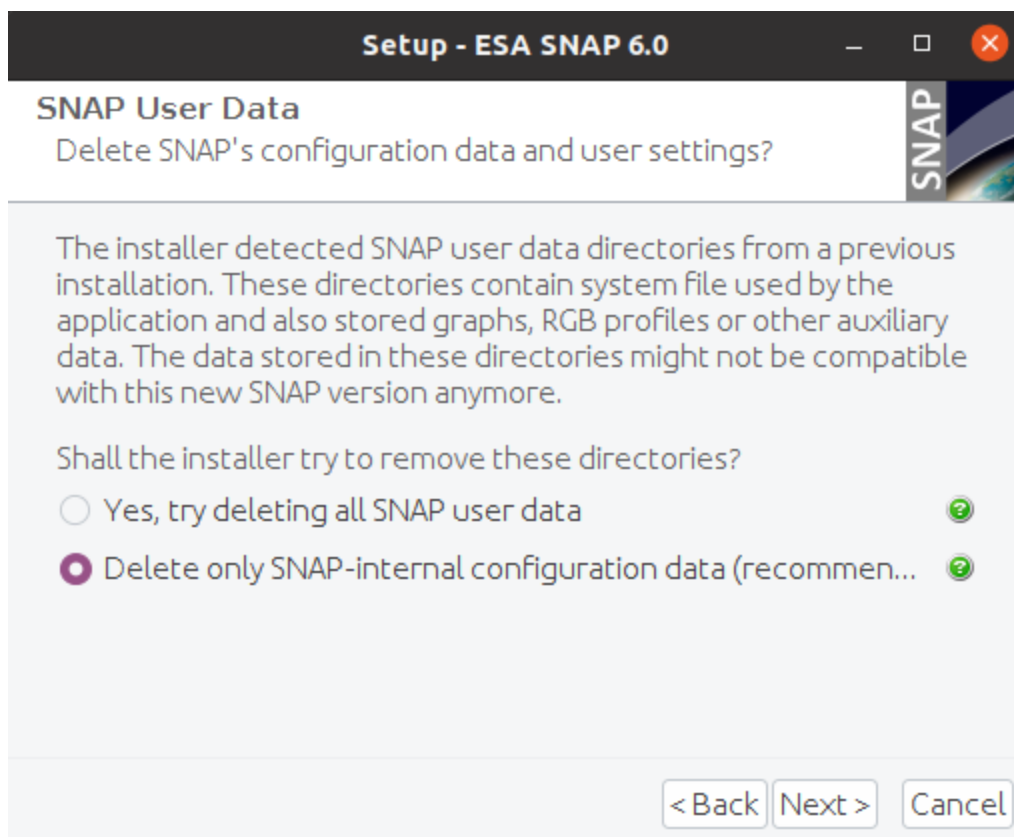
After updating the system, run the following commands.

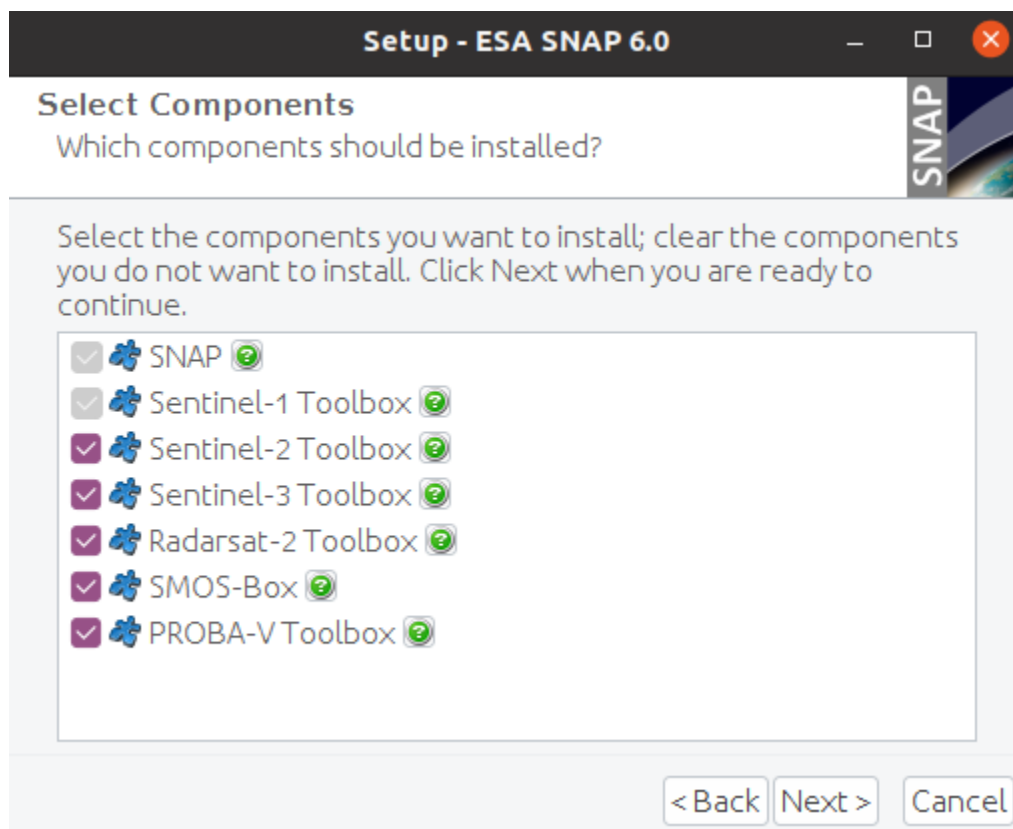
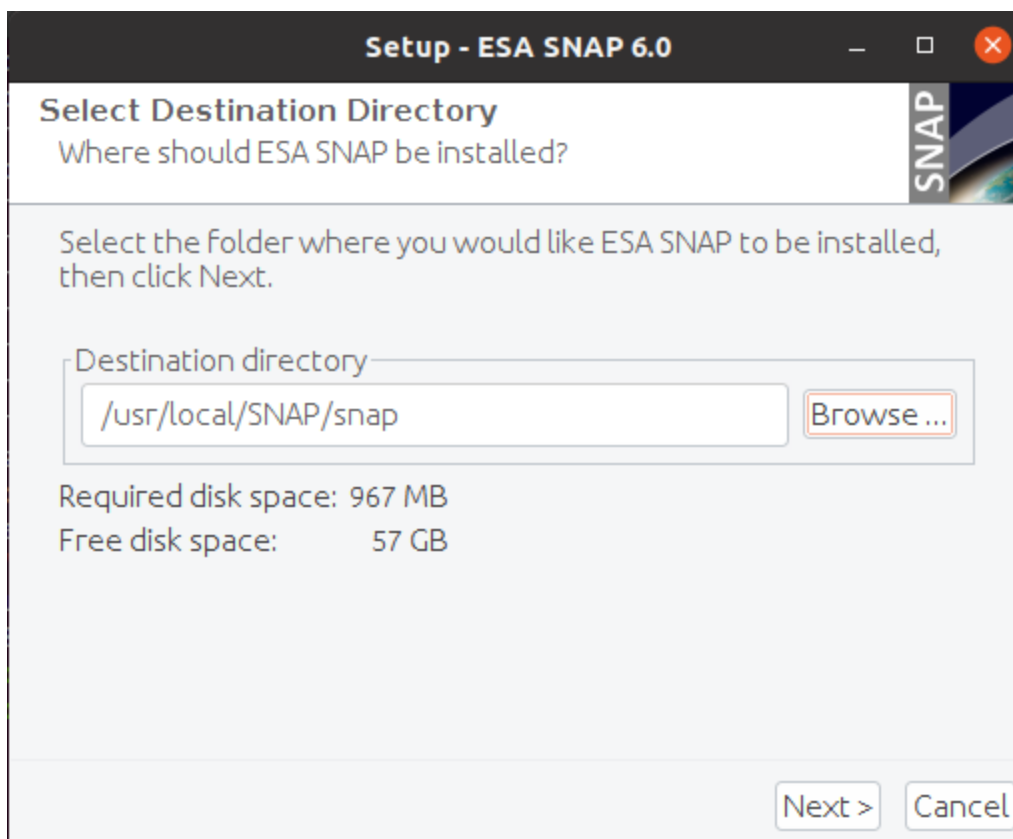
```
# sudo chmod +x esa-snap_all_unix_6_0.sh
# ./esa-snap_all_unix_6_0.sh
```

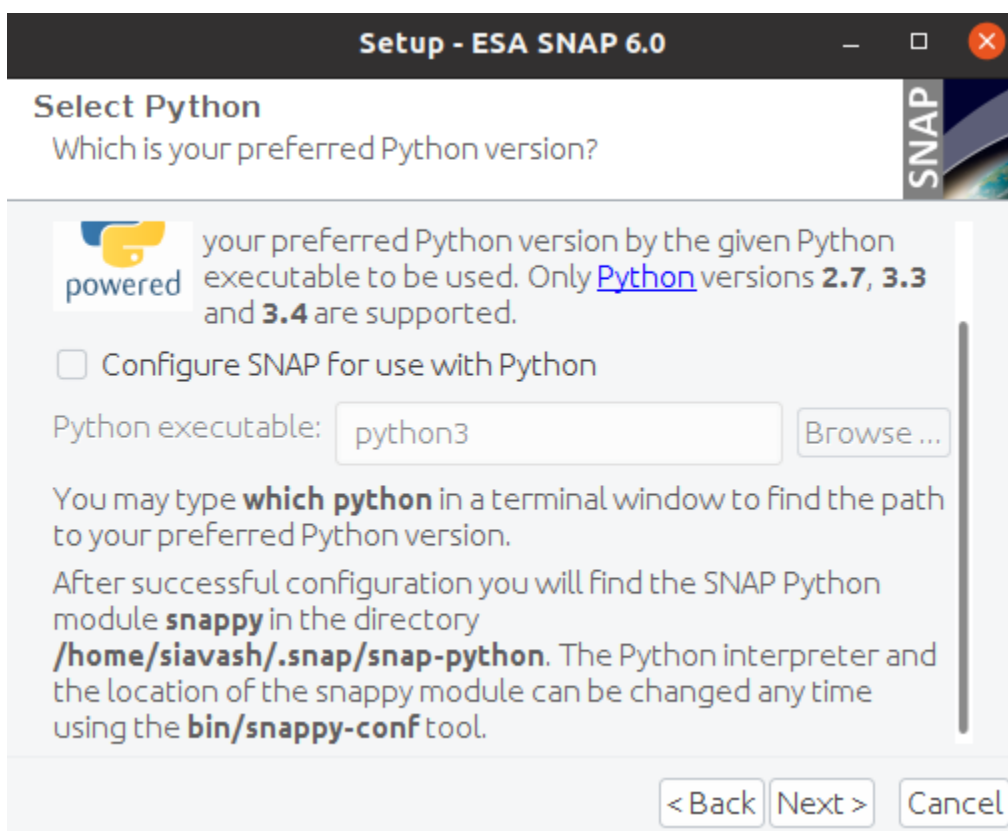
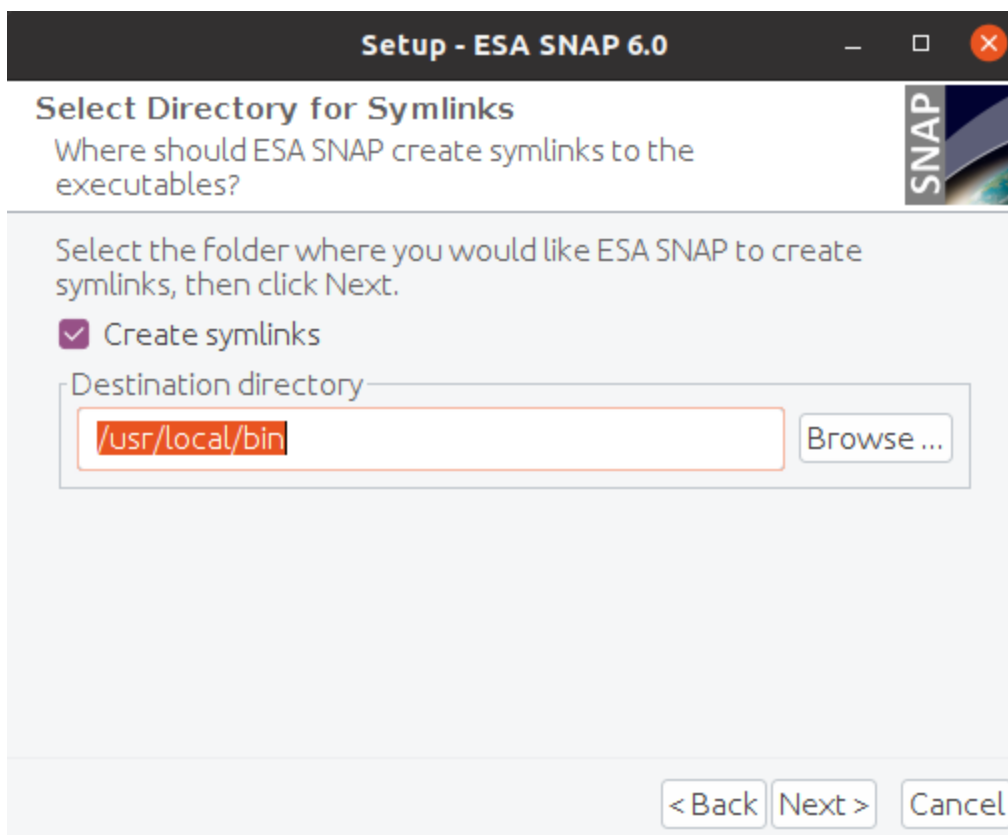
```
siavash@siavash-Aspire-V3-571G:~/Desktop$ sudo chmod +x esa-snap_all_unix_6_0.sh
siavash@siavash-Aspire-V3-571G:~/Desktop$ ./esa-snap_all_unix_6_0.sh
```

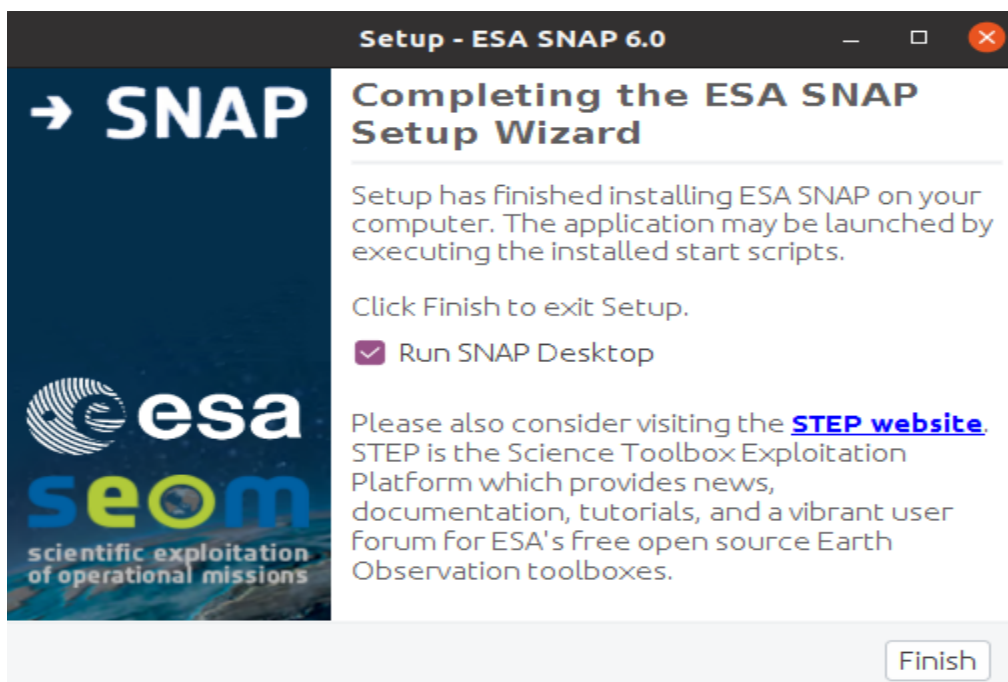
Running the above commands starts the software installation. Follow the installation steps according to the default options to complete the installation process.



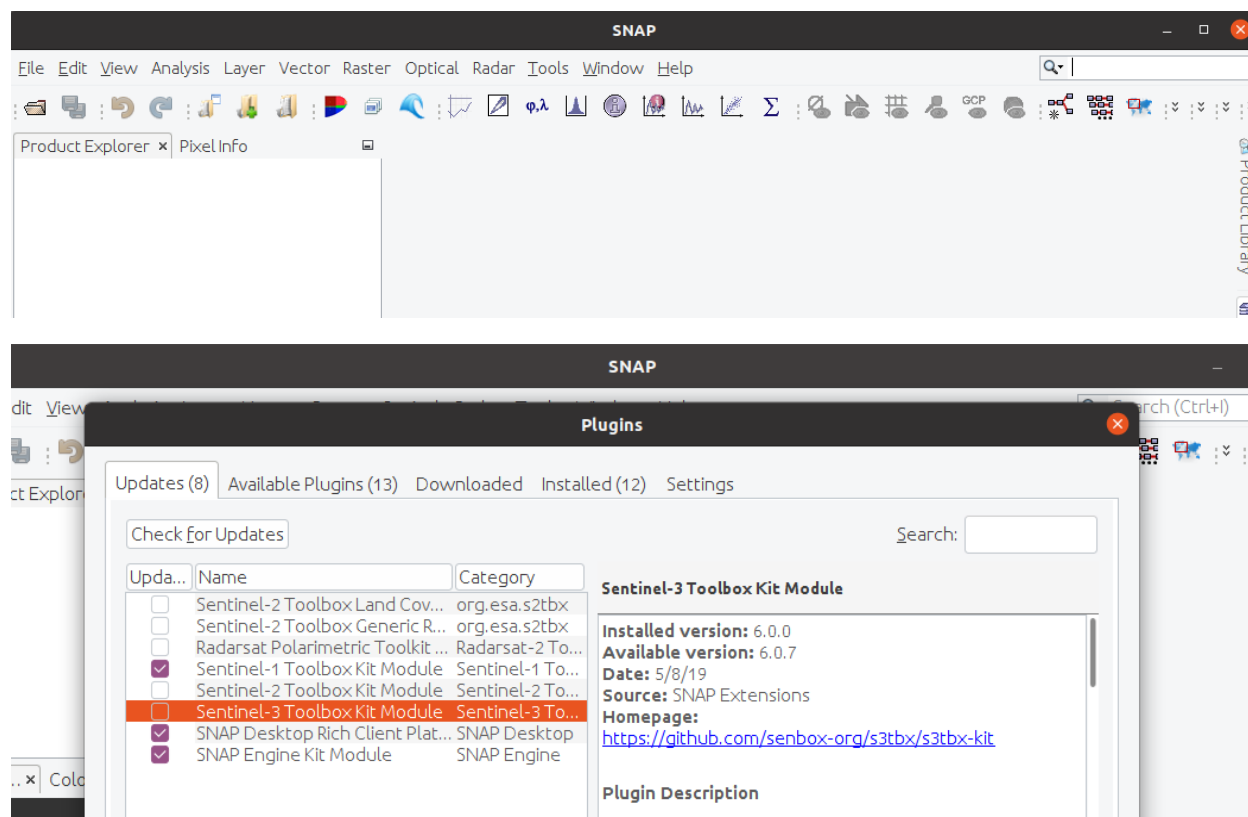








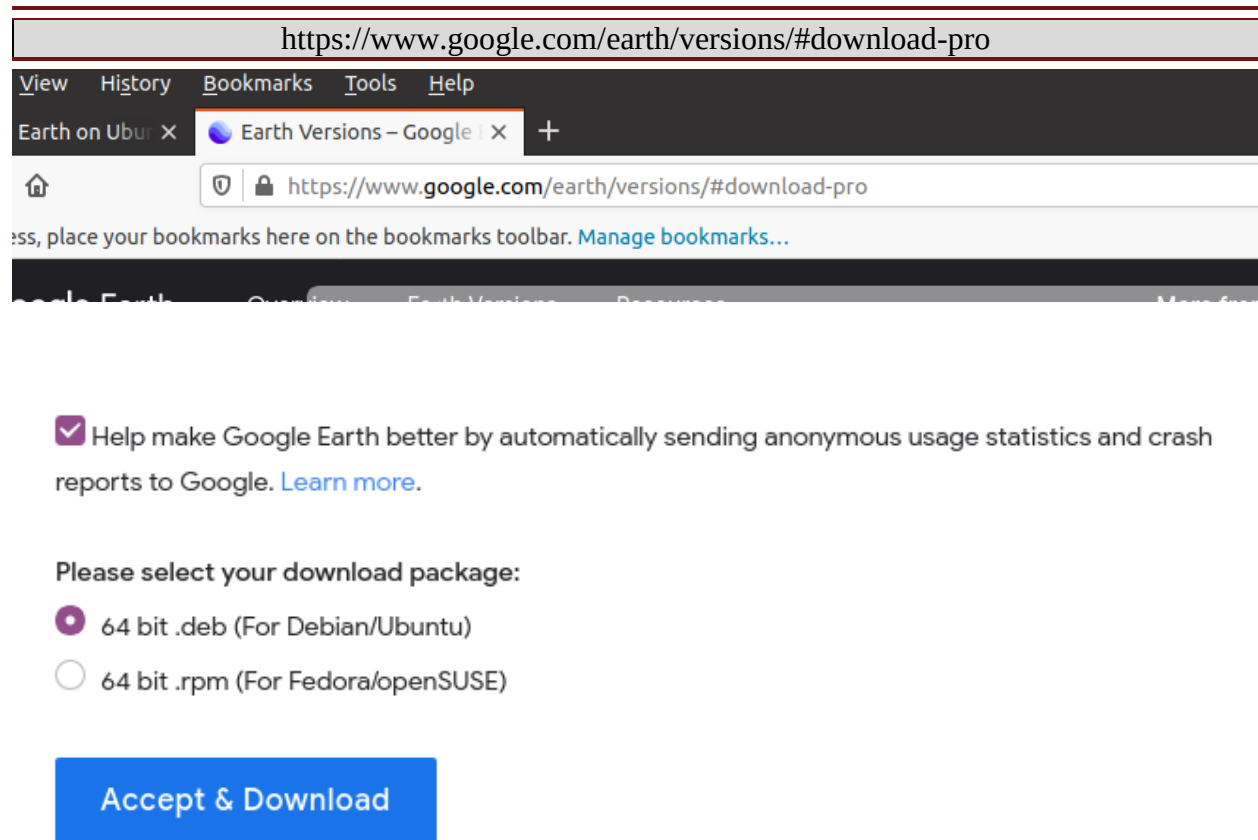
After installing the software, open it and update it from /Tools/Plugins.



- The type of sensor required is Sentinel-1A, so update the option for this sensor and the options for the software itself. (Like the image above, select and update the three specified options)

1.15. Install Google Earth

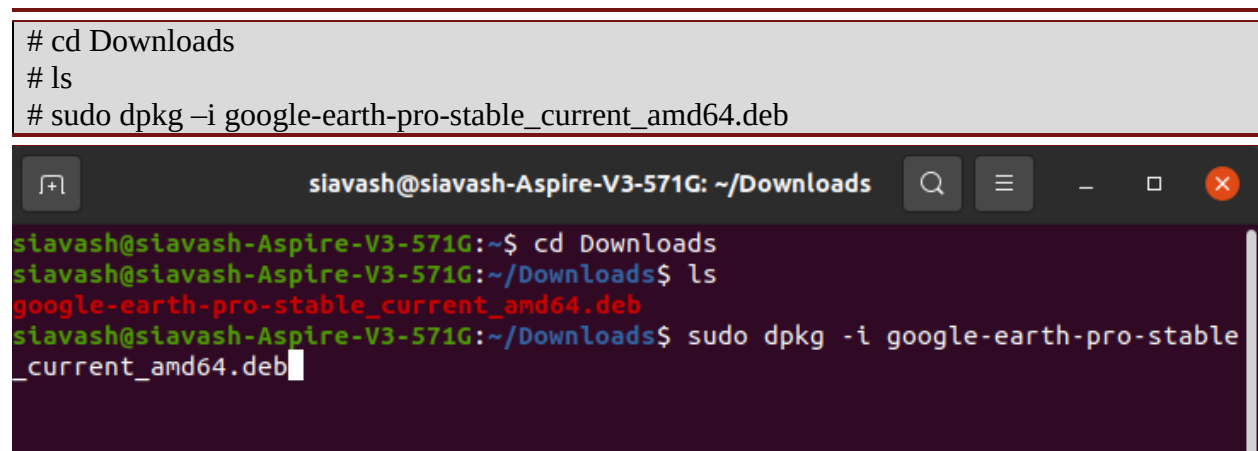
Download the Linux version of Google Earth software by logging in to the site below.



The screenshot shows a web browser window with the address bar displaying `https://www.google.com/earth/versions/#download-pro`. The browser's address bar and tabs are visible. Below the browser window, there is a checkbox labeled "Help make Google Earth better by automatically sending anonymous usage statistics and crash reports to Google. [Learn more.](#)" which is checked. Below this, the text "Please select your download package:" is followed by two radio button options: "64 bit .deb (For Debian/Ubuntu)" which is selected, and "64 bit .rpm (For Fedora/openSUSE)". At the bottom of the form is a blue button labeled "Accept & Download".

Then copy the software to `/home/Downloads/` and run the following commands.

```
# cd Downloads
# ls
# sudo dpkg -i google-earth-pro-stable_current_amd64.deb
```



The screenshot shows a terminal window with the title bar "siavash@siavash-Aspire-V3-571G: ~/Downloads". The terminal output shows the user running the commands: `siavash@siavash-Aspire-V3-571G:~$ cd Downloads`, `siavash@siavash-Aspire-V3-571G:~/Downloads$ ls`, and `siavash@siavash-Aspire-V3-571G:~/Downloads$ sudo dpkg -i google-earth-pro-stable_current_amd64.deb`. The output of the `ls` command is `google-earth-pro-stable_current_amd64.deb`. The cursor is at the end of the last command.

1.16. Sentinel-1

The SENTINEL-1 mission is the European Radar Observatory for the Copernicus joint of the European Commission and the European Space Agency. Sentinel-1A launched in April 2015, followed by Sentinel-1B launched in April 2016 afterward as a polar-orbiting, all-weather, day-and-night radar imaging mission to study monitor sea ice, icebergs, oil spills, marine winds, waves & currents, land-use change, land deformation, among others, and to respond to emergencies such as floods and earthquakes. Both S-1A and S-1B were Established in orbit by a Soyuz rocket from Europe's Spaceport in French Guiana. The instrument of S-1 is a C-band synthetic aperture radar (SAR) at 5.405 GHz, which means S-1 Operates C-band imaging in four different imaging modes, which will be described in follows. The imaging modes have different resolutions down to 5 m and coverage up to 400 km. The combination of S-1A and S-1B provides a revised time of 6 days. The life of the mission is estimated to be more than 7 years. (Sentinel-1 technical fact sheet, ESA)

- Wave (WV)
- Strip map (SM)
- Extra Wide swath (EW)
- Interferometric Wide swath (IW)

The Interferometric Wide (IW) swath mode is the main acquisition mode over land and satisfies the majority of service requirements. It acquires data with a 250 km swath at 5 m by 20 m spatial resolution (single look). IW mode captures three sub-swaths using Terrain Observation with Progressive Scans SAR (TOPSAR). With the TOPSAR technique, in addition to steering the beam in range as in ScanSAR, the beam is also electronically steered from backward to forward in the azimuth direction for each burst, avoiding scalloping and resulting in homogeneous image quality throughout the swath (This part from ESA).

TOPSAR mode replaces the conventional ScanSAR mode, achieving the same coverage and resolution as ScanSAR, but with a nearly uniform Signal-to-Noise Ratio and Distributed Target Ambiguity Ratio.

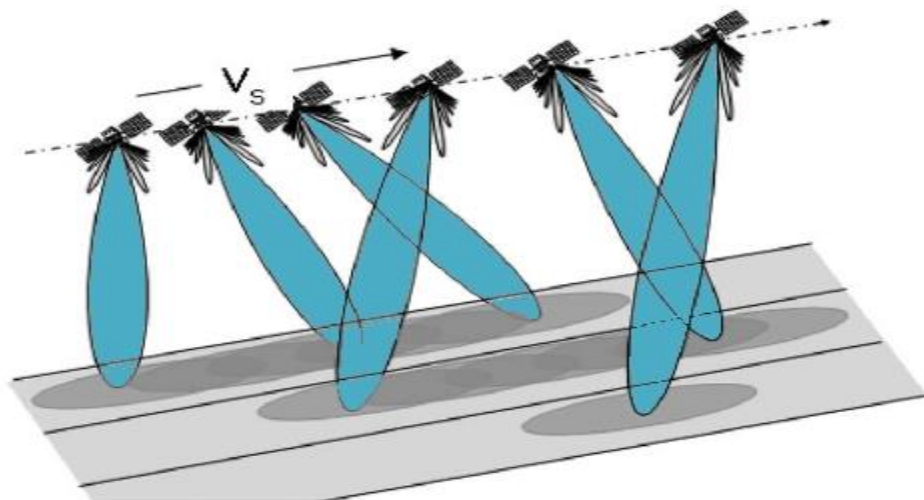


Figure 1: TOPSAR Sub-Swath Acquisition

The azimuth resolution is reduced compared to SM due to the shorter target illumination time of the burst. Using the sweeping azimuth pattern, each target is seen under the same antenna pattern, independently from its azimuth position in the burst image. By shrinking the azimuth antenna pattern, as seen by a target on the ground, scalloping effects on the image can be reduced. Bursts are synchronized from pass to pass to ensure the alignment of interferometric pairs. IW SLC products contain one image per sub-swath and one per polarisation channel, for a total of three (single polarisation) or six (dual polarisation) images in an IW product. Each sub-swath image consists of a series of bursts, where each burst has been processed as a separate SLC image. The individually focused complex burst images are included, in azimuth-time order, into a single sub-swath image with black-fill demarcation in between. There is sufficient overlap between adjacent bursts and between sub-swaths to ensure continuous coverage of the ground as provided in GRD products. The images for all bursts in all sub-swaths are resampled to a common pixel spacing grid in range and azimuth while preserving the phase information.

The table below shows the main characteristics of the Interferometric Wide swath mode.

Table 1. Characteristics of Interferometric Wide swath mode

Characteristic	Value
Swath width	250 km
Incidence angle range	29.1° - 46.0°
Sub-swaths	3
Azimuth steering angle	$\pm 0.6^\circ$
Polarisation options	Dual HH+HV, VV+VH Single HH, VV
Maximum Noise Equivalent Sigma Zero (NESZ)	-22 dB

The table below shows the incidence and off-nadir angles for Interferometric Wide swath mode beams.

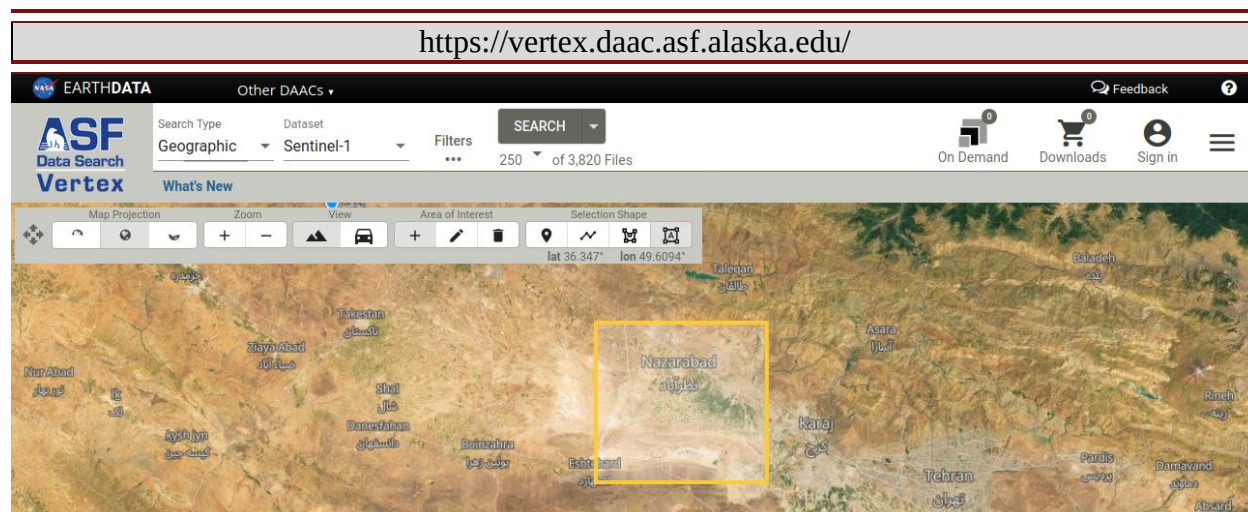
Table 2. Angles for Interferometric Wide sub-swaths

Beam	IW1	IW2	IW3
Off-nadir angles at min orbit altitude [°]	27.53-32.48	32.38-36.96	36.87-40.40
Incident angles at min orbit altitude [°]	30.86-36.59	36.47-41.85	41.75-46.00
Off-nadir angles at max orbit altitude [°]	26.00-30.96	30.86-35.43	35.35-38.88
Incident angles at max orbit altitude [°]	29.16-34.89	34.77-40.15	40.04-44.28

This book uses Sentinel-1 Radar images in SLC format and IW resolution. Alaska and Copernicus can be used to download these images.

1.16.1. Download Images in Alaska

To download images from the Alaska site, enter the following address and select the area of your study.



Then use the filter tool to select the type of images, including SLC format and IW resolution, as well as orbit numbers and other settings, and search for images.

Search Filters

☐ Seasonal Search

Additional Filters

File Type

L1 Single Look Comp...

1/14 file types selected

Beam Mode

IW

1/9 beam modes selected

Polarization

0/8 polarizations selected

Direction

0/2 flight directions selected

Subtype

0/2 subtypes selected

Path and Frame Filters

Path Start

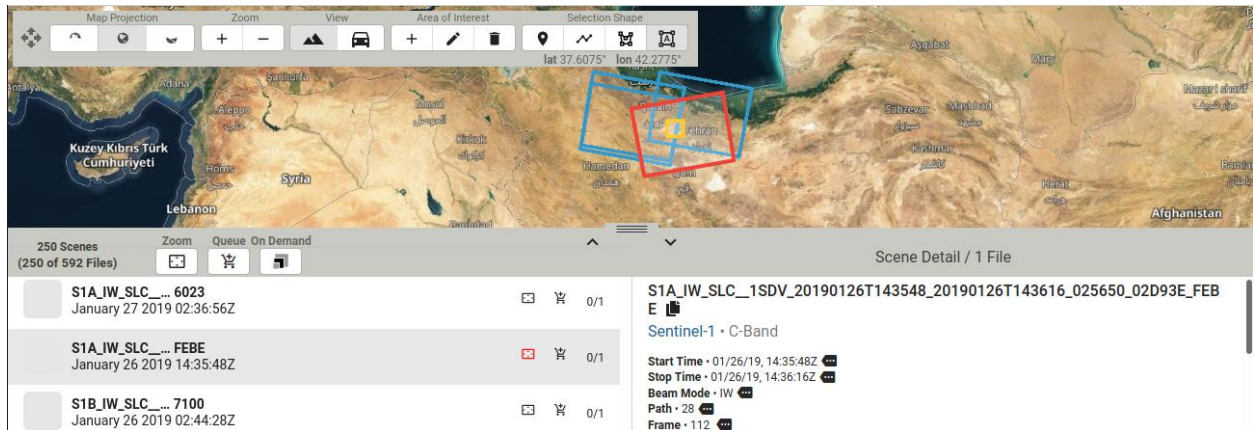
Path End

Frame Start

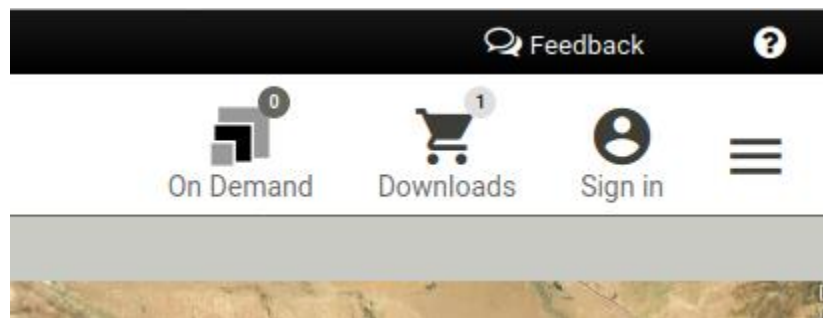
Frame End

Clear

Clear Search Area



Once the available images are specified, specify your desired image to the download section.



Click on the image download option and enter your username and password on the site to download your image.

Alaska Satellite Facility: making remote-sensing data accessible

Username ?

siavashshami

Password

• • • • • • • • • •

☒ **Stay signed in (this is a private workstation)**

LOG IN

REGISTER

[? I don't remember my username](#)

[? I don't remember my password](#)

[? Help](#)

1.16.2. Download Images in Copernicus

Select the Open Hub option by entering the following address and enter the username and password and enter the site.

<https://scihub.copernicus.eu>

Access to all Sentinel missions
Data
through the interactive
Graphical User Interface

Open Hub



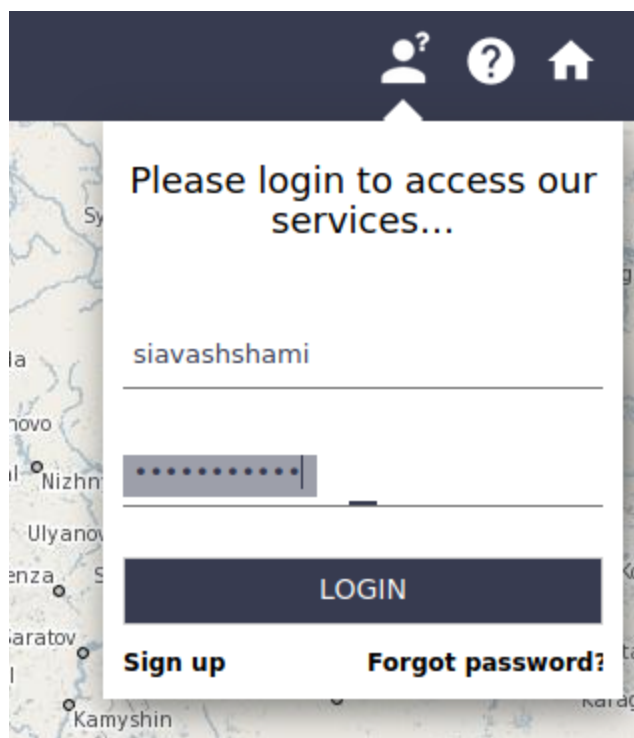
API Hub



S-5P Pre-Ops

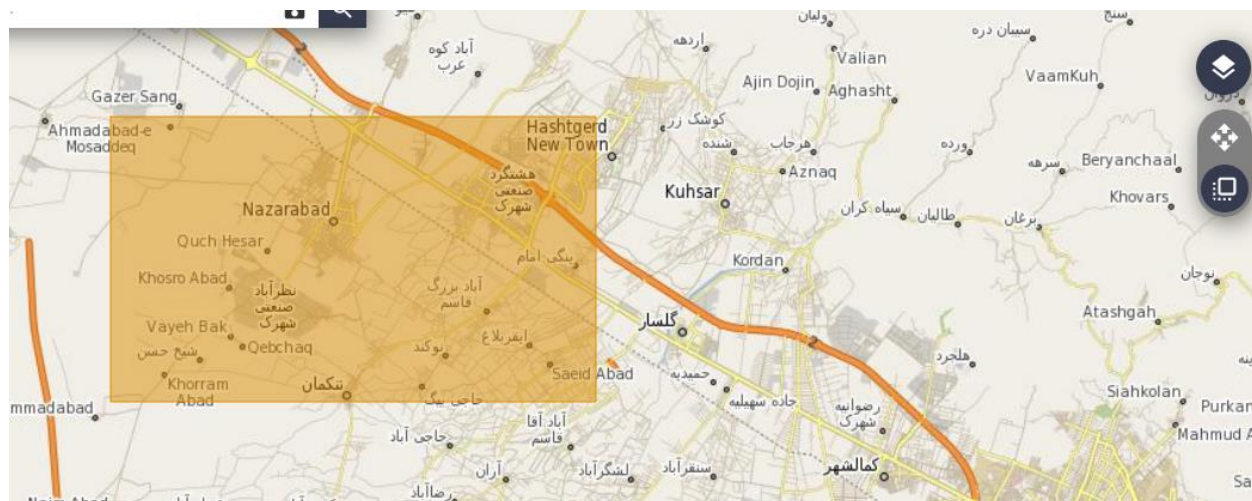


POD Hub



A login screen overlay on a map background. The screen has a dark blue header with icons for a user profile, a question mark, and a home button. The main text reads "Please login to access our services...". Below this is a text input field containing the username "siavashshami". Underneath the username field is a password field represented by a series of dots. At the bottom of the login form is a dark blue button labeled "LOGIN". Below the "LOGIN" button are two links: "Sign up" and "Forgot password?".

Then, like the Alaska site, select the area of your choice and search for images.



Insert search criteria...

» Ingestion period

☒ **Mission: Sentinel-1**

Satellite Platform:

Polarisation:

Relative Orbit Number (from 1 to 175):

☐ **Mission: Sentinel-2**

Satellite Platform:

Product Type:

Display 1 to 25 of 590 products.

Order By: Ingestion Date ↓

1 product selected

Request Done: (footprint:"Intersects(POLYGON((50.505191174953474 35.888426352510024,50.73162933537566 35.888426352510024,50.73162933537566 35.888426352510024,50.505191174953474 35.888426352510024,50.505191174953474 35.888426352510024)))")

S1A | SAR-C | S1A_IW_SLC_1SDV_20210322T024526_202103... ☒

Download URL: <https://scihub.copernicus.eu/dhus/odata/v1/>
Mission: Sentinel-1 Instrument: SAR-C Sensing Date: 2021-03-22

S1A | SAR-C | S1A_IW_SLC_1SDV_20210317T023708_202103... ☐

Download URL: <https://scihub.copernicus.eu/dhus/odata/v1/>
Mission: Sentinel-1 Instrument: SAR-C Sensing Date: 2021-03-17

S1A | SAR-C | S1A_IW_SLC_1SDV_20210316T143600_202103... ☐

To download offline images, announce your request through the download product view option and then click on download to get the images ready for download in a few hours.

1.17. Download Sentinel POE

To download the orbit information of Sentinel-1 images, enter the following address.

http://step.esa.int/auxdata/orbits/Sentinel-1/POEORB/

First select the type of radar image according to the image below S1A and then click on the year of the image and then on the month.

 2016-10.zip	2016-12-30 17:53 35M
 2016-11.zip	2016-12-30 17:57 34M
 2016-12.zip	2016-12-30 17:59 10M
 S1A/	2021-03-02 19:23 -
 S1B/	2021-03-02 19:24 -
 remote_contentVersion.txt	2016-12-30 17:41 491

<u>Name</u>	<u>Last modified</u>	<u>Size</u>	<u>Description</u>
 Parent Directory		-	
 2014/	2016-09-22 16:25	-	
 2015/	2016-09-22 16:26	-	
 2016/	2017-03-29 16:01	-	
 2017/	2017-03-29 16:01	-	
 2018/	2018-12-22 16:15	-	
 2019/	2019-12-22 16:15	-	
 2020/	2020-12-22 16:15	-	
 2021/	2021-03-26 16:15	-	

<u>Name</u>	<u>Last modified</u>	<u>Size</u>	<u>Description</u>
 Parent Directory		-	
 01/	2019-02-26 23:18	-	
 02/	2019-03-21 16:15	-	
 03/	2019-04-21 16:15	-	
 04/	2019-05-21 16:15	-	
 05/	2019-06-21 16:15	-	
 06/	2019-07-21 16:15	-	
 07/	2019-08-21 16:15	-	
 08/	2019-09-24 23:18	-	
 09/	2019-10-21 16:15	-	
 10/	2019-11-21 16:15	-	
 11/	2019-12-21 16:15	-	
 12/	2020-01-21 16:15	-	

 [S1A_OPER_AUX_POEORB_OPOD_20191113T120731_V20191023T225942_20191025T005942.EOF.zip](#) 2019-11-14 16:15 584K
 [S1A_OPER_AUX_POEORB_OPOD_20191114T120702_V20191024T225942_20191026T005942.EOF.zip](#) 2019-11-14 16:15 587K
 [S1A_OPER_AUX_POEORB_OPOD_20191115T120656_V20191025T225942_20191027T005942.EOF.zip](#) 2019-11-15 16:15 588K
 [S1A_OPER_AUX_POEORB_OPOD_20191116T120635_V20191026T225942_20191028T005942.EOF.zip](#) 2019-11-16 16:15 587K
 [S1A_OPER_AUX_POEORB_OPOD_20191117T121347_V20191027T225942_20191029T005942.EOF.zip](#) 2019-11-17 16:15 585K
 [S1A_OPER_AUX_POEORB_OPOD_20191118T120657_V20191028T225942_20191030T005942.EOF.zip](#) 2019-11-18 16:15 583K
 [S1A_OPER_AUX_POEORB_OPOD_20191119T120713_V20191029T225942_20191031T005942.EOF.zip](#) 2019-11-19 16:15 581K
 [S1A_OPER_AUX_POEORB_OPOD_20191120T120719_V20191030T225942_20191101T005942.EOF.zip](#) 2019-11-20 16:15 580K
 [S1A_OPER_AUX_POEORB_OPOD_20191121T120822_V20191031T225942_20191102T005942.EOF.zip](#) 2019-11-21 16:15 579K

To download the orbit file of the image to 20191028, you must select a file from the files in the month of that image, which is between the dates 20191027 and 20191029. For example, the orbit file for the image dated 20191028 is equal to:

S1A_OPER_AUX_POEORB_OPOD_20191117T121347_v20191027T225942_20191029T005942.EOF.zip

- Depending on the number of images, you must download the orbit file for each image.

1.18. Download Digital Elevation Model

Also, in the process of processing Radar images, a Digital Elevation Model (DEM) is required. To download the DEM with an accuracy of 30 meters, download the DEM of the study area by entering the site.

<http://topex.ucsd.edu/gmtsar/demgen>

Check status

e.g.
 31.9
 -115 -113
 27.3

north

west

east

south

After entering the site, first, select the type of DEM in terms of accuracy SRTM3 or SRTM1, and then enter the desired latitude and longitude and download the generated DEM.

Check status

e.g.
 31.9
 -115 -113
 27.3

north

west

east

south

Opening 1617015700.2.186.79.82.tar

You have chosen to open:

1617015700.2.186.79.82.tar
 which is: Tar archive (188 MB)
 from: https://topex.ucsd.edu

What should Firefox do with this file?

☐ Open with Archive Manager (default)

☒ **Save File**

☐ Do this automatically for files like this from now on.

ID	1617015700.2.186.79.82
Status	Successfully completed query
Data	Download

© 2010 GMTSAR

- SRTM3 = 90 m
- SRTM1 = 30 m

Chapter 2: InSAR Processing

The accuracy of the spatial resolution of the SAR data depends on the length of the satellite antenna or sensor. For example, for a sensor that is in the C band of electromagnetic waves, i.e. a wavelength of 5 cm, in order to produce a spatial resolution of 10 meters, an antenna with a length of about 4250 meters is required. This is not practical in space. That's why engineers created the Synthetic Aperture Radar (SAR) system. That is, using a small antenna to simulate a large length antenna to achieve high resolution. Unlike optical imaging systems, Synthetic Aperture Radar imaging systems are active systems that have an antenna mounted on their platform that sends a Radar signal to the ground in the direction of suffering. The signal is reflected after colliding with ground surface features and returns to the antenna as an echo. Return signals from the ground are recorded as amplitudes and phases in the antenna, which are used in the imaging process. In Synthetic Aperture Radar, a sensor or satellite sends pulses in a Range direction toward the ground by moving forward in the Azimuth direction. In other words, imaging in star systems is inclined. In Synthetic Aperture Radar, a sensor or satellite sends pulses in a Range direction toward the ground by moving forward in the Azimuth direction. In other words, imaging in star systems is inclined.

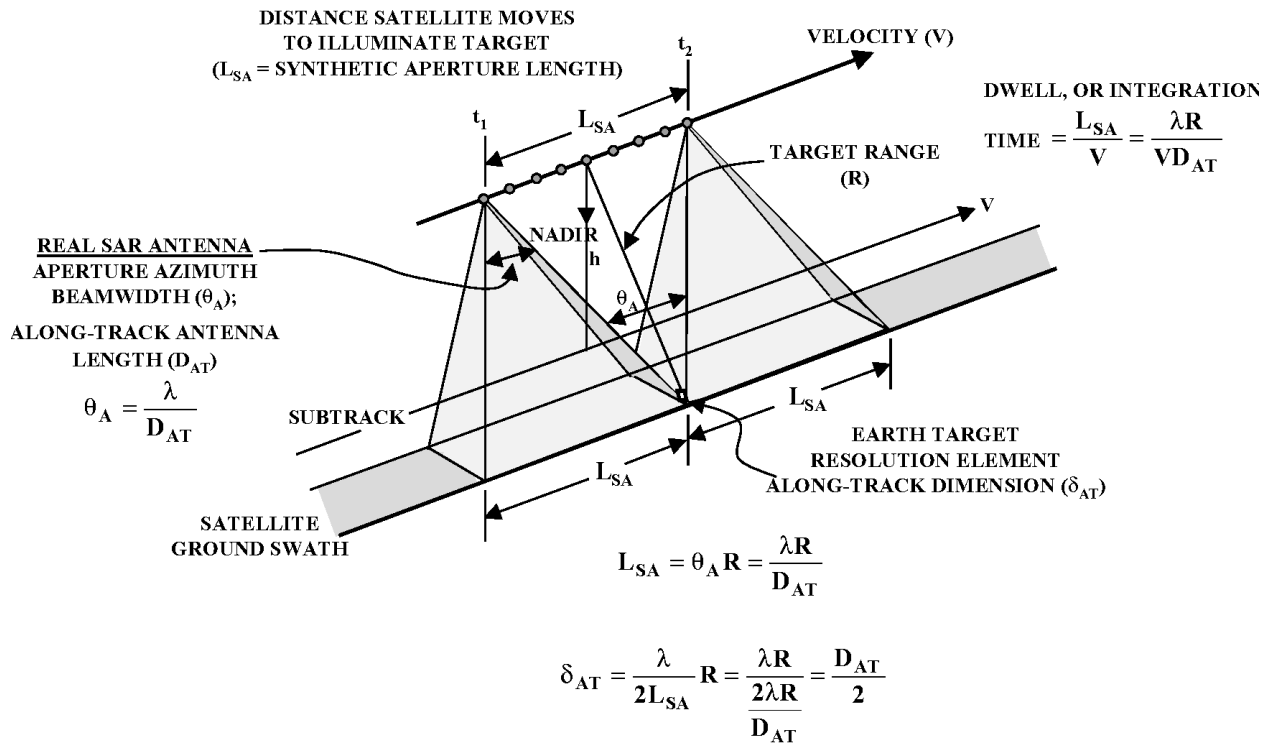


Figure 2. Imaging geometry of sensors SAR

Radar sensors use Electromagnetic wavelengths from 1 cm to 1 m, i.e. they use microwave wavelengths of the electromagnetic spectrum. They also have the ability to imaging day and night and are and all weather conditions. The different wavelengths of SAR are often referred to as bands, with letter designations such as X, C, L, and P. The table below notes the band with associated frequency, wavelength, and the application typical for that band.

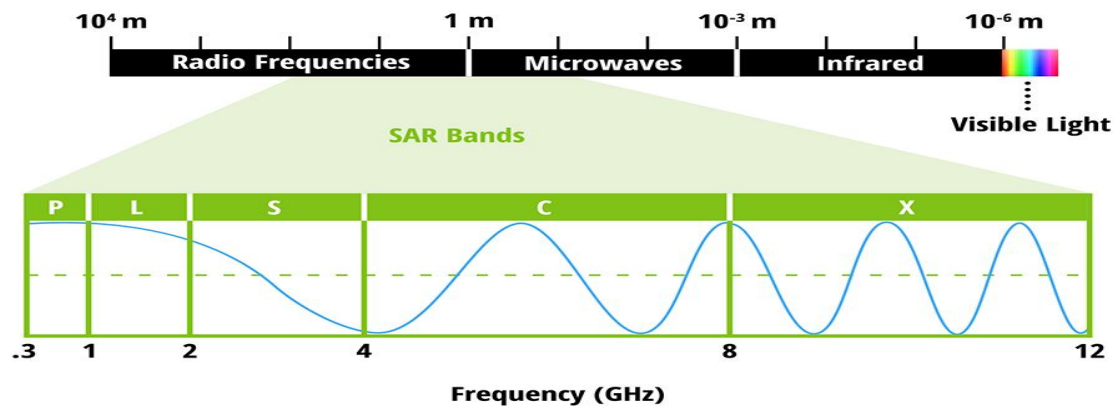


Figure 3. Electromagnetic wavelengths used In SAR

The ability of a radar signal to penetrate the surface is determined by the wavelength. For example, an X-band radar, which operates at a wavelength of about 3 cm, has very little capability to penetrate into the broadleaf forest, and thus mostly interacts with leaves at the top of the tree canopy. An L-band signal, on the other hand, has a wavelength of about 23 cm, achieving greater penetration into a forest and allowing for more interaction between the radar signal.

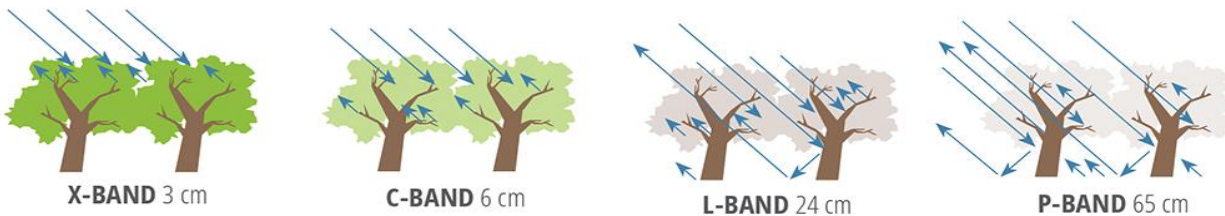


Figure 4. Sensitivity of SAR measurements to forest structure and penetration into the canopy at different wavelengths. Credit: NASA SAR Handbook.

Table 3. Applications of SAR images at different wavelengths

Band	Frequency	Wavelength	Typical Application
Ka	27–40 GHz	1.1–0.8 cm	Rarely used for SAR
K	18–27 GHz	1.7–1.1 cm	rarely used
Ku	12–18 GHz	2.4–1.7 cm	satellite altimetry
X	8–12 GHz	3.8–2.4 cm	High resolution SAR (urban monitoring,; ice and snow, little penetration into vegetation cover; fast coherence decay in vegetated areas)
C	4–8 GHz	7.5–3.8 cm	global mapping; change detection; monitoring of areas with low to moderate penetration; higher coherence, ice, ocean maritime navigation
S	2–4 GHz	15–7.5 cm	agriculture monitoring (NISAR will carry an S-band channel)
L	1–2 GHz	30–15 cm	Medium resolution SAR (geophysical monitoring; biomass and vegetation mapping; high penetration, InSAR)
P	0.3–1 GHz	100–30 cm	Biomass. First p-band spaceborne SAR will be launched ~2020; vegetation mapping and assessment. Experimental SAR.

2.1. Interferometric Synthetic Aperture Radar (InSAR)

One of the applications of the SAR is SAR interferometry or Interferometric Synthetic Aperture Radar (InSAR) (Gabriel et al., 1989; Rosen 1998; Antonio, 1992). InSAR involves measurements by two antennas. Antonio, 2017 considers three possibilities for the relation of the two involved satellites:

- Across-track interferometry is based on the acquisition of the image at the same time and from different positions.
- Along-track interferometry where the acquisitions are at different times but from the same position.
- Repeat-pass across-track interferometry where image acquisitions are at different times and from different orbital positions.

SAR data can also enable an analysis method called interferometry, or InSAR. InSAR uses the phase information recorded by the sensor to measure the distance from the sensor to the target. When at least two observations of the same target are made, the distance, with additional geometric information from the sensor, can be used to measure changes in land surface topography. These measurements are very accurate (up to the centimeter level!) and can be used to identify areas of deformation from events like volcanic eruptions and earthquakes. A SAR signal contains amplitude and phase information. The amplitude is the strength of the radar response, and the phase is the fraction of one complete sine wave cycle. The distance between the satellite antenna and the ground targets helps to determine the phase of the SAR image. Combining the phase of two images produces an interferogram. The phase of this interferogram is highly correlated to the terrain topography (Rosen 1998). In simple words, an interferogram is a colorful representation of topography and change. The ability of InSAR to map ground deformation with sub-centimeter over a large area, has made it possible, have rapid progress in methodology development. One of the first studies that used InSAR application to observe the earth was done by Massonnet et al., 1993.

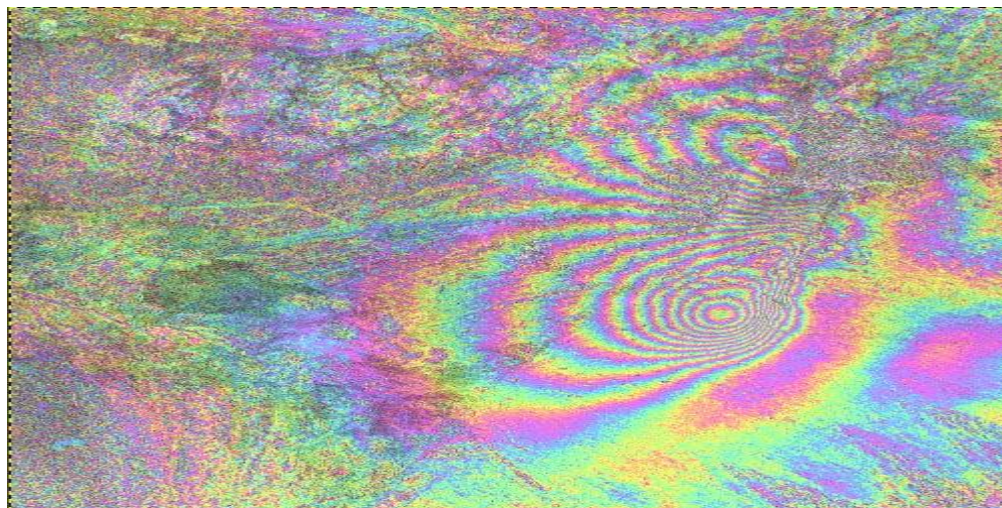
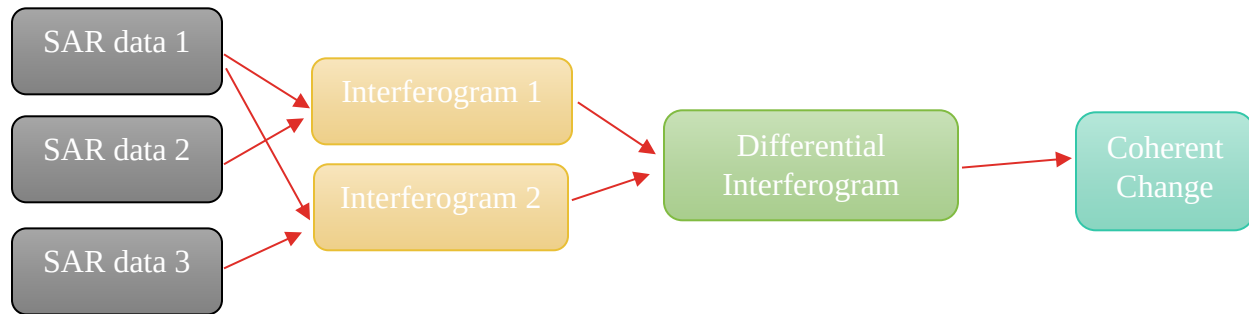


Figure 5. Processing ENVISAT SAR data 2003/12/03 and 2004/01/07 using Doris Software, shows Interferogram earthquake Bam, Iran.

2.2. DInSAR Processing by GMTSAR

for monitoring of geodynamic phenomena, in which the target has changed position at a relatively slow pace, we can be using interferometry. Also for analysing the results of earthquakes.



2.2.1. interferogram

The interferogram is obtained by multiplying a SAR image by the conjugate of the second image. Therefore, the phase of the interferogram, the difference between the phase of the image pair and its amplitude, is the product of the amplitudes of the image pair.

$$y = |a| e^{i\phi}$$

The mathematical principles of SAR Interferometry are that SAR images can be written as follows:

$$y_m \bar{y}_s = |a_m| |a_s| e^{i(\phi_m - \phi_s)}$$

In this regard, respectively $|a|$, i , ϕ , phase, imaginary unit and amplitude

As a result, interferogram is generated. The main components involved in the production interference phase are:

$$\Delta\phi = \Delta\phi_{\text{def}} + \Delta\phi_{\text{topo}} + \Delta\phi_{\text{fe}} + \Delta\phi_{\text{atm}} + \Delta\phi_{\text{orb}} + \Delta\phi_{\text{noise}}$$

$\Delta\phi_{\text{def}}$ Deformation phase, $\Delta\phi_{\text{topo}}$ topographic phase, $\Delta\phi_{\text{fe}}$ flat Earth phase, $\Delta\phi_{\text{atm}}$ atmospheric phase, $\Delta\phi_{\text{orb}}$ orbital phase, and $\Delta\phi_{\text{noise}}$ residual noise phase.

Among the above effects, topography and deformation are the most effective factors. The main purpose of differential interferometry is to extract the amount of deformation from the overall measured phase by eliminating or minimizing the effect of other components.

On 7 November 2019 (22:47 UTC), a Mw 5.9 earthquake took place in the East-Azerbaijan (hereinafter referred to as E-Azerbaijan earthquake), in northwestern Iran, about 100 km east of Tabriz, the fourth largest city of Iran, with over two million citizens (Figure 1a). The event was

clearly felt in most parts of the E-Azerbaijan province, killing at least five people, injuring hundreds, and causing widespread damage to the surrounding villages. During the next days, the mainshock was followed by more than 170 events with $2.5 \leq M_w \leq 4.8$, recorded by the Broadband Iranian Network (BIN). The mainshock was recorded by different institutions: the Iranian Seismological Center (IRSC), the United States Geological Survey (USGS), the GFZ-Potsdam (GEOFON), and the Harvard Global CMT catalog.

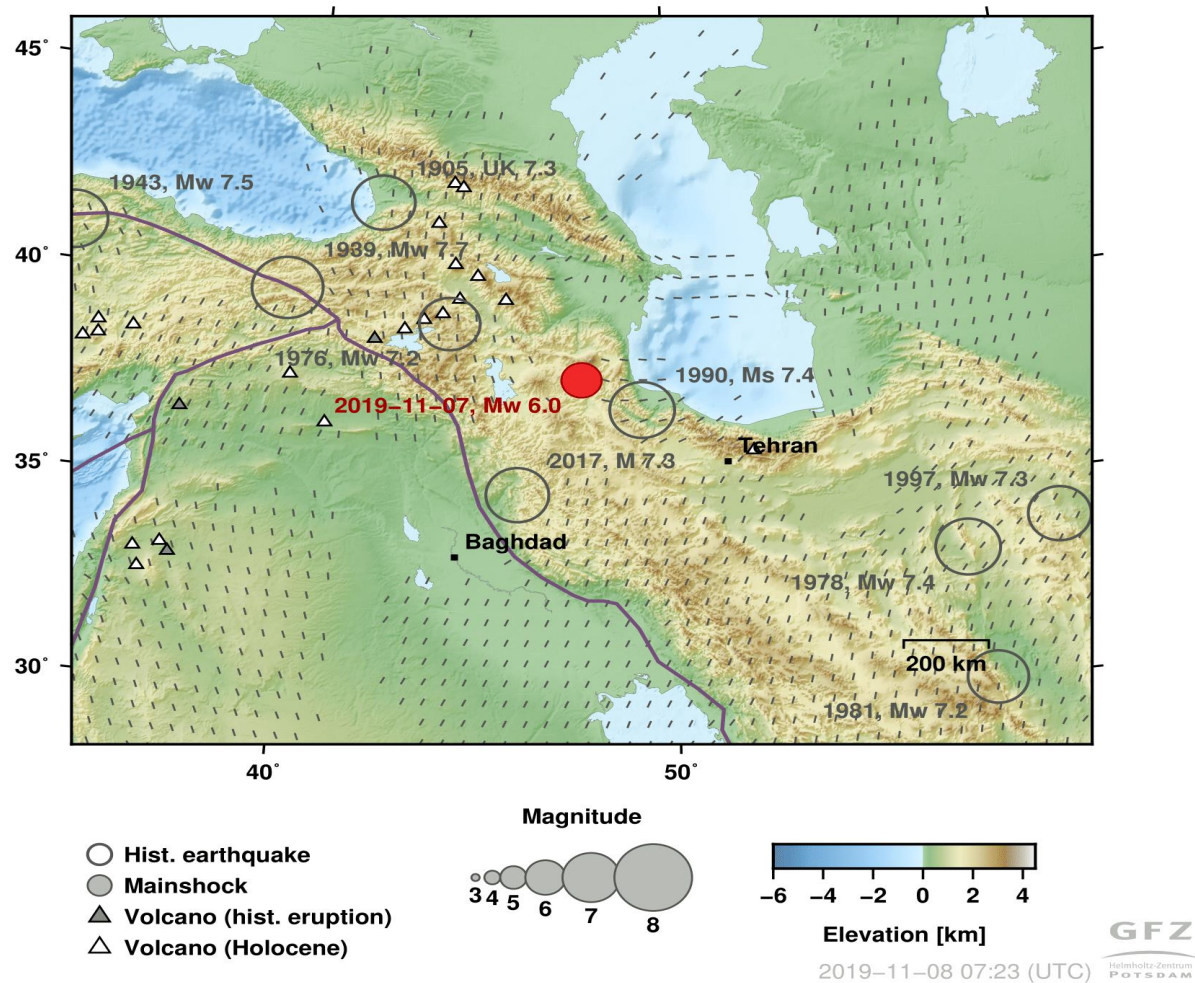


Figure 6. Detailed structural map of the seismogenic area

This earthquake with a left-lateral strike-slip mechanism struck the Bozgush mountain in NW Iran, marking the largest event in the mountain area in the past 44 years. Because no surface rupture associated with the earthquake has been reported, the causative fault of the earthquake is unknown. This event provides an opportunity to gain insight into the active tectonics of the region. Here, we acquire the coseismic deformation associated with this earthquake through Sentinel-1A interferometric synthetic aperture radar (InSAR) data and further invert.

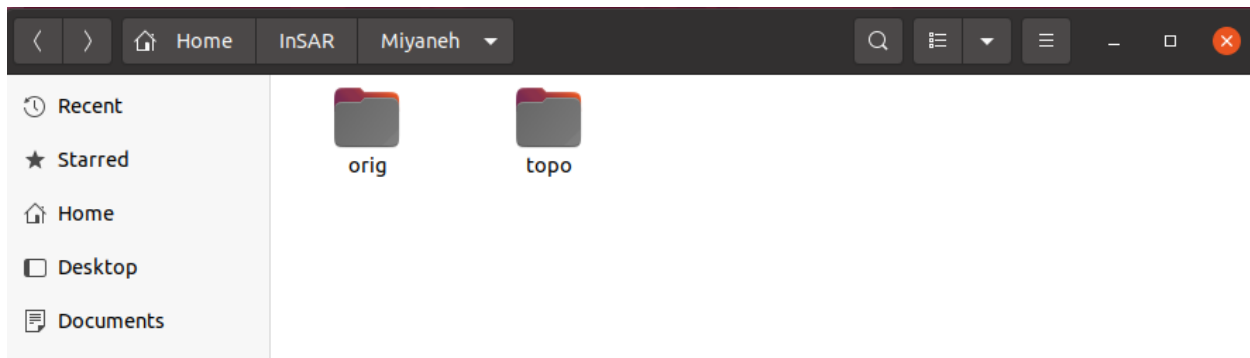
Then, using two Sentinel-1 images in Descending orbit with date 20191028 and orbit number 29653 and date 20191109 and orbit number 29828 and GMTSAR software, we performed DInSAR processing.

Table 4. Main characteristics of the interferometric pairs considered for of the E-Azerbaijan earthquake

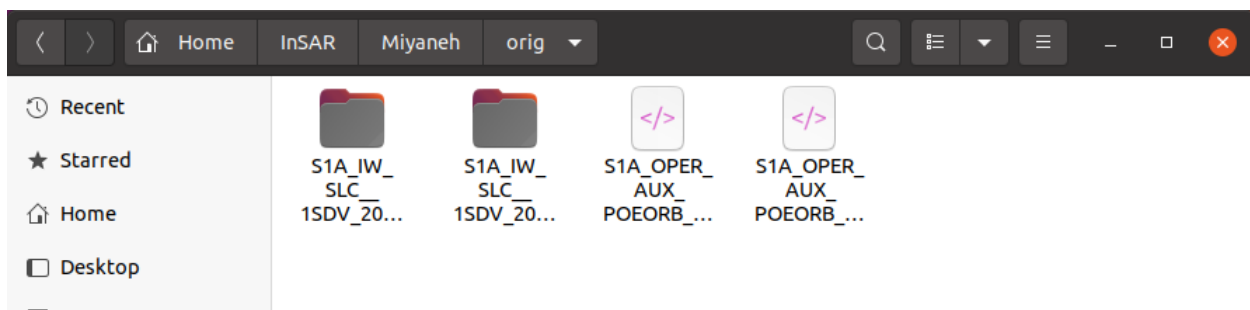
Sensor	Perpendicular Baseline (m)	Temporal Baseline (day)	Track
Sentinel-1	15.16	12	6

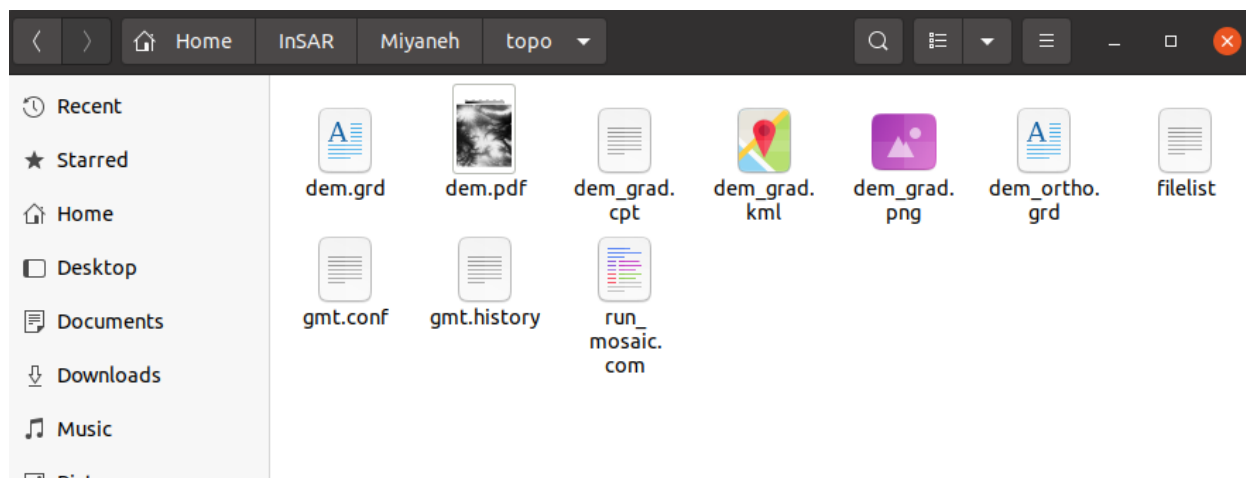
From this stage onwards, the process of processing DInSAR and extracting the amount of displacement resulting from the above earthquake will be taught with the help of software GMTSAR.

For each process, first, create your own folder. For example, in this section, the processing is related to the Miyaneh City earthquake. We create the Miyaneh folder in /home/InSAR/ and then create two folders named orig and topo inside the Miyaneh folder.(Miyaneh folder can be any name)

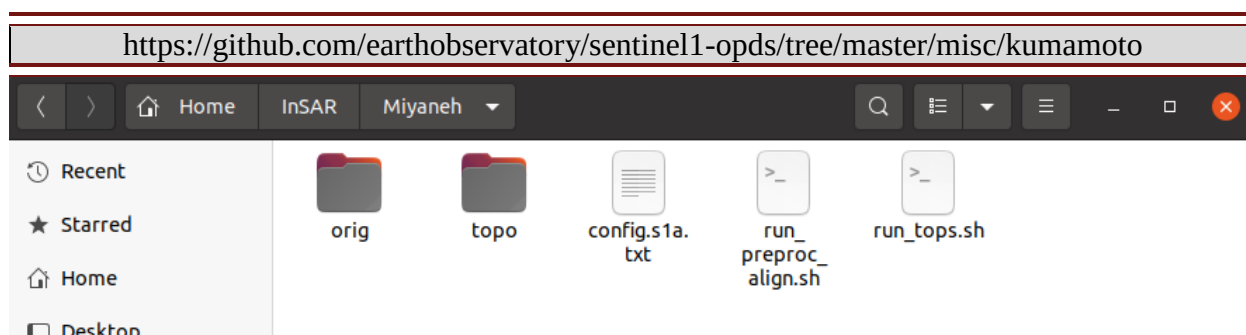


Then extract the two downloaded images related to the earthquake of the .SAFE format along with their orbit file in the .EOF format and copy them to the orig folder. Also, decompress the DEM file and copy it to the topo folder.

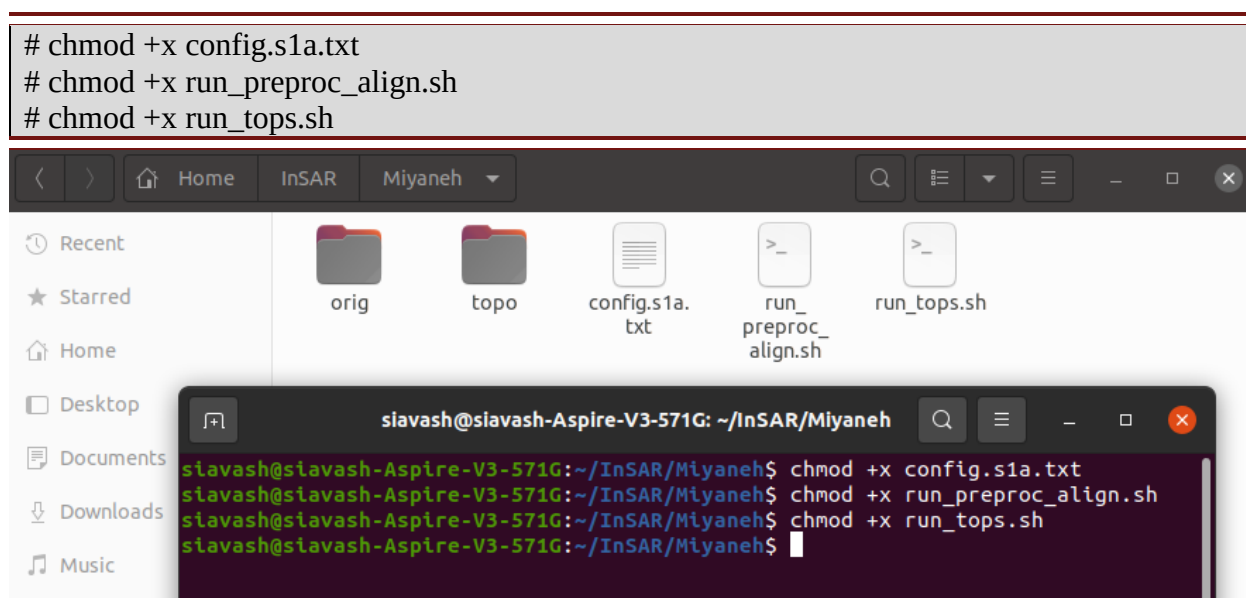




Then download the three files config.s1a.txt, run_preproc_align.sh, run_tops.sh from the following address and copy them into the Miyaneh folder next to the topo and orig folders.

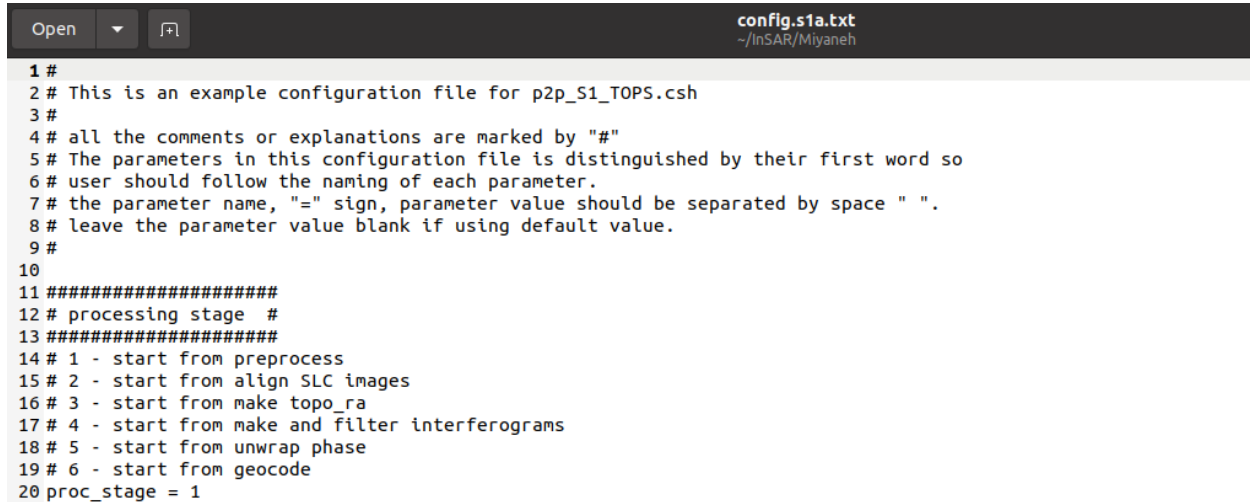


Then open the Terminal in the Mianeh folder and run the following commands.



Before executing the processing commands, you must first open and edit the config.s1a.txt file. First, you need to make sure that the first row is set for processing Sentinel-1 Radar images: p2p_S1A_TOPS.csh

Set the processing start step to option 1 to start the processing process from step 1: proc_stage = 1



```

1 #
2 # This is an example configuration file for p2p_S1_TOPS.csh
3 #
4 # all the comments or explanations are marked by "#"
5 # The parameters in this configuration file is distinguished by their first word so
6 # user should follow the naming of each parameter.
7 # the parameter name, "=" sign, parameter value should be separated by space " ".
8 # leave the parameter value blank if using default value.
9 #
10
11 #####
12 # processing stage #
13 #####
14 # 1 - start from preprocess
15 # 2 - start from align SLC images
16 # 3 - start from make topo_ra
17 # 4 - start from make and filter interferograms
18 # 5 - start from unwrap phase
19 # 6 - start from geocode
20 proc_stage = 1

```

The wavelength of the filter in meters is set to 200: filter_wavelength = 200

How to select an appropriate number of looks

The number of looks is a function of

- pixel spacing in azimuth
- pixel spacing in slant range
- incidence at scene center

ground range resolution = pixel spacing range/sin(incidence angle)

The following are two examples for ENVISAT and Sentinel-1 images to illustrate how to select multiple Look values. SNAP software can be used to find pixel spacing in Azimuth and pixel spacing in slant Range. To do this, first, open the image with SNAP software and write down these values from the Abstracted_Metadata section.



range_spacing	7.80397367	Float64	m	Range sample spacing
azimuth_spacing	4.05184698	Float64	m	Azimuth sample spacing

In ENVISAT images, the value is range_spacing = 7.80, azimuth_spacing = 4.05 and incidence is= 23 Degree.

according to the formula:

ground range resolution = pixel spacing range/sin(incidence angle)

ground range resolution = 7.80/sin(23)=7.80/0.39=20.01

20.01/4.05=4.93

So, in ENVISAT images, the multi-looks is:

1→5, 2→10, 3→15, 4→20, ...

range_spacing	2.3295619999999997	Float64	m	Range sample spacing
azimuth_spacing	13.934946666666667	Float64	m	Azimuth sample spacing

In the Sentinel-1 images, the value range_spacing = 2.32 and azimuth_spacing = 1393

incidence at scene center:

Beam ID	IW1	IW2	IW3
Incidence angle °	32.9	38.3	43.1

According to the formula in IW2 is equal to:

ground range resolution = pixel spacing range/sin(incidence angle)

ground range resolution = $2.32/\sin(38.3)=2.32/0.61=3.80$

$13.93/3.80=3.66$

So, in Sentinel-1 images, the multi-looks is:

1→4, 2→8, 3→12, 4→16, ...

```

55 #####
56 #   parameters for make and filter interferograms #
57 #   - intf.csh                                     #
58 #   - filter.csh                                   #
59 #####
60 # switch the master and slave when doing intf.
61 # put "1" if assume master as repeat and slave as reference
62 # put "0" if assume master as reference and slave as repeat [Default]
63 # phase = repeat phase - reference phase
64 switch_master = 0
65
66 # filters
67 # look at the filter/ folder to choose other filters
68 # for tops processing, to force the decimation factor
69 # recommended range decimation to be 8, azimuth decimation to be 2
70 filter_wavelength = 200
71 range_dec = 8
72 azimuth_dec = 2
73
74 # decimation of images
75 # decimation control the size of the amplitude and phase images. It is either 1 or 2.
76 # Set the decimation to be 1 if you want higher resolution images.
77 # Set the decimation to be 2 if you want images with smaller file size.
78 #
79 dec_factor = 2
80 #
81 #

```

threshold_snaphu indicates the degree of coherence. This value is between 0 and 1. The closer the value is to 1, the higher the coherence.

region_cut using this option, you can crop the image and leave it blank if needed. Image cropping in the unwrapping reduces processing time.

Region_cut in Descending image: 0/N/0/N = Right/Left/Up/Down

In the unwrapping in interferograms with obvious phase mutations, the maximum deformation can be adjusted using the defomax value.

threshold_geocode Displays the value of the correlation threshold in the image reference field.

```

82 #####
83 #   parameters for unwrap phase           #
84 #   - snaphu.csh                         #
85 #####
86 # correlation threshold for snaphu.csh (0~1)
87 # set it to be 0 to skip unwrapping.
88 threshold_snaphu = 0.15
89
90 # interpolate masked or low coherence pixels with their nearest neighbors, 1 me:
91 # others or blank means using original phase, see snaphu.csh and snaphu_interp.c
92 # this could be very slow in case a large blank area exist
93 near_interp = 1
94
95 # region to unwrap in radar coordinates (leave it blank if unwrap the whole reg
96 # example 300/5900/0/25000
97 region_cut =
98
99 # mask the wet region (Lakes/Oceans) before unwrapping (1 -- yes; else -- no)
100 switch_land = 0
101
102 #
103 # Allow phase discontinuity in unwrapped phase. This is needed for interferogram:
104 # defo_max = 0 - used for smooth unwrapped phase such as interseismic deformati
105 # defo_max = 65 - will allow a phase jump of 65 cycles or 1.82 m of deformation
106 #
107 defomax = 0.001
108 #####
109 #   parameters for geocode                 #
110 #   - geocode.csh                         #
111 #####
112 # correlation threshold for geocode.csh (0~1)
113 threshold_geocode = .10
114

```

Finally, save the config.s1a.txt file after applying your desired settings.

Then open the Terminal in the Miyaneh folder and run the following commands.

```

# tcsh
# ./run_preproc_align.sh name master image nama slove image
name orbit master image nama orbit slove image

```



```

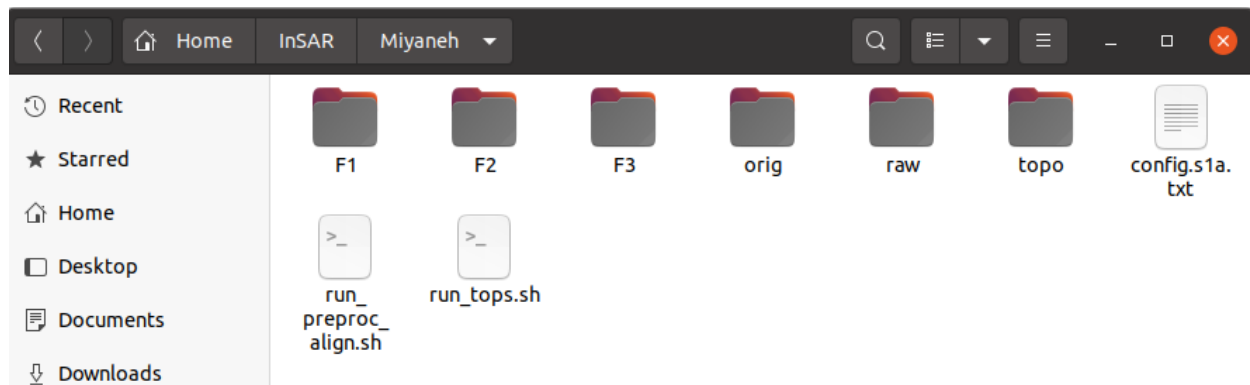
siavash@siavash-Aspire-V3-571G: ~/InSAR/Miyaneh
siavash@siavash-Aspire-V3-571G:~/InSAR/Miyaneh$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Miyaneh> ./run_preproc_align.sh S1A_IW_SLC__1SDV_20191028T025310_2
0191028T025338_029653_03608A_CEB7.SAFE S1A_IW_SLC__1SDV_20191109T025310_20191109T025338_029828_03
66BC_3A7B.SAFE S1A_OPER_AUX_POEORB_OPOD_20191117T121347_V20191027T225942_20191029T005942.EOF S1A_
OPER_AUX_POEORB_OPOD_20191129T120726_V20191108T225942_20191110T005942.EOF
S1_20191028_025312_F1
S1_20191109_025312_F1
Reading in 4976 lines of info from XML...
Writing 17 lines for orbit...
Reading in 4962 lines of info from XML...
Writing 17 lines for orbit...
Writing 52 lines of precise orbit for the LED file...
Writing 52 lines of precise orbit for the LED file...
Successfully opened S1_20191028_025312_F1.PRM
Successfully opened S1_20191109_025312_F1.PRM

Successfully opened S1_20191109_025312_F1.PRM
[1] 3551
[2] 3552
[2] + Done SAT_llt2rat S1_20191109_025312_F1.PRM 1 < topo.llt > aligned.ratll
[1] + Done SAT_llt2rat S1_20191028_025312_F1.PRM 1 < topo.llt > master.ratll
[1] 3566
[2] 3567
[2] Done gmt surface atmp.xyz -bi3d -R0/21292/0/12196 -I16/8 -T0.3 -Gatmp.grd -N1000 -r
[1] + Done gmt surface rtmp.xyz -bi3d -R0/21292/0/12196 -I16/8 -T0.3 -Grtmp.grd -N1000 -r
Reading in 4976 lines of info from XML...
Writing 17 lines for orbit...
Writing SLCL & SLCH..

```

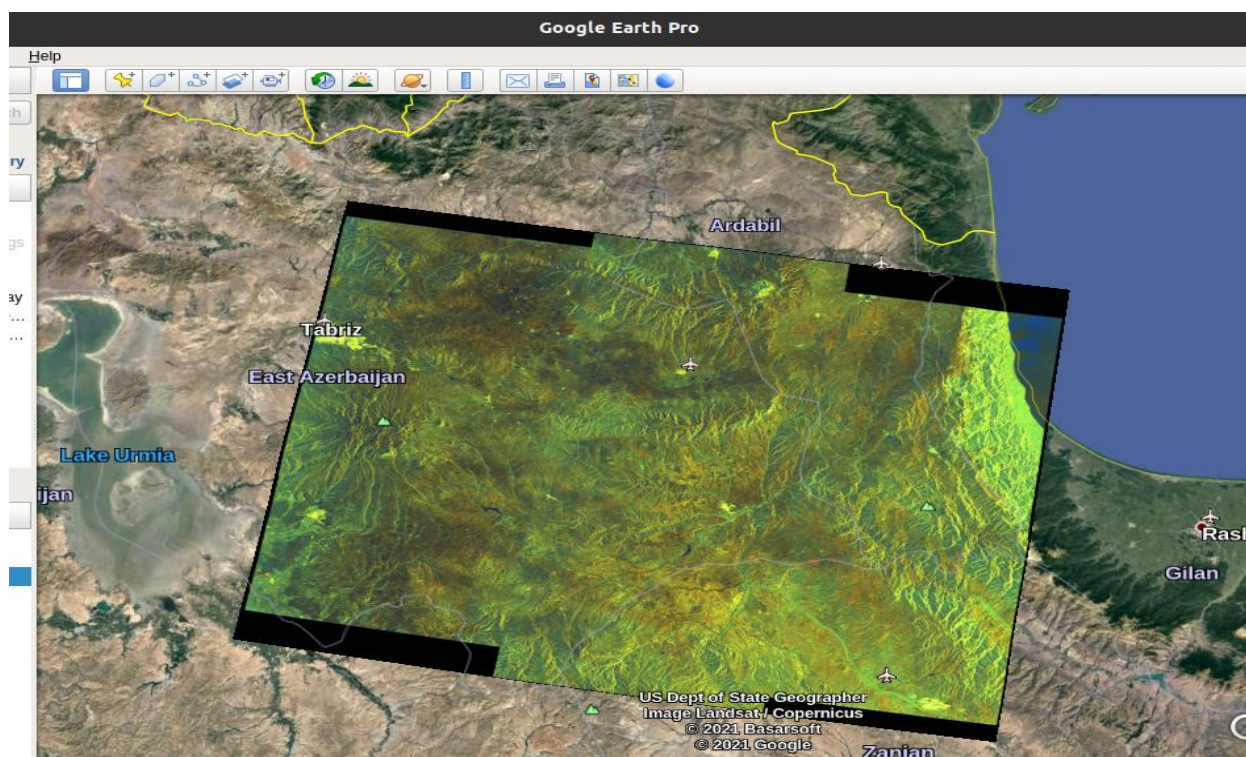
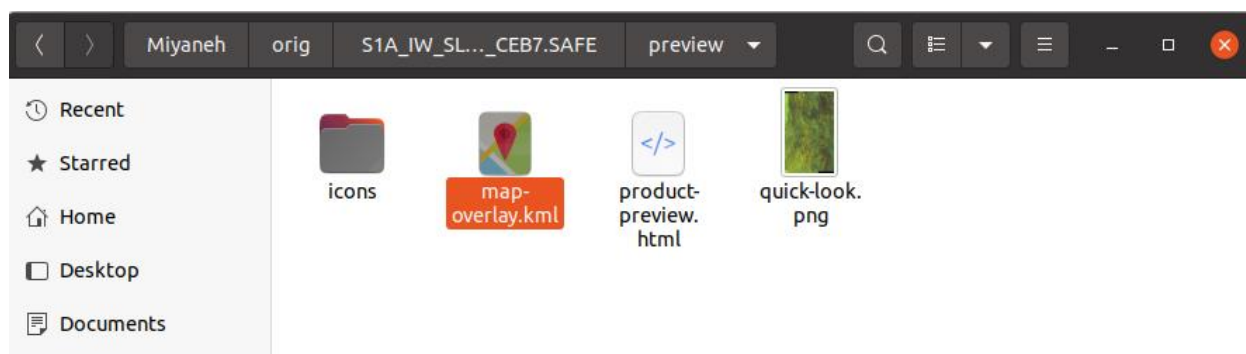
- Shows the number of pixels in the Azimuth and Range direction.

After processing, folders F1, F2, F3, raw are created. F1, F2, F3 = IW1, IW2, IW3



Since the study area is often located in one of the IWs, in order to reduce the processing time of the images, select your study area from the IWs and delete the rest .

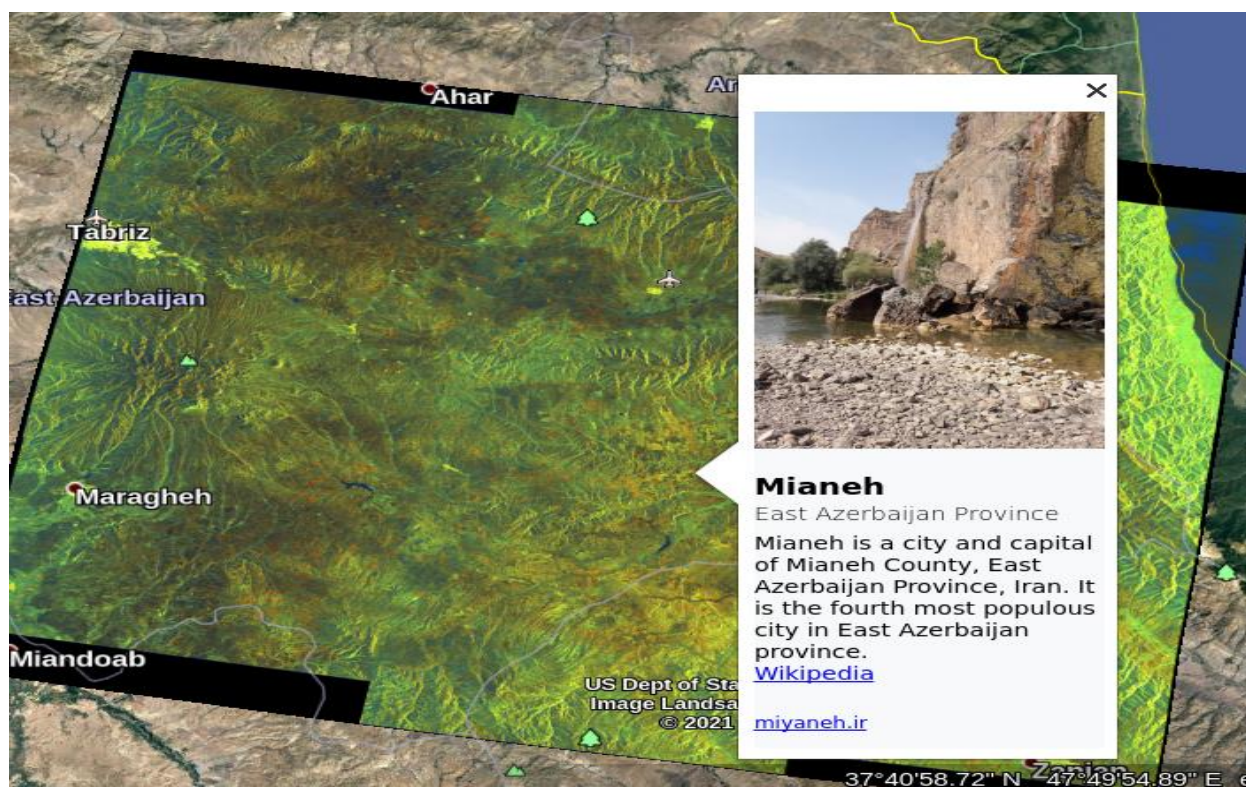
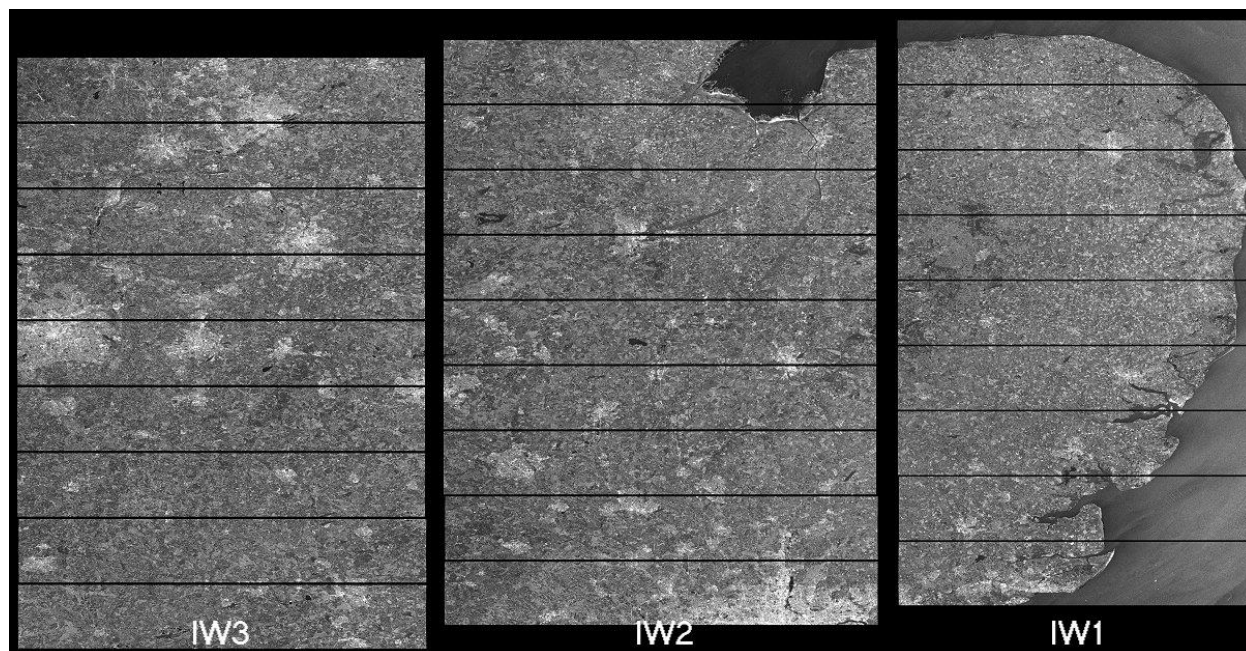
To find out where the study area is located in the image, open the Google Earth image file in the folder of each image.



- IWs 1 to 3 in the image

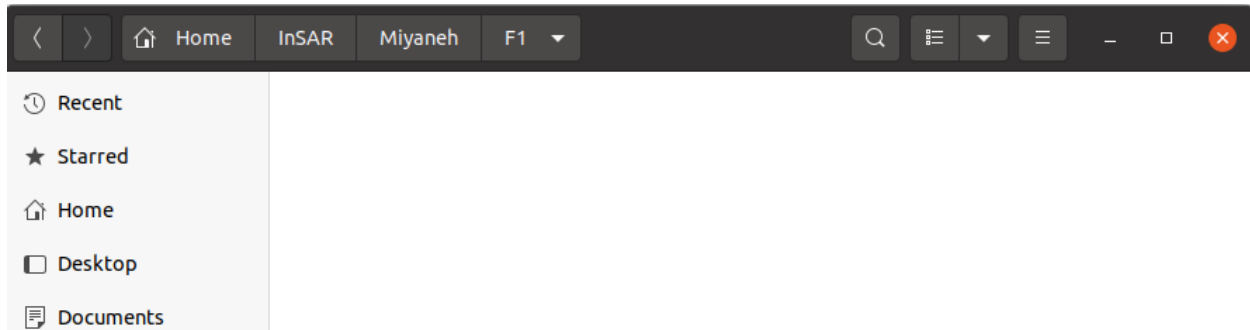
The image below shows IW1 on the right, IW2 in the middle, and IW3 on the left of the image.

In the Ascending images, the location of IW1 and IW3 is opposite to the Descending image.

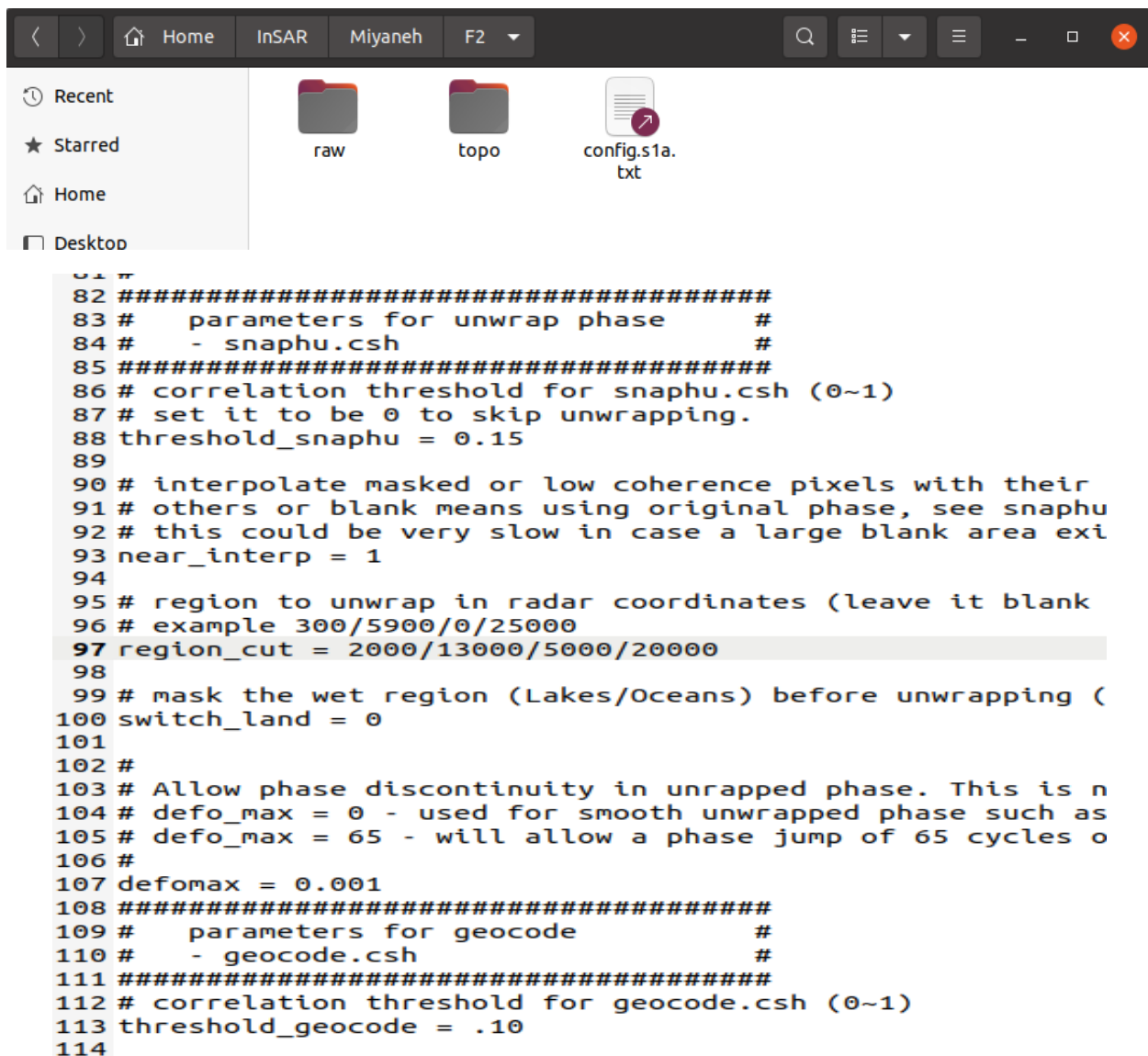


- In the example of this book, Mianeh city is in the IW2 = F2 section of the image .

This way, we delete the files inside the F1 and F3 folders.



Then configure and save the config.s1a.txt file in the F2 folder as shown above.

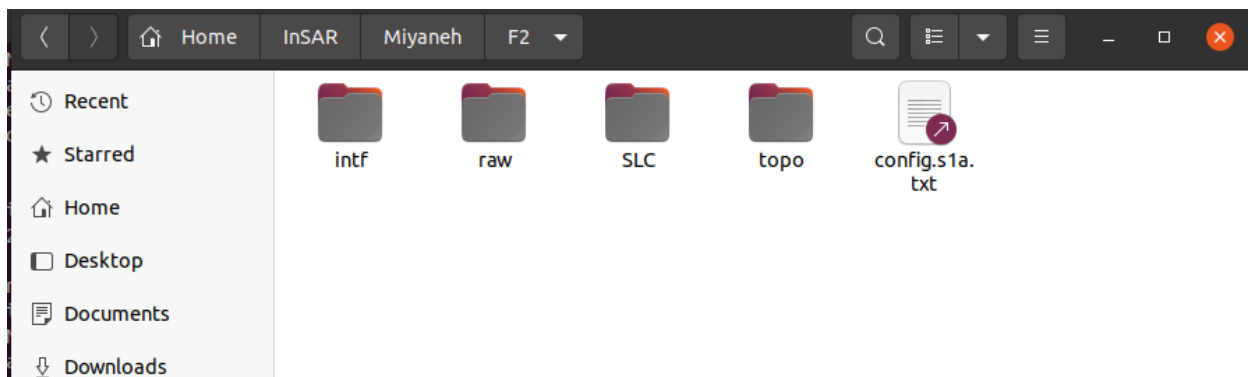


Then open the Terminal in the Miyaneh folder and run the following commands.

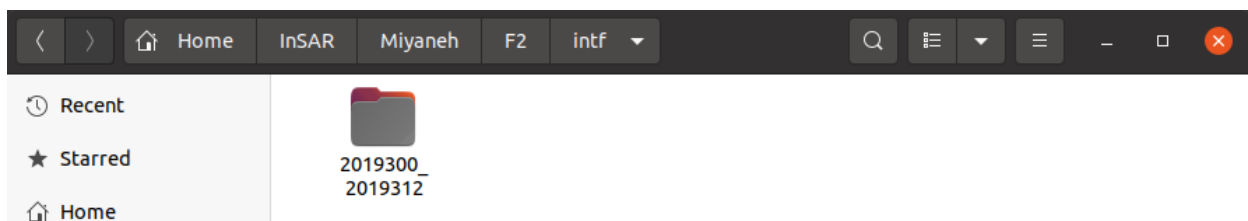
```
# tcsh
# bash run_tops.sh
```

```
siavash-Aspire-V3-571G:~/InSAR/Miyaneh> bash run_tops.sh
/home/siavash/InSAR/Miyaneh
/home/siavash/InSAR/Miyaneh/F1
basename: missing operand
Try 'basename --help' for more information.
*.SLC
raw/*.PRM: No match.
/home/siavash/InSAR/Miyaneh
/home/siavash/InSAR/Miyaneh/F2
S1_20191028_025310_F2.SLC
200

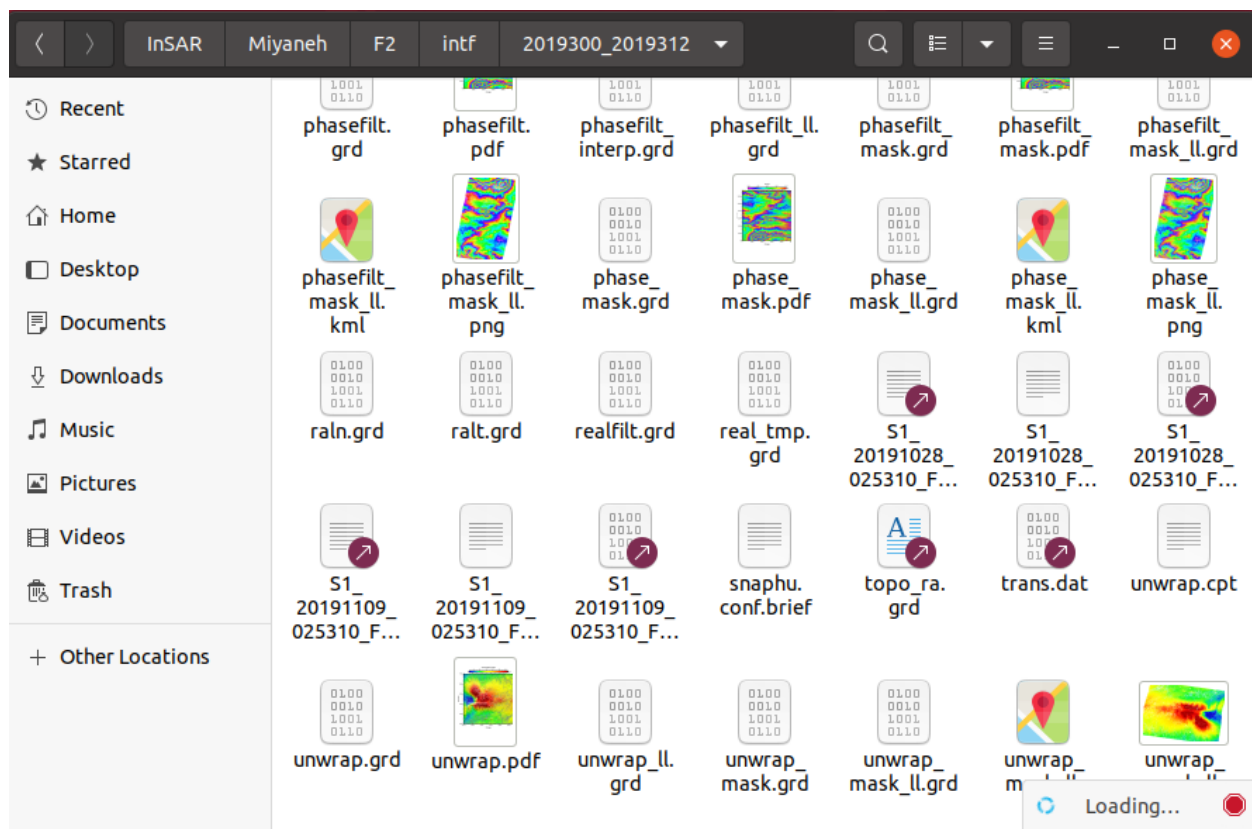
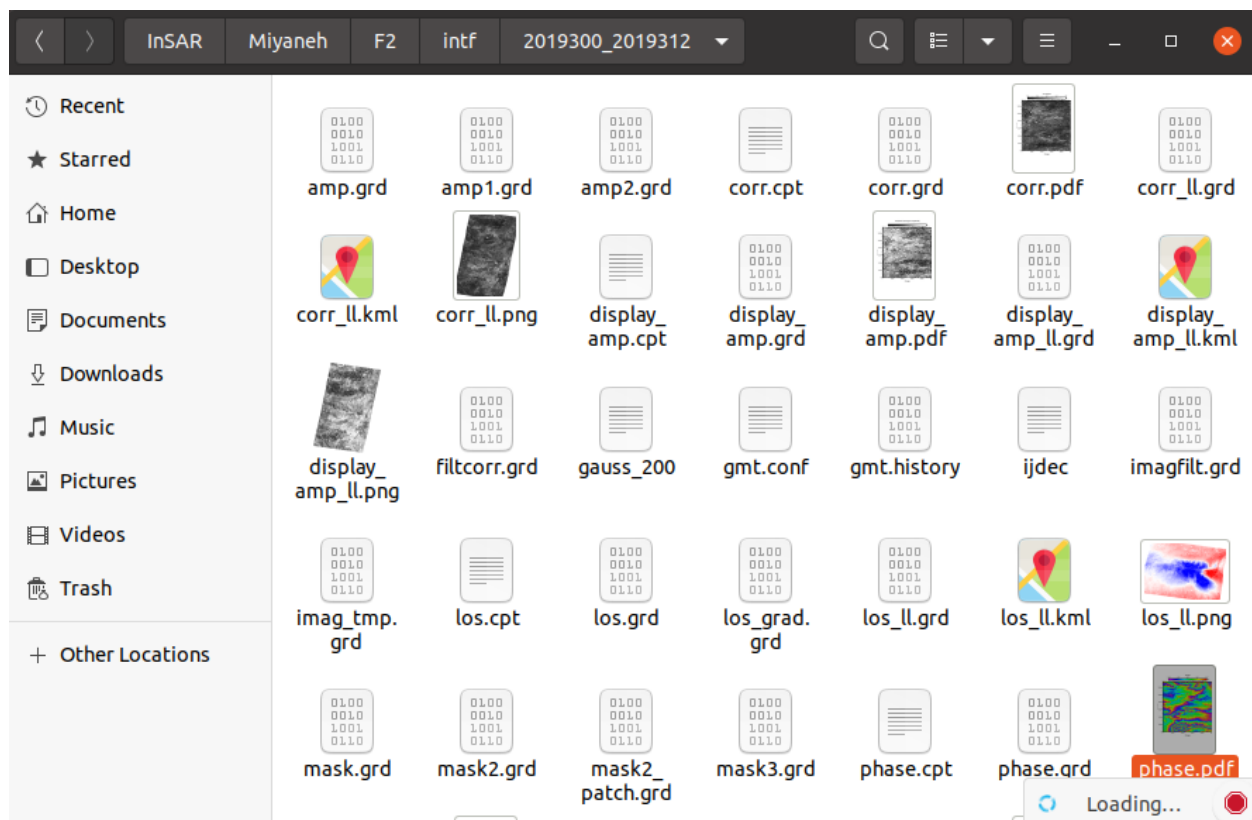
PREPROCESS - START
rm: No match.
```



- After processing is complete, the intf, SLC folder is created in F2



Inside the folder 20019300-2019312, you can see the interferogram, the displacement map, the KML file, and other files created in the interferogram folder:



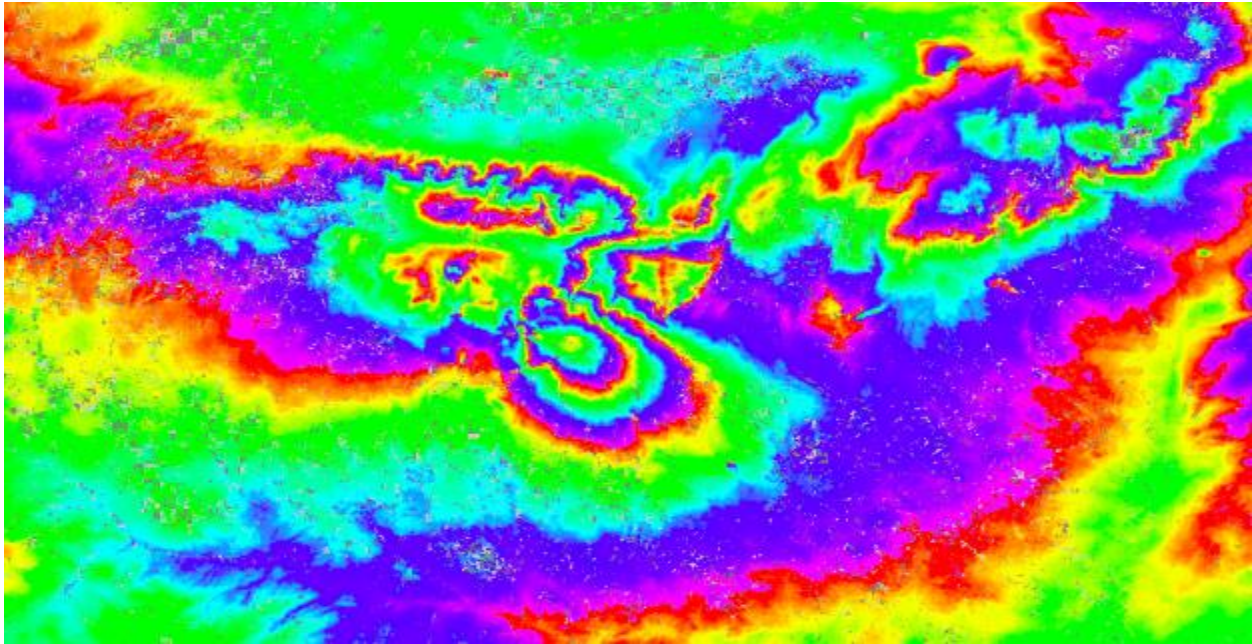


Figure 7. Torkamanchay (Iran) earthquake: Sentinel-1 interferometry

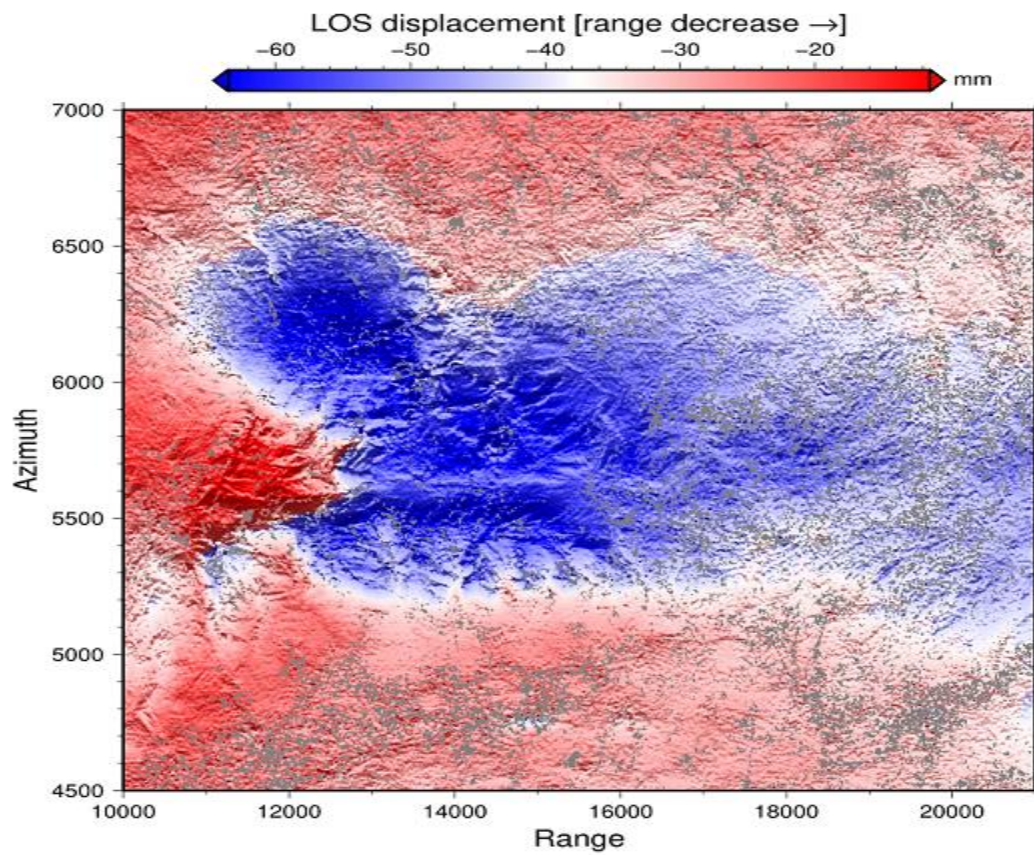


Figure 8. Torkamanchay (Iran) earthquake: LOS displacement

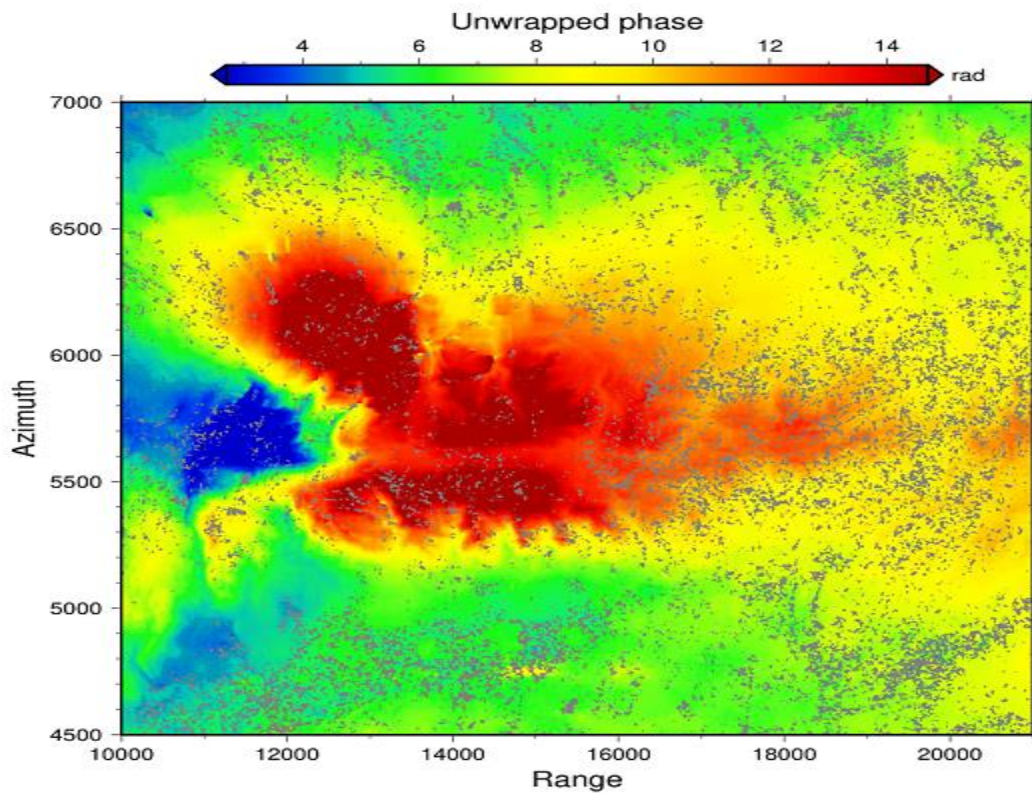
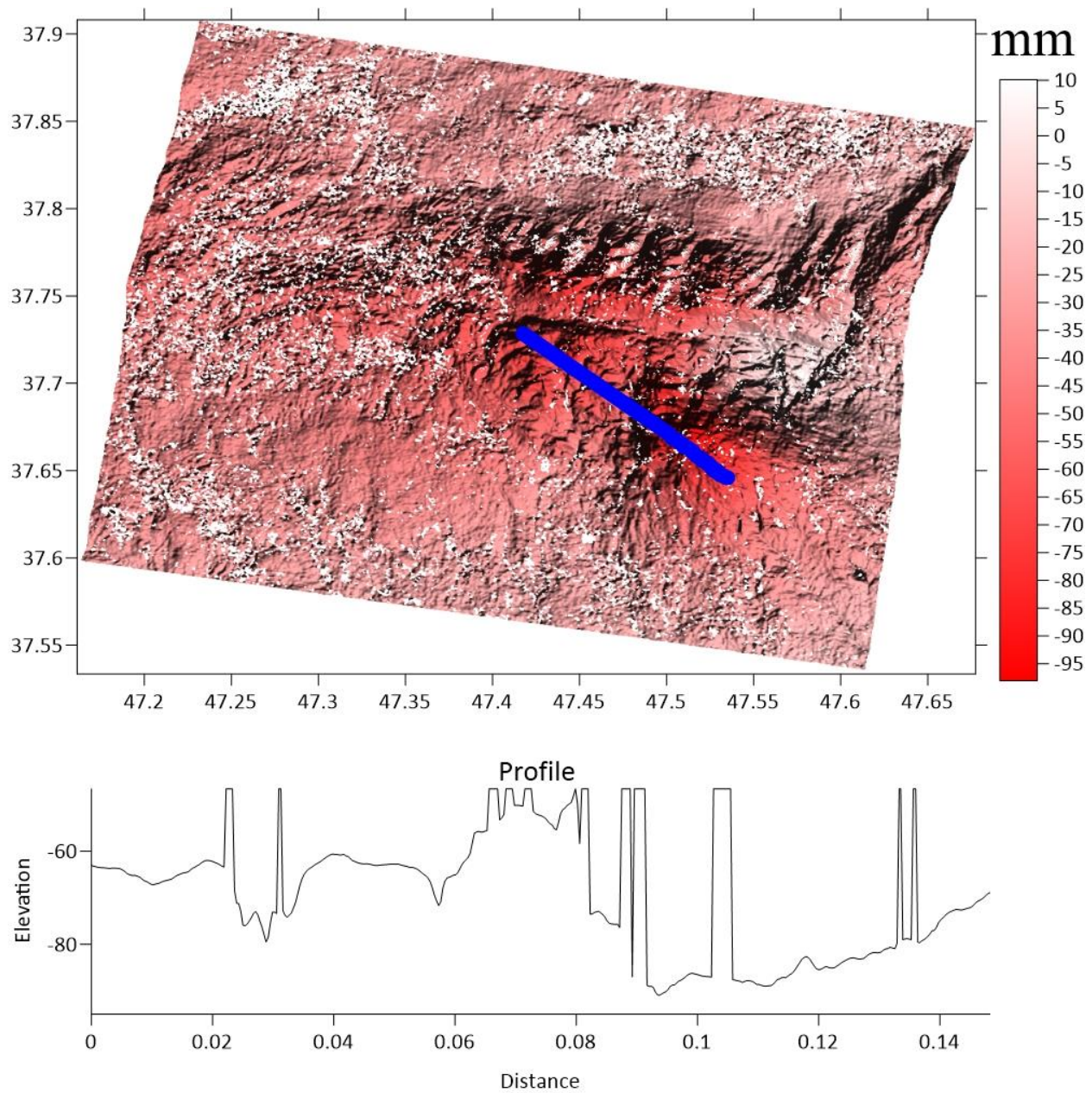


Figure 9. Torkamanchay (Iran) earthquake: phase unwrapping

Also, you can use software such as surfer and QGIS to do your graphic work on file `los_ll` (displacement file) or files with extension `..._ll`



You can compare the results of your processing with the results of two published articles about this earthquake that are available in the references and make sure that your processing is correct.

2.3. Multi-Temporal InSAR methods

The limitations of the DInSAR technique, such as the non-correlation in the production of interferograms and the residual topographic error, led to the development of Persistent (Permanent) Scatterers (PS) and Time-series. Because a useful way to study the behavior of a phenomenon during the time is the generation of time series analysis (Berardino et al., 2002). Recent improvements in multi-temporal InSAR (MTI) methods and sensors increased the quality of monitoring land deformations with precision in millimeters (Colesanti et al., 2003; Ferretti et al., 2001). Several analysis methods in remote sensing, especially with radar data, rely on processing information over time. Time series analysis of InSAR data is an important tool for monitoring and measuring the displacement of the Earth's surface. Different algorithms have been developed for the time series analysis of InSAR. The best methods are as follows:

- SBAS: Small Baseline Subset (Berardino et al., 2002)
- DePSI: Delft Persistent Scatterer Interferometry (Kampes, 2005, 2006)
- PSInSAR: Permanent Scatterer InSAR (Ferretti et al., 2000, 2001)
- SqueeSAR (Ferretti et al., 2009, 2011)
- StaMPS: Stanford Method for Persistent Scatterers (Hooper et al., 2004; Hooper, 2008)

2.3.1. Small Baseline Subset (SBAS)

Berardino et al., 2002 introduced Small Baseline Subset (SBAS), as a Multi-Temporal InSAR methodology. Despite PS relying on a single master interferogram, SBAS is based on multiple-master interferograms. To solve decorrelation and increase temporal and spatial sampling, the focus of SBAS is on small and short sizes of spatial and temporal baselines (Berardino et al., 2002). The ability of SBAS to detect and investigate has been approved in different applications, mostly on SAR data (Lanari et al., 2002; Lanari et al., 2004a; Lanari et al., 2004b; Manzo et al., 2006; Cascini et al., 2006). As mentioned before, PS is based on a common master. If data pairs have a large baseline, especially larger than the critical baseline, the results will be affected by the decorrelation of baselines. In this method, all the available data acquisitions are used by the price of the imaged pixel density (Berardino et al., 2002). As a result SBAS is proposed as a development of the techniques proposed by Lundgren et al., 2001. The focus is on multi-short-based line acquisition subsets by easily and effectively combining all the short baseline interferograms. Applying the Singular Value Decomposition (SVD) obtains the combination based on a minimum norm of the velocity deformation. SBAS is reposing two requirements. First, one is using all the acquisitions in the different SB subsets to increase temporal sampling rate and the second one is spatially dense deformation maps. (Berardino et al., 2002). Having described SBAS, the following figure provides the spatial-temporal interferogram network, based on SBAS schematically. As seen, SBAS performs based on multi-master and slave SAR images.

2.3.2. SBAS Processing by GMTSAR

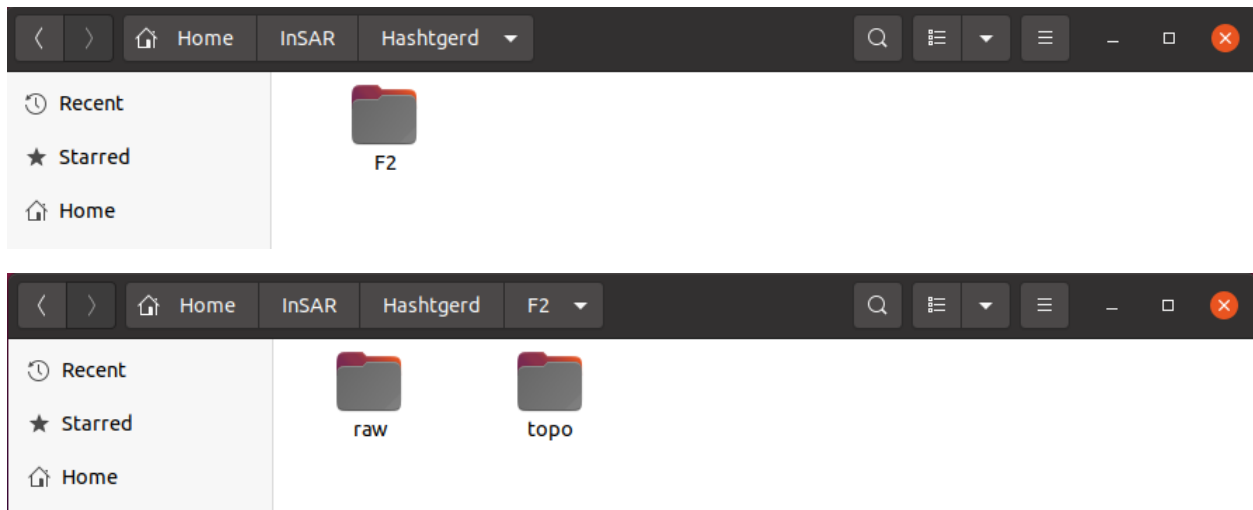
In the following, using 13 Ascending images, the information of which is shown in the table below. Hashtgerd plain subsidence time-series was calculated using SBAS Algorithm and GMTSAR.

Table 5. Main characteristics of the Sentinel-1 images considered for of the Hashtgerd plain subsidence

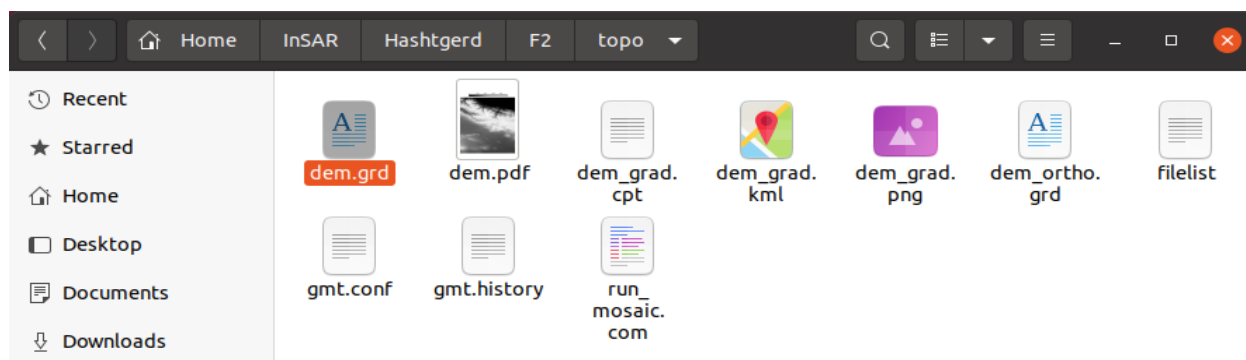
Image date	Master/Slave	Track	Orbit	Bprep (m)	Btemp (days)	Modeled Coherence	Height Ambg (m)	Delta fDC (Hz)
۲۰۲۰/۱۲/۱۰	Master	28	35625	0.00	0.00	1.00	∞	0.00
۲۰۲۱/۱۰/۳	Slave	28	35975	-10.12	-24.00	0.97	1552.03	-2.62
۲۰۲۰/۱۲/۲۲	Slave	28	35800	-15.75	-12.00	0.98	996.95	-1.23
۲۰۲۱/۱۱/۵	Slave	28	36150	13.91	-36.00	0.96	-1129.39	-0.62
۲۰۲۱/۱۲/۷	Slave	28	36325	45.56	-48.00	0.92	-344.83	-0.14
۲۰۲۰/۹/۲۹	Slave	28	34575	6.52	72.00	0.93	-2410.05	-3.13
۲۰۲۱/۲/۰۸	Slave	28	36500	23.27	-60.00	0.93	-675.01	-0.87
۲۰۲۱/۲/۲۰	Slave	28	36675	-3.06	-72.00	0.93	5136.13	-1.21
۲۰۲۰/۱۰/۲۳	Slave	28	34925	-67.97	48.00	0.90	231.13	-1.48
۲۰۲۰/۱۱/۴	Slave	28	35100	-3.42	36.00	0.96	4593.81	-2.34
۲۰۲۰/۱۰/۱۱	Slave	28	34750	-132.96	60.00	0.84	118.16	-2.57
۲۰۲۰/۱۱/۱۶	Slave	28	35275	13.30	24.00	0.97	-1181.26	0.55
۲۰۲۰/۱۱/۲۸	Slave	28	35450	112.21	12.00	0.90	-140.00	-0.60

First create a folder named F2 and create raw and topo folders in it according to the images below.

(Our study area is at IW2 = F2 and if it is at IW1/3, you must create F1/3)

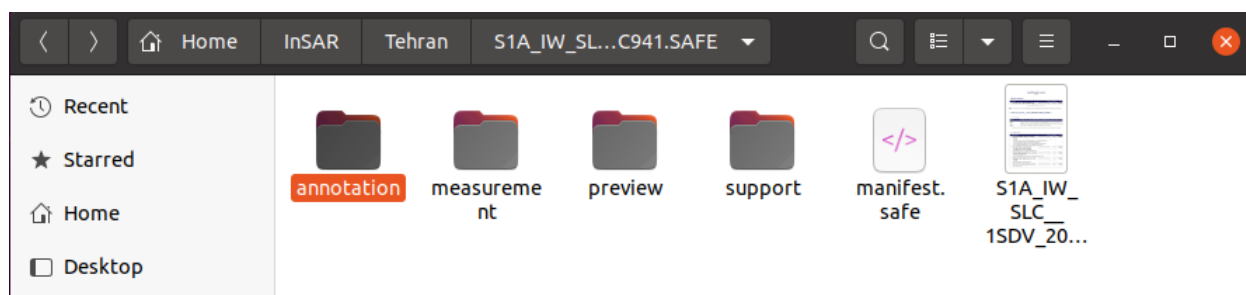
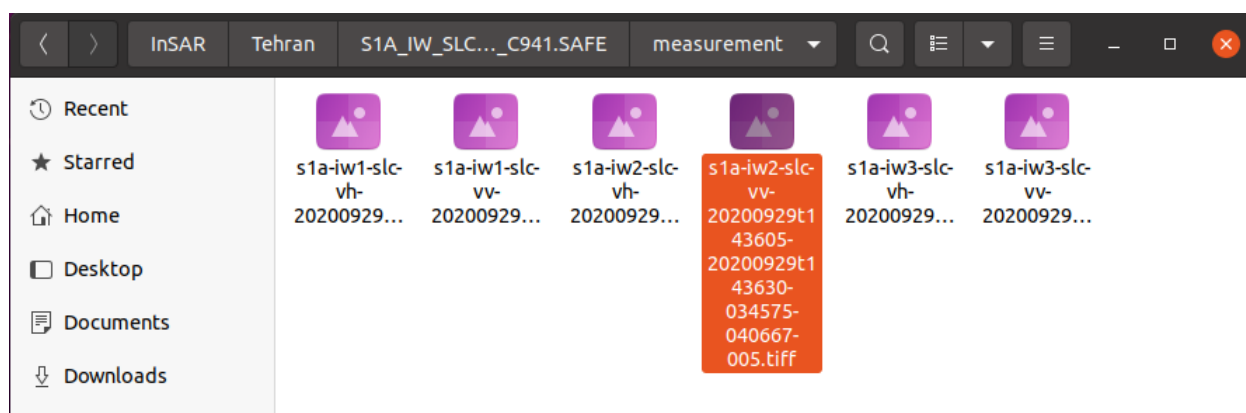


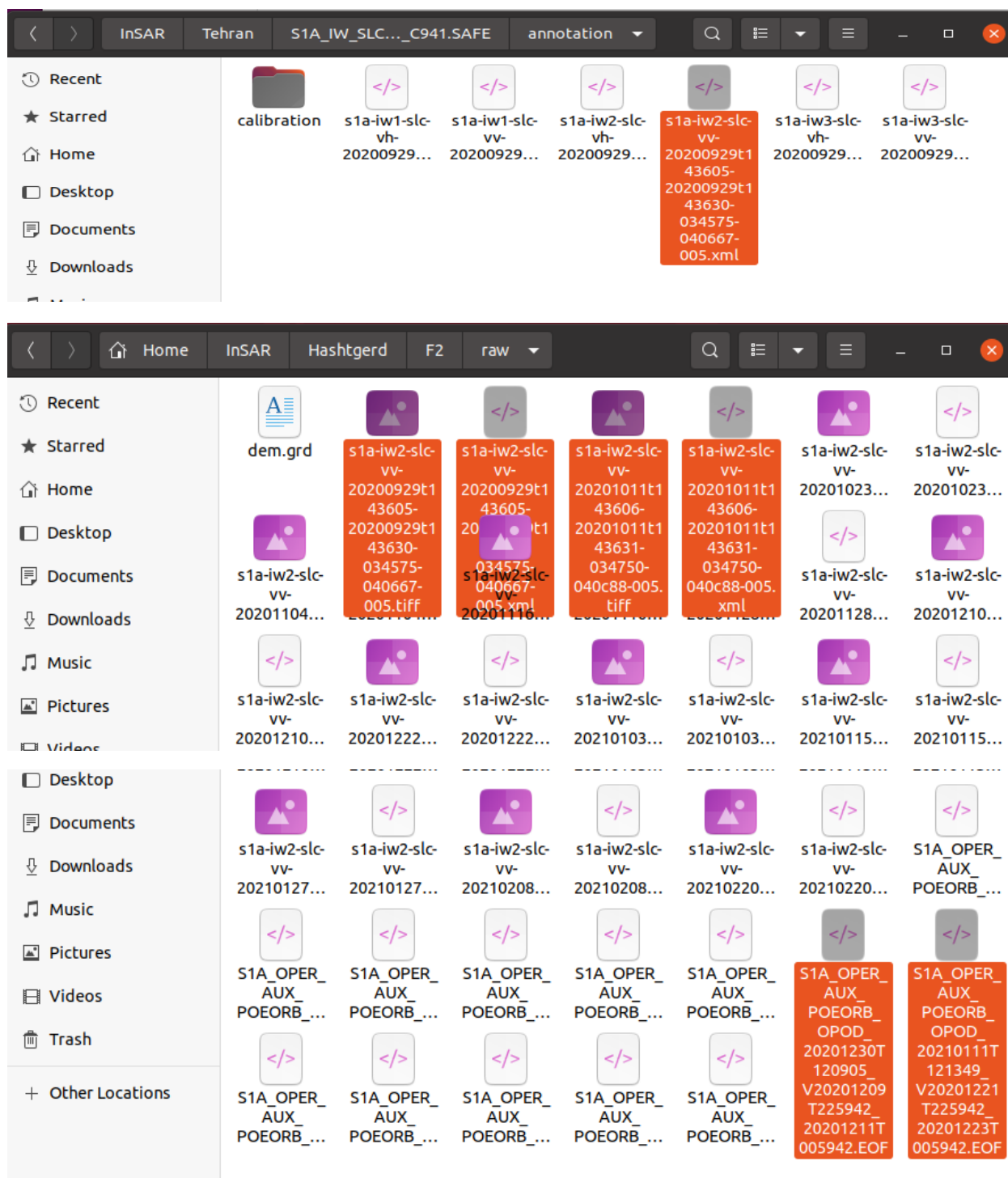
Then unzip the downloaded DEM file and copy it into the topo folder.



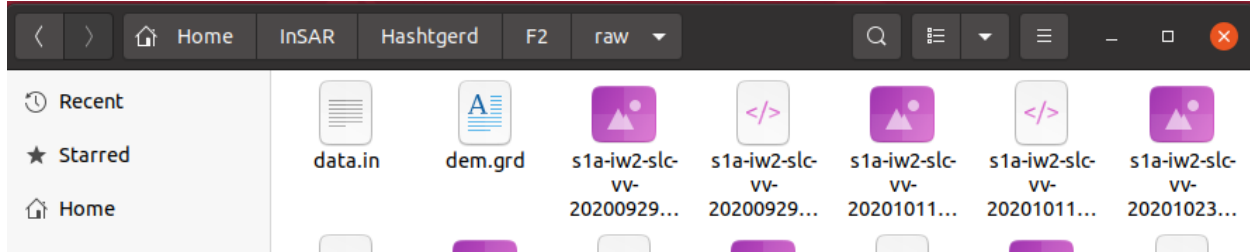
Then SENTINEL_1A images in .xml format And .tiff with vv polarization from inside the measurement and annotation folders and orbital data files in EOF format. And copy the dem.grd file in the topo folder into the raw folder .

(IW2 images with VV or VH polarization should be copied and both images should not be copied in .tiff format)



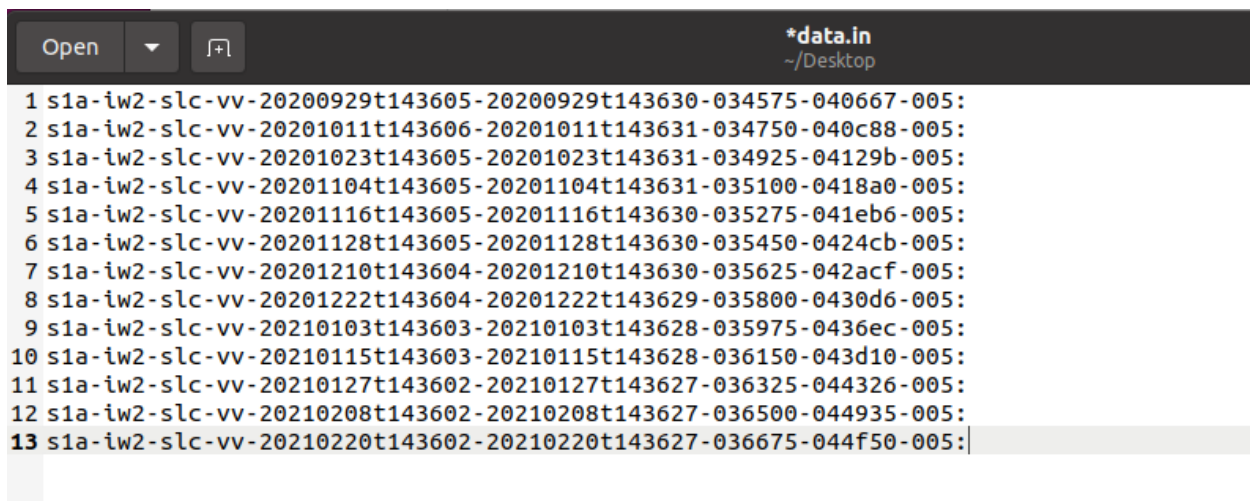
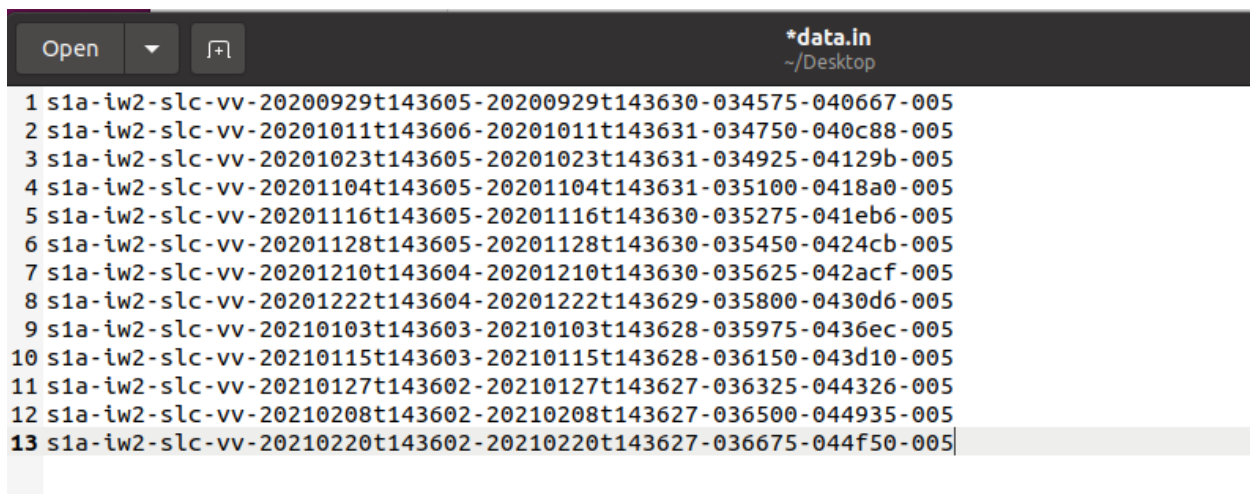


Then create a file named data.in and put it in the raw folder.



Each row of the data.in file contains the image name SENTINEL1 without the .tiff format and the orbit image file name.

(Use two dots in each row to separate the image name and orbital file :)




```
*data.in
~/Desktop

667-005:S1A_OPER_AUX_POEORB_OPOD_20201019T121408_V20200928T225942_20200930T005942.EOF
c88-005:S1A_OPER_AUX_POEORB_OPOD_20201031T121318_V20201010T225942_20201012T005942.EOF
29b-005:S1A_OPER_AUX_POEORB_OPOD_20201112T121152_V20201022T225942_20201024T005942.EOF
8a0-005:S1A_OPER_AUX_POEORB_OPOD_20201124T121218_V20201103T225942_20201105T005942.EOF
eb6-005:S1A_OPER_AUX_POEORB_OPOD_20201206T121208_V20201115T225942_20201117T005942.EOF
4cb-005:S1A_OPER_AUX_POEORB_OPOD_20201218T121325_V20201127T225942_20201129T005942.EOF
acf-005:S1A_OPER_AUX_POEORB_OPOD_20201230T120905_V20201209T225942_20201211T005942.EOF
0d6-005:S1A_OPER_AUX_POEORB_OPOD_20210111T121349_V20201221T225942_20201223T005942.EOF
6ec-005:S1A_OPER_AUX_POEORB_OPOD_20210123T121622_V20210102T225942_20210104T005942.EOF
d10-005:S1A_OPER_AUX_POEORB_OPOD_20210204T122413_V20210114T225942_20210116T005942.EOF
326-005:S1A_OPER_AUX_POEORB_OPOD_20210216T121817_V20210126T225942_20210128T005942.EOF
935-005:S1A_OPER_AUX_POEORB_OPOD_20210228T121823_V20210207T225942_20210209T005942.EOF
f50-005:S1A_OPER_AUX_POEORB_OPOD_20210312T121633_V20210219T225942_20210221T005942.EOF
```

Then open the Terminal in the raw folder and run the following commands with mode = 1

```
# tcsh
# preproc_batch_tops.csh data.in dem.grd 1
```

```
siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/raw
siavash@siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/raw$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/raw> preproc_batch_tops.csh data.in dem.grd 1
rm: No match.
rm: cannot remove 'baseline_table.dat': No such file or directory
Reading in 5379 lines of info from XML...
Writing 16 lines for orbit...
Writing 52 lines of precise orbit for the LED file...
Successfully opened junk1
```

Recent

Starred

Home

baseline.ps

baseline_table.dat

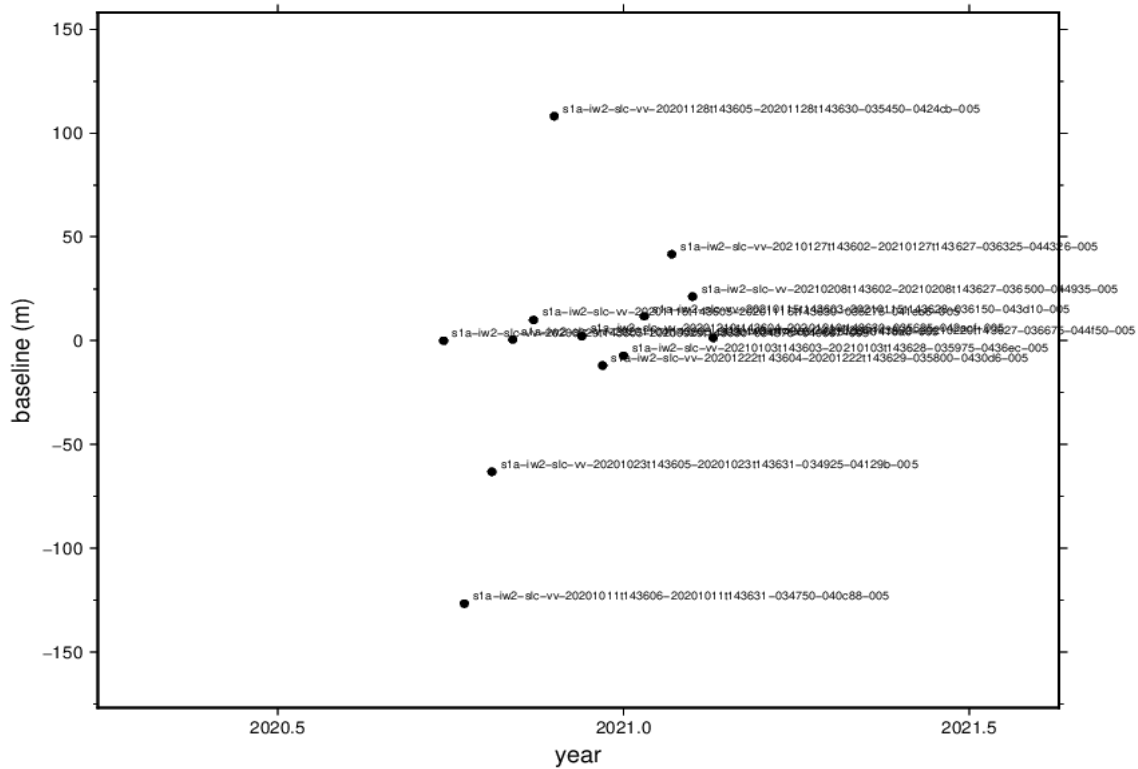
data.in

dem.grd

gmt.history

s1a-iw2-slc-vv-20200929...

s1a-iw2-slc-vv-20200929...



- The baseline_table.dat and baseline.ps files are created, ie the temporal and spatial baseline information of the images

In this step, one of the available images must be selected as the Master image. The Master image should have the least amount of correlation. For this purpose, the selected image should have the least amount of temporal, spatial, and Doppler baseline compared to other images.

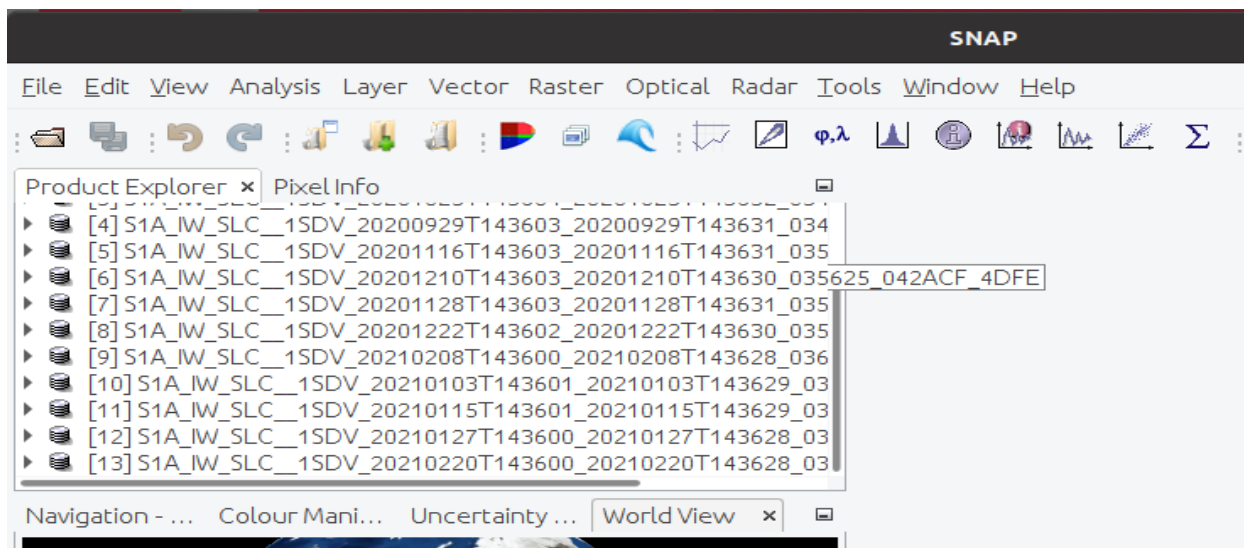
$$\rho_{\text{total}} = \rho_{\text{temporal}} + \rho_{\text{spatial}} + \rho_{\text{doppler}} + \rho_{\text{thermal}} \quad \square$$

$$\left[1 - f\left(\frac{T}{T_{\perp}}\right) \right] \left[1 - f\left(\frac{B_{\perp}}{B_{\perp}^c}\right) \right] \left[1 - f\left(\frac{F_{DC}}{F_{DC}^c}\right) \right] \rho_{\text{thermal}}$$

The above formula is the relation of determining the Master image from the value of temporal, spatial, and Doppler base lines of images in which T components of temporal baseline, $B_{\perp}$ vertical baseline lines, F_{DC} Doppler, ρ correlation indicators, and c index represent the critical value. Therefore, an image is selected as the Master image that has the highest value for N of the existing images.

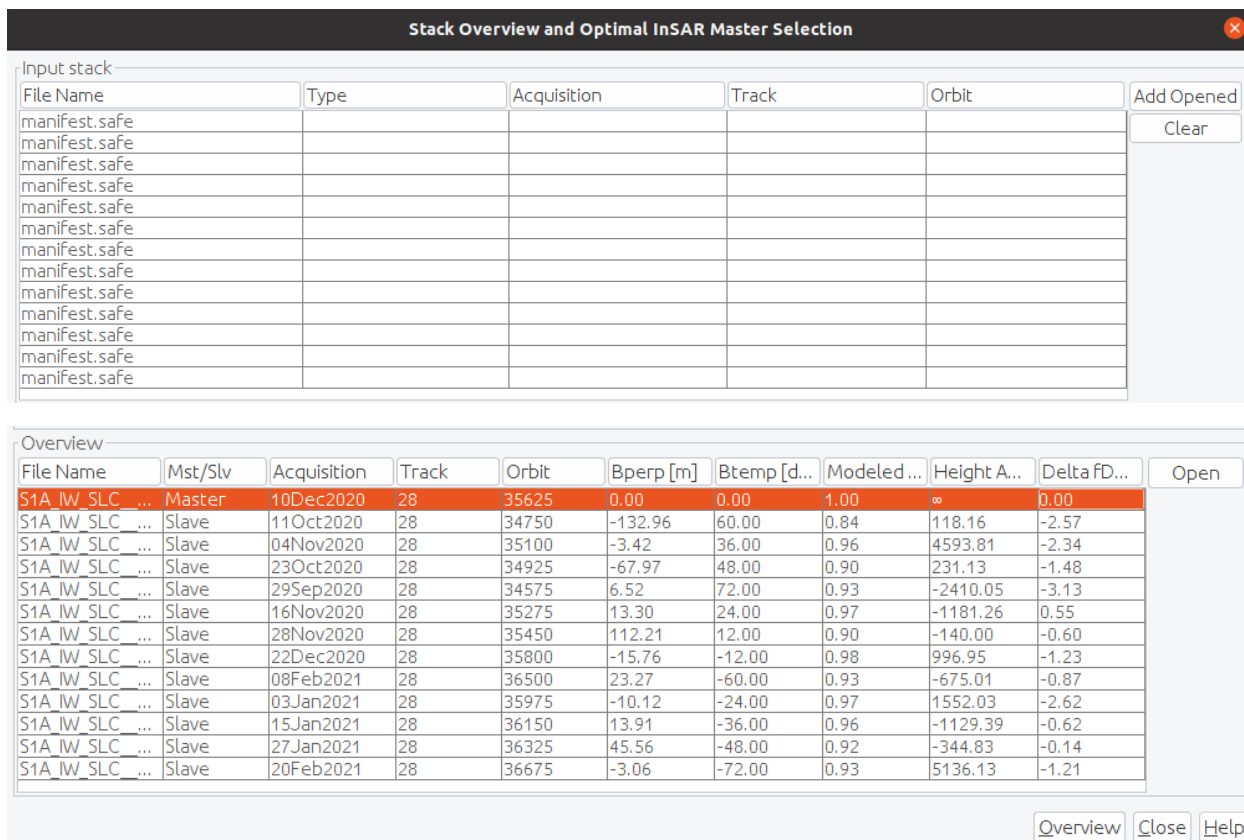
You can also use SNAP software to select the Master image automatically. How to select the Master image with SNAP software is described below.

First, select all the images.



Then follow the steps below in the software to automatically select the Master image.

Radar → Interferometric → InSAR Stack Overview → Add opened → Overview

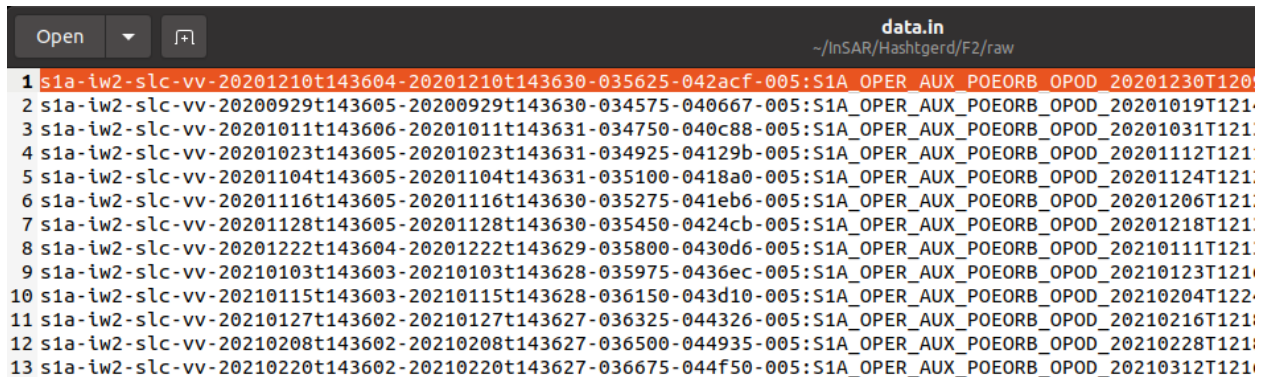
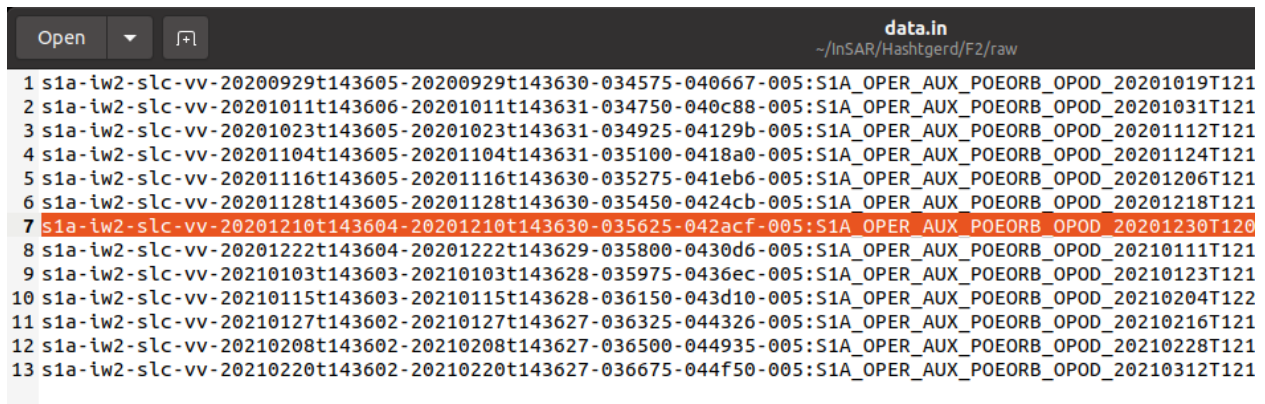


- The image with the date 20201210 has been selected as the Master image.

After copy the baseline_table.dat file in the F2 folder to the raw folder.



Then open the data.in file and move the image selected as the Master image (here is the image with the Master image 20201210) to the first row and save the file.



Then open the Terminal in the raw folder and run the following command.



```

siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/raw
siavash@siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/raw$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/raw> preproc_batch_tops.csh data.in dem.grd 2

```

- Running of this command is long

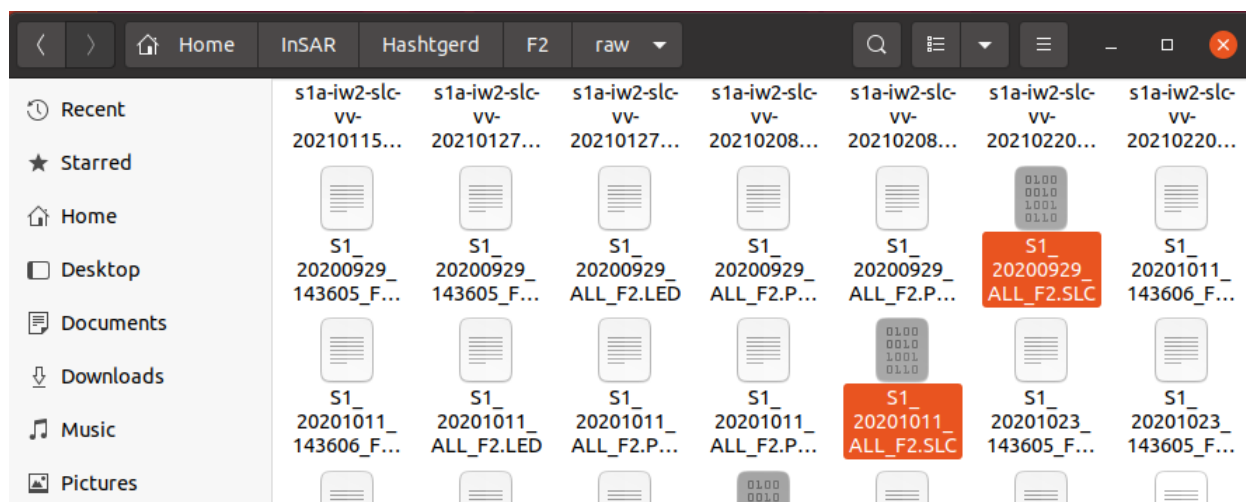
Then open the baseline_table.dat file in the F2 folder and replace the first column with the name of the images created with the SLC format images in the raw folder.



Open baseline_table.dat

~/InSAR/Hashtgerd/F2

1	s1a-iw2-slc-vv-20200929t143605-20200929t143630-034575-040667-005	2020272.6084004175	2462	0.000000
2	s1a-iw2-slc-vv-20201011t143606-20201011t143631-034750-040c88-005	2020284.6084043179	2474	-119.262
3	s1a-iw2-slc-vv-20201023t143605-20201023t143631-034925-04129b-005	2020296.6084029265	2486	-87.5201
4	s1a-iw2-slc-vv-20201104t143605-20201104t143631-035100-0418a0-005	2020308.6084017954	2498	-45.3900
5	s1a-iw2-slc-vv-20201116t143605-20201116t143630-035275-041eb6-005	2020320.6083996973	2510	-43.0586
6	s1a-iw2-slc-vv-20201128t143605-20201128t143630-035450-0424cb-005	2020332.6083941546	2522	34.81749
7	s1a-iw2-slc-vv-20201210t143604-20201210t143630-035625-042acf-005	2020344.6083901611	2534	-54.7312
8	s1a-iw2-slc-vv-20201222t143604-20201222t143629-035800-0430d6-005	2020356.6083831915	2546	-62.5499
9	s1a-iw2-slc-vv-20210103t143603-20210103t143628-035975-0436ec-005	2021002.6083762727	2557	-66.5200
10	s1a-iw2-slc-vv-20210115t143603-20210115t143628-036150-043d10-005	2021014.6083701088	2569	-42.1245
11	s1a-iw2-slc-vv-20210127t143602-20210127t143627-036325-044326-005	2021026.6083658494	2581	-15.3412
12	s1a-iw2-slc-vv-20210208t143602-20210208t143627-036500-044935-005	2021038.6083615348	2593	-32.5932
13	s1a-iw2-slc-vv-20210220t143602-20210220t143627-036675-044f50-005	2021050.6083573883	2605	-52.2226



baseline_table.dat ~/InSAR/Hashtgerd/F2									
1	S1_20200929_ALL_F2	2020272.6084004175	2462	0.000000000000	0.000000000000				
2	S1_20201011_ALL_F2	2020284.6084043179	2474	-119.262191565033	-126.590393757482				
3	S1_20201023_ALL_F2	2020296.6084029265	2486	-87.520154409299	-63.074508293494				
4	S1_20201104_ALL_F2	2020308.6084017954	2498	-45.390081258401	0.631124700164				
5	S1_20201116_ALL_F2	2020320.6083996973	2510	-43.058648834251	9.982721062489				
6	S1_20201128_ALL_F2	2020332.6083941546	2522	34.817492990522	108.076064181767				
7	S1_20201210_ALL_F2	2020344.6083901611	2534	-54.731291062388	2.246383288969				
8	S1_20201222_ALL_F2	2020356.6083831915	2546	-62.549959829139	-11.907525069547				
9	S1_20210103_ALL_F2	2021002.6083762727	2557	-66.520039997138	-7.217992401164				
10	S1_20210115_ALL_F2	2021014.6083701088	2569	-42.124551852898	11.765894256297				
11	S1_20210127_ALL_F2	2021026.6083658494	2581	-15.341273060346	41.656535439103				
12	S1_20210208_ALL_F2	2021038.6083615348	2593	-32.593291069440	21.298088096057				
13	S1_20210220_ALL_F2	2021050.6083573883	2605	-52.222688507093	1.355018403001				

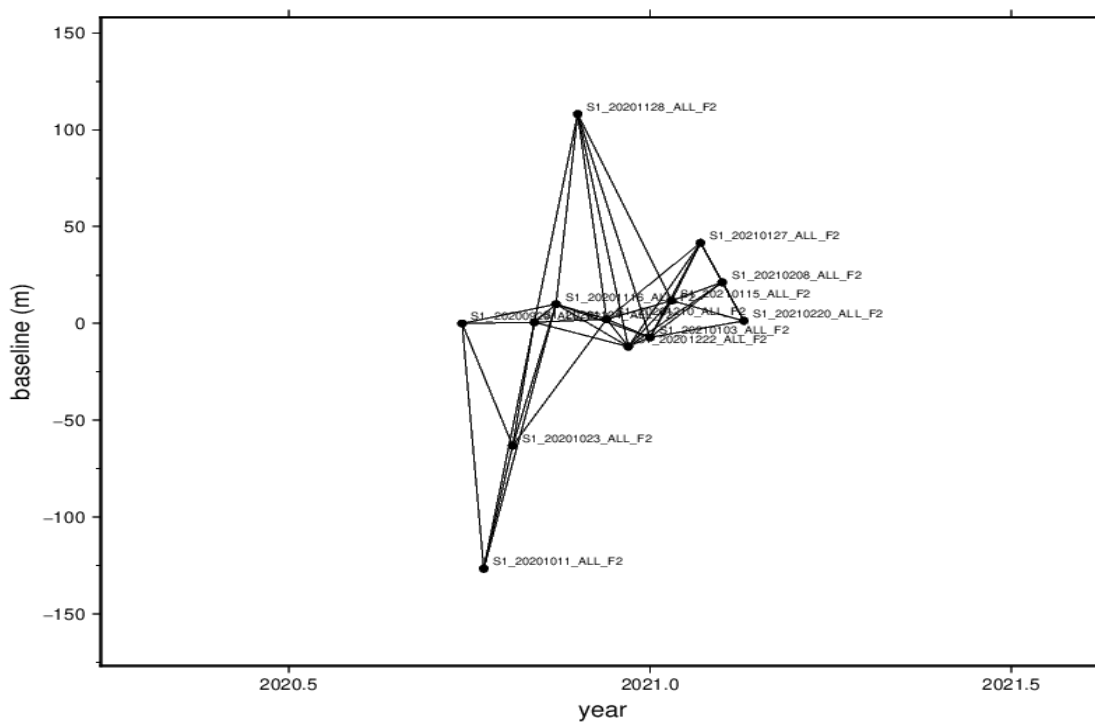
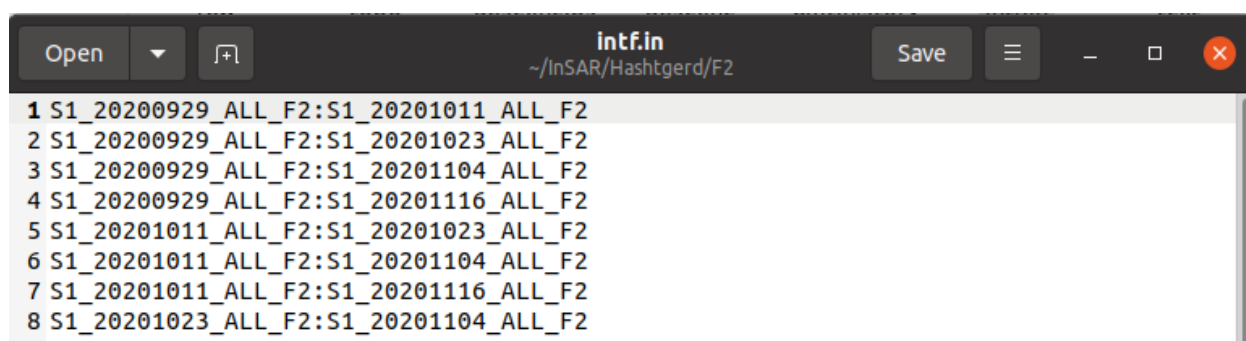
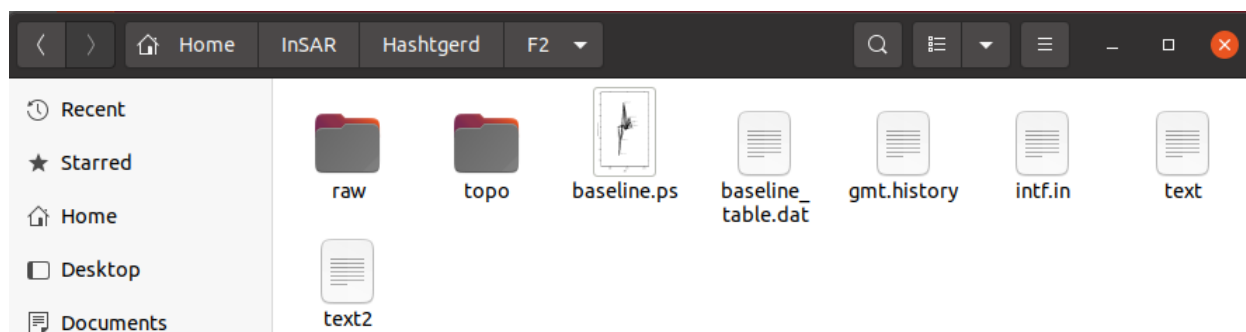
Create a blank text file named intf.in in the F2 folder, then open the Terminal in the F2 folder and run the following commands.



```
# tcsh
# select_pairs.csh baseline_table.dat 50 150
```

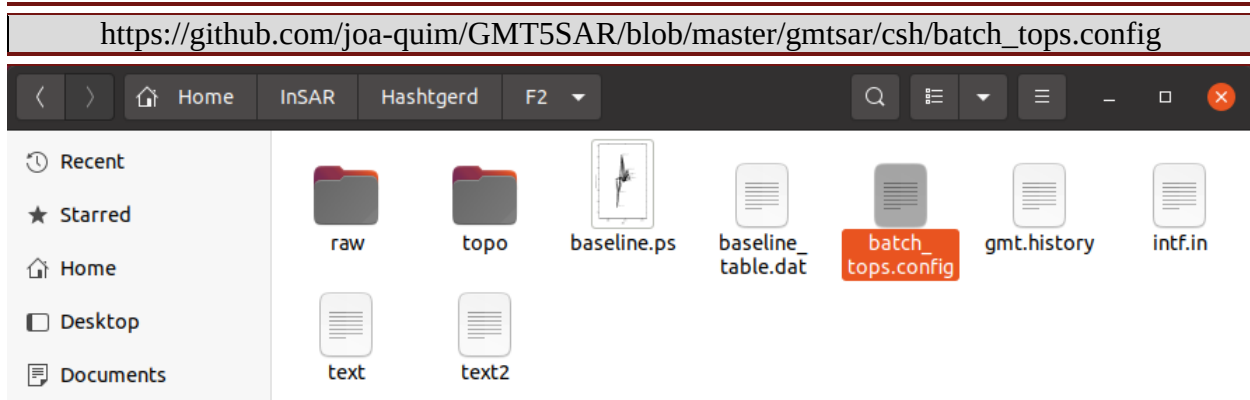
```
siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2
siavash@siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2> select_pairs.csh baseline_table.dat 50 150
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2>
```

- The number 50 is the temporal baseline (day), and 100 is the spatial baseline (m) and the output file intf.in contains a pair of interferograms with a temporal baseline of fewer than 50 days and the spatial baseline is less than 100 meters.



- The intf.in file shows the list of interferograms, and the baseline.ps file shows the network of interferogram graphs.

Then download the batch_tops.config file from the following address and place it in the F2 folder.



Then open the batch_tops.config file and set the following parameters and finally save the file.

master_image = master image name

proc_stage = 1

filter_wavelength = 200

range_dec = 8

azimuth_dec = 2

threshold_snaphu = 0

threshold_geocode = 0

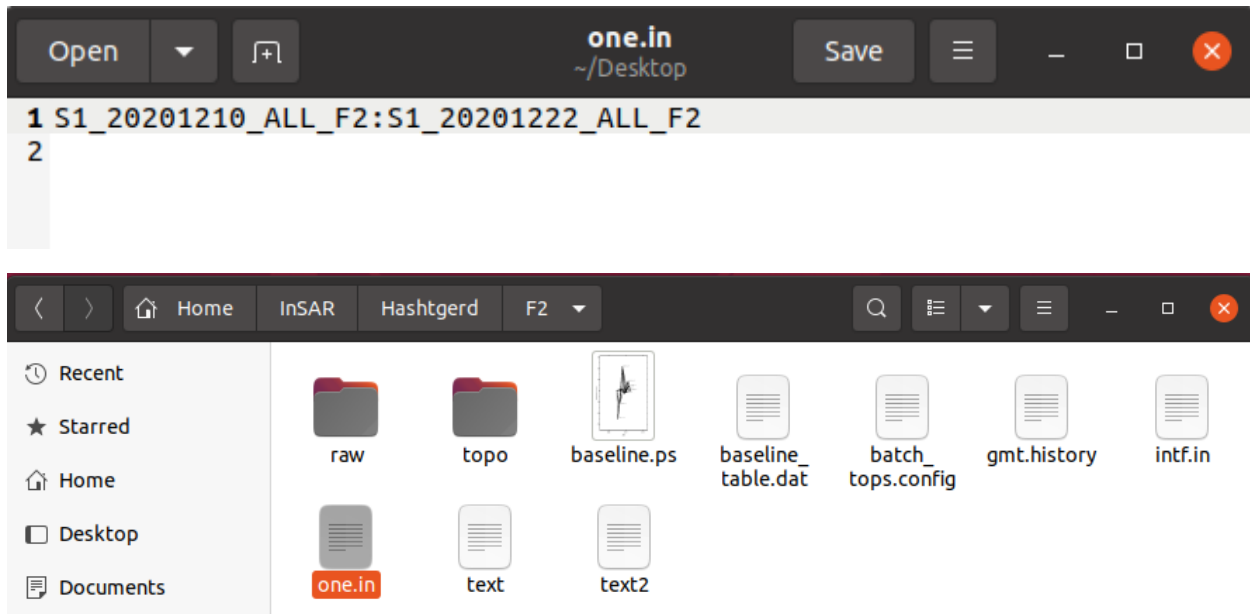
```

11 #####
12 # processing stage for intf_batch.csh #
13 #####
14 # 1 - start from make topo_ra
15 # 2 - start from make and filter interferograms, unwrap and geocode
16 proc_stage =1
17
18 # the namestem of the master image - REQUIRED
19 master_image = S1_20201210_ALL_F2
20
21 #####
22 #   parameters for preprocess           #
23 #   - pre_proc_batch.csh                #
24 #   following 4 parameters are OPTIONAL #
25 #####

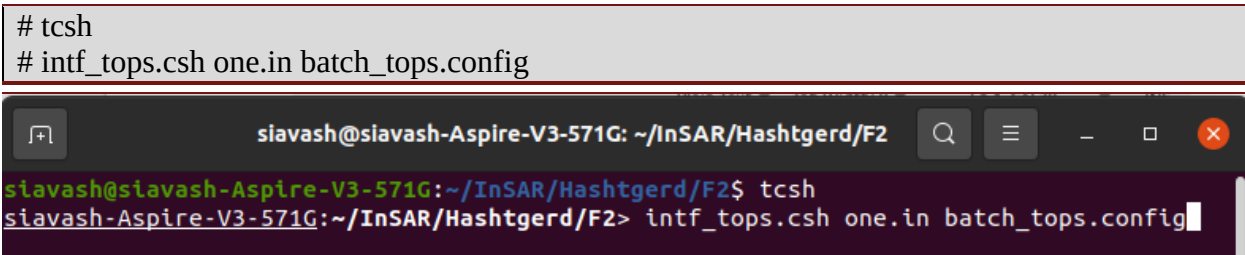
```

- The value of the above parameters can vary depending on the purpose of the study, the study area, the data used, etc., and in the example, in this book, these numbers are hypothetically selected.

Then create a text file called one.in and copy and save the Master image name as one of the Slave images and place it in the F2 folder.

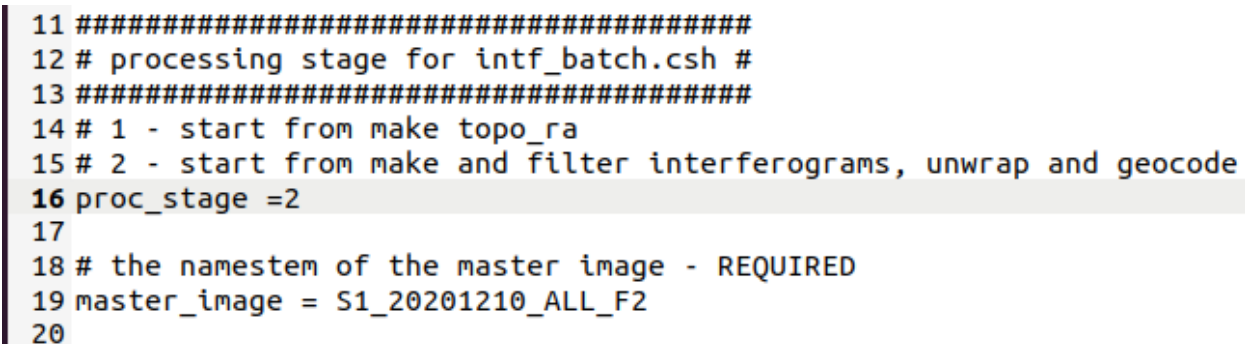


Then open the terminal in folder F2 and run the following commands.



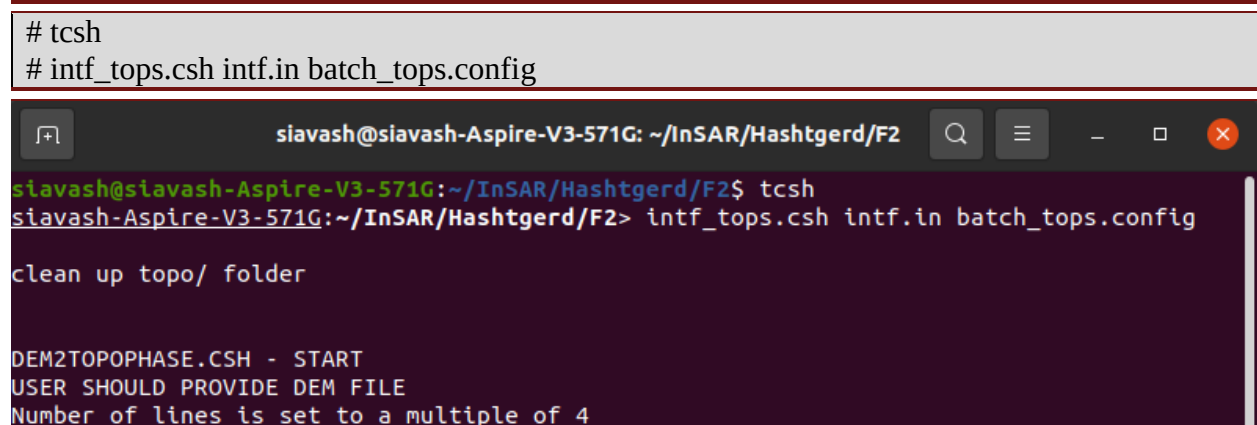
- In this part, the value of the DEM or topography is converted to phase.

After running the above command, open the batch_tops.config file again and set and save proc_stage = 2



Then open the Terminal in the F2 folder and run the following commands.

```
# tcsh
# intf_tops.csh intf.in batch_tops.config
```

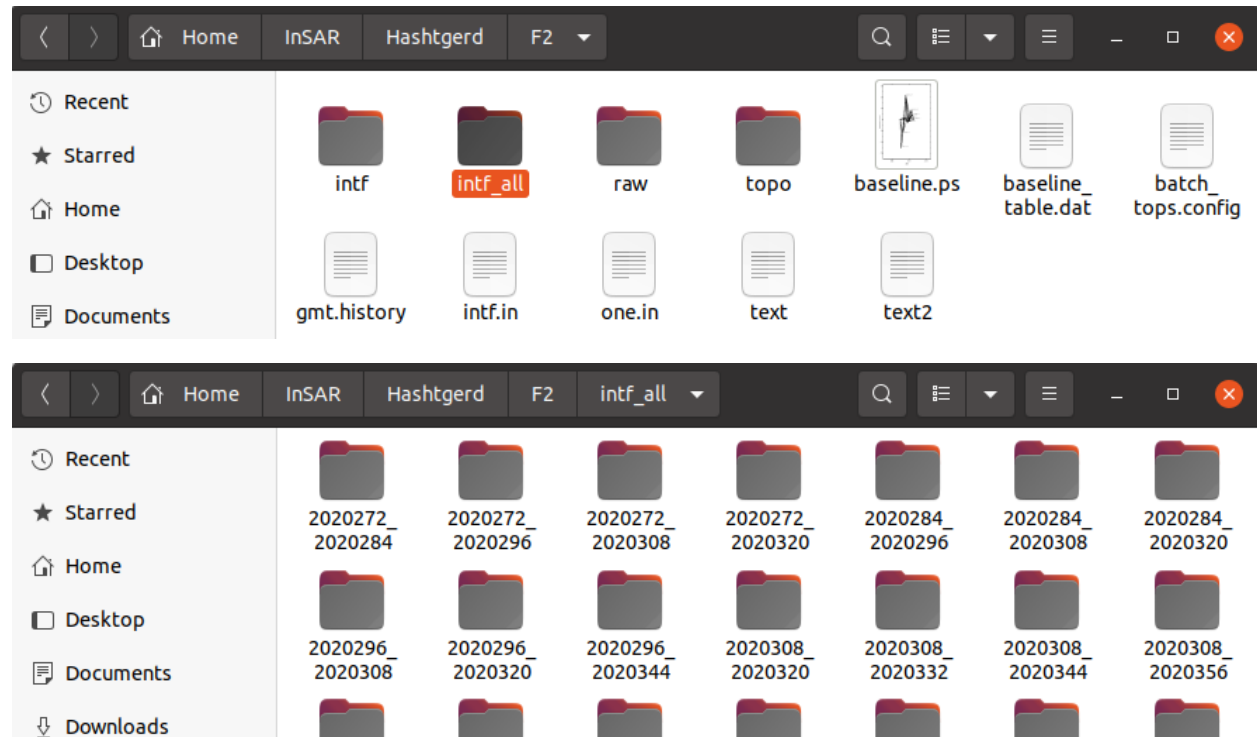


```
siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2> intf_tops.csh intf.in batch_tops.config

clean up topo/ folder

DEM2TOPOPHASE.CSH - START
USER SHOULD PROVIDE DEM FILE
Number of lines is set to a multiple of 4
```

- In this step, the filter is applied on the interferogram logs. You can also generate masked files by setting the geocode and threshold in the file above.



- The output of the above command includes the generated interferograms folder.

Then open the Terminal and run the following commands.

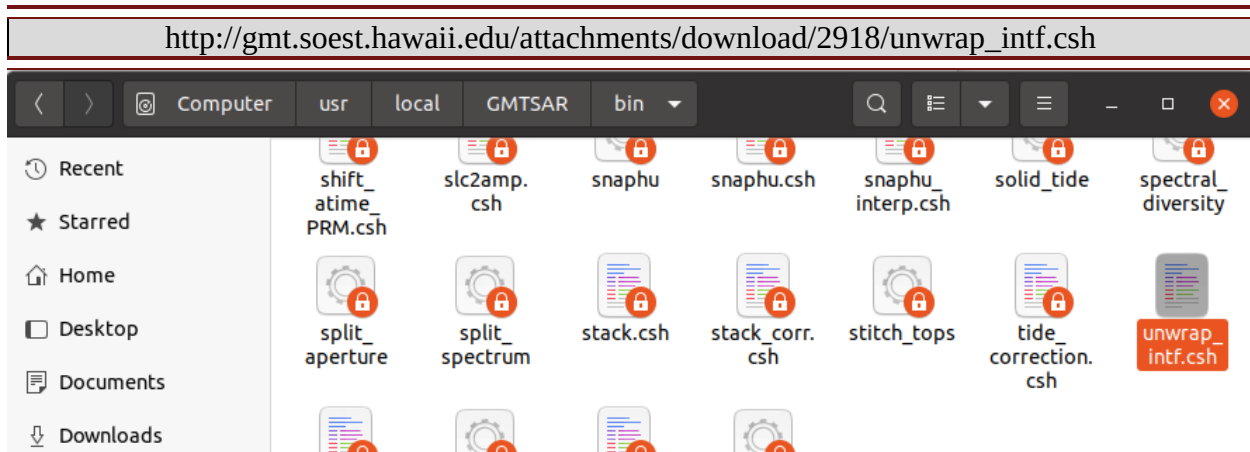
```
# cd /usr/local/GMTSAR
# sudo chown siavsh bin
```

```

siavash@siavash-Aspire-V3-571G: /usr/local/GMTSAR
siavash@siavash-Aspire-V3-571G:~$ cd /usr/local/GMTSAR
siavash@siavash-Aspire-V3-571G:/usr/local/GMTSAR$ sudo chown siavash bin
[sudo] password for siavash:
siavash@siavash-Aspire-V3-571G:/usr/local/GMTSAR$

```

Then download the `unwrap_intf.csh` file from the following URL and copy it to the GMTSAR/bin path.



Then open the Terminal in path `/usr/local/GMTSAR/bin/` and run the following commands.

```

# tcsh
# chmod +x unwrap_intf.csh

```

```

siavash@siavash-Aspire-V3-571G: /usr/local/GMTSAR/bin
siavash@siavash-Aspire-V3-571G:/usr/local/GMTSAR/bin$ tcsh
siavash-Aspire-V3-571G:/usr/local/GMTSAR/bin> chmod +x unwrap_intf.csh
siavash-Aspire-V3-571G:/usr/local/GMTSAR/bin>

```

The code inside the `unwrap_intf.csh` file should look exactly like the code inside the image. Otherwise, modify and save the file as shown.

```

1 #!/bin/csh -f
2 # intflist contains a list of all date1_date2 directories.
3 foreach line (`awk '{print $1}' intflist`)
4 cd $line
5 snaphu_interp.csh 0.1 0.01
6 cd ..
7 end

```

- Unwrap threshold = 0.4 and smoothing factor = 0.01, and their value depends on the area and purpose of the user's study.

The unwrap_intf.csh file contains an intflist to prepare this file. Go to the intf_all folder and open the Terminal and run the **ls** command.

```

slavash@slavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/intf_all$ ls
2020272_2020284 2020308_2020320 2020332_2021002 2021002_2021014
2020272_2020296 2020308_2020332 2020332_2021014 2021002_2021026
2020272_2020308 2020308_2020344 2020344_2020356 2021002_2021038
2020272_2020320 2020308_2020356 2020344_2021002 2021002_2021050
2020284_2020296 2020320_2020332 2020344_2021014 2021014_2021026
2020284_2020308 2020320_2020344 2020344_2021026 2021014_2021038
2020284_2020320 2020320_2020356 2020356_2021002 2021014_2021050
2020296_2020308 2020320_2021002 2020356_2021014 2021026_2021038
2020296_2020320 2020332_2020344 2020356_2021026 2021026_2021050
2020296_2020344 2020332_2020356 2020356_2021038 2021038_2021050
slavash@slavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/intf_all$

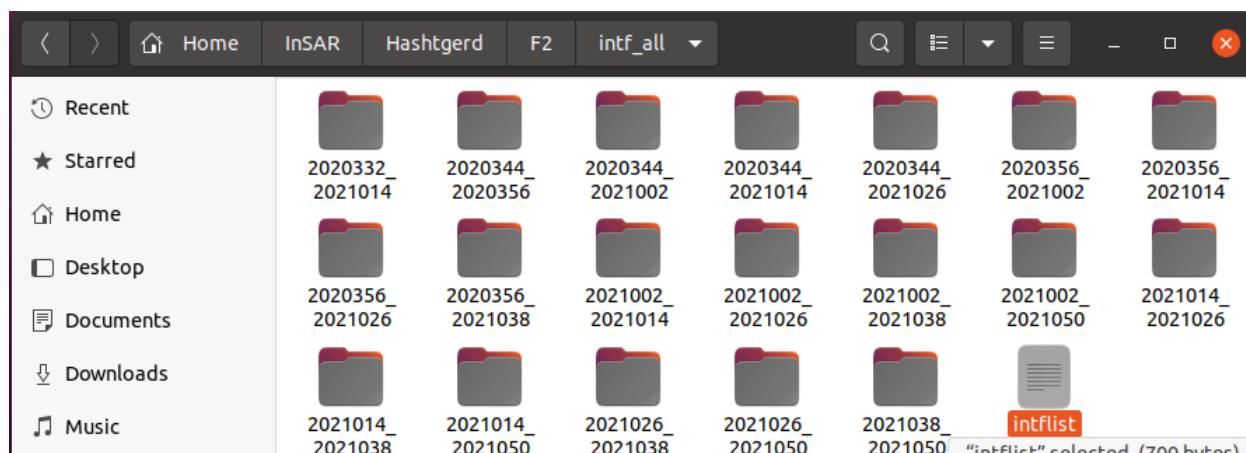
```

Then save the list of interferograms shown in the Terminal to a file named intflist and place it in the intf_all folder.

```

1 2020272_2020284
2 2020308_2020320
3 2020332_2021002
4 2021002_2021014
5 2020272_2020296
6 2020308_2020332
7 2020332_2021014
8 2021002_2021026

```

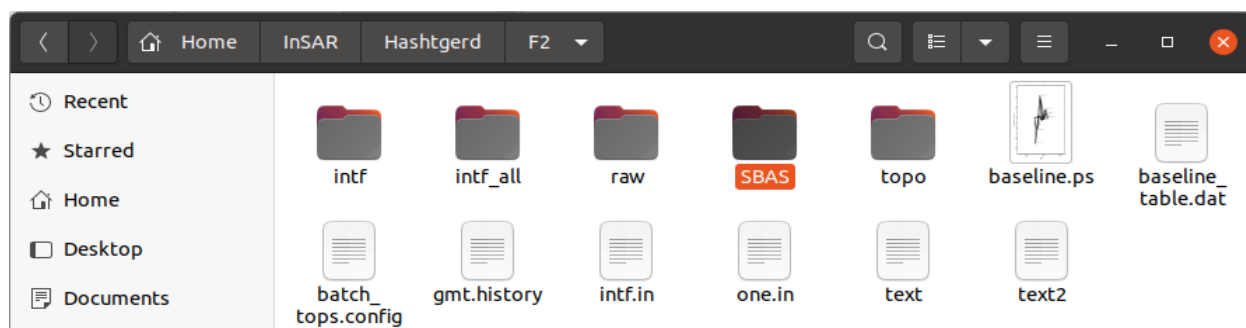



Then open the Terminal in the `intf_all` folder and run the following commands.

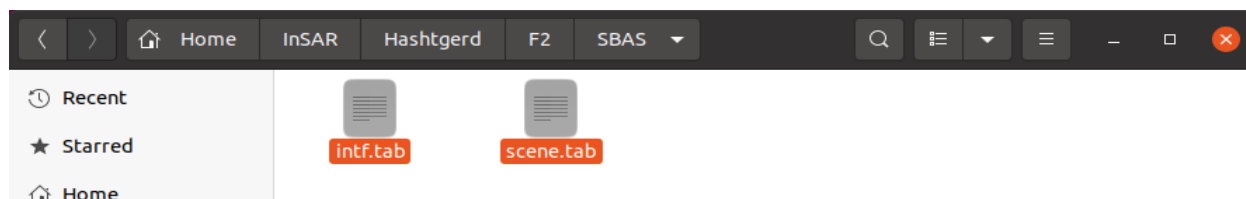
```
# tcsh
# unwrap_intf.csh intflist
```

```
siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/intf_all$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/intf_all> unwrap_intf.csh intflist
Setting search radius to be 300 ...
Reading in original grid...
Interpolating to nearest neighbour...
Working on line 1000 ...
1671202 number of searches used ...
WRITING GRID IMAGE: Width x Height = 812 x 1000
```

Then create a folder called SBAS in folder F2



Then create two scenes.tab and intf.tab and put them in the SBAS folder.



The scene.tab file contains two columns: the first column is the date of the image and the second column is the number of days in the baseline_table.dat file (third column) in the F2 folder.

scene.tab		
~/InSAR/Hashtgerd/F2/SBAS		
1	20200929	2462
2	20201011	2474
3	20201023	2486
4	20201104	2498
5	20201116	2510
6	20201128	2522
7	20201210	2534
8	20201222	2546
9	20210103	2557
10	20210115	2569
11	20210127	2581
12	20210208	2593
13	20210220	2605

baseline_table.dat						
~/InSAR/Hashtgerd/F2						
1	S1_20200929_ALL_F2	2020272.6084004175	2462	0.000000000000	0.000000000000	
2	S1_20201011_ALL_F2	2020284.6084043179	2474	-119.262191565033	-126.590393757482	
3	S1_20201023_ALL_F2	2020296.6084029265	2486	-87.520154409299	-63.074508293494	
4	S1_20201104_ALL_F2	2020308.6084017954	2498	-45.390081258401	0.631124700164	
5	S1_20201116_ALL_F2	2020320.6083996973	2510	-43.058648834251	9.982721062489	
6	S1_20201128_ALL_F2	2020332.6083941546	2522	34.817492990522	108.076064181767	
7	S1_20201210_ALL_F2	2020344.6083901611	2534	-54.731291062388	2.246383288969	
8	S1_20201222_ALL_F2	2020356.6083831915	2546	-62.549959829139	-11.907525069547	
9	S1_20210103_ALL_F2	2021002.6083762727	2557	-66.520039997138	-7.217992401164	
10	S1_20210115_ALL_F2	2021014.6083701088	2569	-42.124551852898	11.765894256297	
11	S1_20210127_ALL_F2	2021026.6083658494	2581	-15.341273060346	41.656535439103	
12	S1_20210208_ALL_F2	2021038.6083615348	2593	-32.593291069440	21.298088096057	
13	S1_20210220_ALL_F2	2021050.6083573883	2605	-52.222688507093	1.355018403001	

The intf.tab file contains the contents of a pair of interferograms as follows.

The first column: the path unwrap.grd or unwrap_mask.grd located in the intf_all folder

*intf.tab	
~/InSAR/Hashtgerd/F2/SBAS	
1	file:///home/siavash/InSAR/Hashtgerd/F2/intf_all/2020272_2020284/unwrap.grd
2	file:///home/siavash/InSAR/Hashtgerd/F2/intf_all/2020272_2020296/unwrap.grd
3	file:///home/siavash/InSAR/Hashtgerd/F2/intf_all/2020272_2020308/unwrap.grd
4	file:///home/siavash/InSAR/Hashtgerd/F2/intf_all/2020272_2020320/unwrap.grd

*intf.tab	
~/InSAR/Hashtgerd/F2/SBAS	
1	../intf_all/2020272_2020284/unwrap.grd
2	../intf_all/2020272_2020296/unwrap.grd
3	../intf_all/2020272_2020308/unwrap.grd

- Replace the address before the intf_all folder with ..

Column 2: The corr.grd path is located in the intf_all folder.

(If the area is cut, corr_cut.grd should be used and renamed to corr.grd)

The first screenshot shows a text editor window titled `*intf.tab` with the path `~/InSAR/Hashtgerd/F2/SBAS`. It contains three lines of file paths:

```
1 ../intf_all/2020272_2020284/unwrap.grd file:///home/siavash/InSAR/Hashtgerd/F2/intf_all/2020272_2020284/corr.grd
2 ../intf_all/2020272_2020296/unwrap.grd file:///home/siavash/InSAR/Hashtgerd/F2/intf_all/2020272_2020296/corr.grd
3 ../intf_all/2020272_2020308/unwrap.grd file:///home/siavash/InSAR/Hashtgerd/F2/intf_all/2020272_2020308/corr.grd
```

The second screenshot shows the same file after editing, where the `file://` paths have been replaced with relative paths starting with `../`:

```
1 ../intf_all/2020272_2020284/unwrap.grd ../intf_all/2020272_2020284/corr.grd
2 ../intf_all/2020272_2020296/unwrap.grd ../intf_all/2020272_2020296/corr.grd
3 ../intf_all/2020272_2020308/unwrap.grd ../intf_all/2020272_2020308/corr.grd
```

- Replace the address before the intf_all folder with `..`

Column 3: The date of the first image in each interferogram

Column 4: The date of the second image in each interferogram

The first screenshot shows the text editor window with the file paths from the previous step, now with dates added at the end of each line:

```
1 ../intf_all/2020272_2020284/unwrap.grd ../intf_all/2020272_2020284/corr.grd S1_20200929_ALL_F2:S1_20201011_ALL_F2
2 ../intf_all/2020272_2020296/unwrap.grd ../intf_all/2020272_2020296/corr.grd S1_20200929_ALL_F2:S1_20201023_ALL_F2
3 ../intf_all/2020272_2020308/unwrap.grd ../intf_all/2020272_2020308/corr.grd S1_20200929_ALL_F2:S1_20201104_ALL_F2
```

The second screenshot shows the same file with the dates formatted as separate columns:

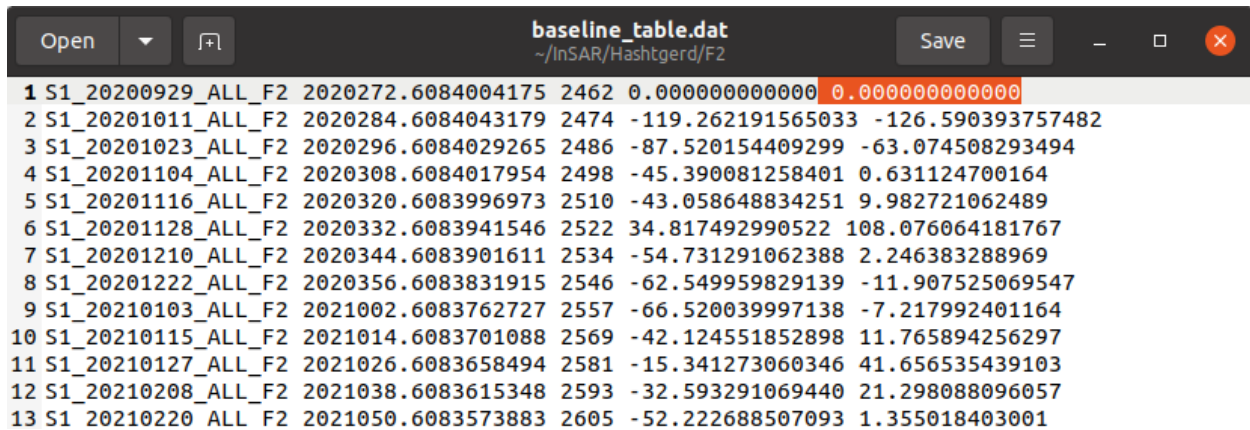
```
1 ../intf_all/2020272_2020284/unwrap.grd ../intf_all/2020272_2020284/corr.grd 20200929 20201011
2 ../intf_all/2020272_2020296/unwrap.grd ../intf_all/2020272_2020296/corr.grd 20200929 20201023
3 ../intf_all/2020272_2020308/unwrap.grd ../intf_all/2020272_2020308/corr.grd 20200929 20201104
```

The screenshot shows a text editor window titled `intf.in` with the path `~/InSAR/Hashtgerd/F2`. It contains a list of dates, each preceded by a line number:

```
1 S1_20200929_ALL_F2:S1_20201011_ALL_F2
2 S1_20200929_ALL_F2:S1_20201023_ALL_F2
3 S1_20200929_ALL_F2:S1_20201104_ALL_F2
4 S1_20200929_ALL_F2:S1_20201116_ALL_F2
5 S1_20201011_ALL_F2:S1_20201023_ALL_F2
6 S1_20201011_ALL_F2:S1_20201104_ALL_F2
7 S1_20201011_ALL_F2:S1_20201116_ALL_F2
8 S1_20201023_ALL_F2:S1_20201104_ALL_F2
9 S1_20201023_ALL_F2:S1_20201116_ALL_F2
10 S1_20201023_ALL_F2:S1_20201210_ALL_F2
```

- Use the intf.in file to find the date of the first and second images in each interferogram and delete the rest except the date

Column 5: The value of `b_perp` is based on calculating the difference between the spatial baseline of the first date, from which you subtract the spatial baseline of the second date. This value is in the `baseline_table.dat` file in path F2



Line	Image 1	Image 2	Baseline (m)	b_perp (m)
1	S1_20200929_ALL_F2	2020272.6084004175	2462	0.000000000000
2	S1_20201011_ALL_F2	2020284.6084043179	2474	-119.262191565033
3	S1_20201023_ALL_F2	2020296.6084029265	2486	-87.520154409299
4	S1_20201104_ALL_F2	2020308.6084017954	2498	-45.390081258401
5	S1_20201116_ALL_F2	2020320.6083996973	2510	-43.058648834251
6	S1_20201128_ALL_F2	2020332.6083941546	2522	34.817492990522
7	S1_20201210_ALL_F2	2020344.6083901611	2534	-54.731291062388
8	S1_20201222_ALL_F2	2020356.6083831915	2546	-62.549959829139
9	S1_20210103_ALL_F2	2021002.6083762727	2557	-66.520039997138
10	S1_20210115_ALL_F2	2021014.6083701088	2569	-42.124551852898
11	S1_20210127_ALL_F2	2021026.6083658494	2581	-15.341273060346
12	S1_20210208_ALL_F2	2021038.6083615348	2593	-32.593291069440
13	S1_20210220_ALL_F2	2021050.6083573883	2605	-52.222688507093

Open

intf.tab

~/InSAR/Hashtgerd/F2/SBAS

Save

1	../intf_all/2020272_2020284/unwrap.grd	../intf_all/2020272_2020284/corr.grd	20200929	20201011	126.590393757482
2	../intf_all/2020272_2020296/unwrap.grd	../intf_all/2020272_2020296/corr.grd	20200929	20201023	63.074508293494
3	../intf_all/2020272_2020308/unwrap.grd	../intf_all/2020272_2020308/corr.grd	20200929	20201104	-0.631124700164
4	../intf_all/2020272_2020320/unwrap.grd	../intf_all/2020272_2020320/corr.grd	20200929	20201116	-9.982721062489

For example, for row 1 in column 5, the value of image 20200929 must be subtracted from the value of image 20201011.

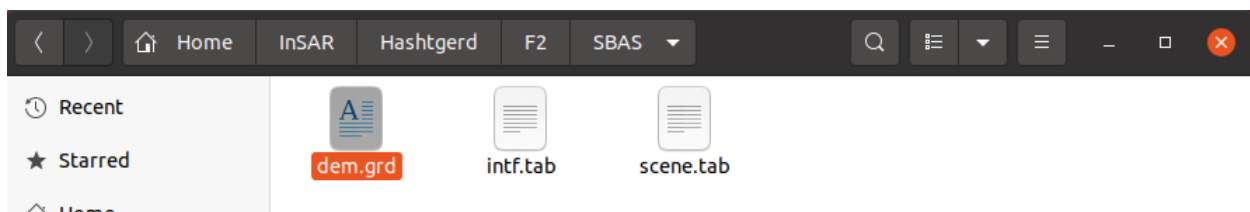
$$20200929 = 0.000000000000$$

$$20201011 = -126.590393757482$$

$$20200929 - 20201011 = 0.000000000000 - (-126.590393757482) = 126.590393757482$$

- In this way, subtract the value of the first image from the second image in each row .

Also, copy the `dem.grd` file in the topo folder to the SBAS path.



Then open the Terminal in the sbas folder and run the following commands.

```
# tcsh
# sbas
```

```

siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/SBAS$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> sbas

USAGE: sbas intf.tab scene.tab N S xdim ydim [-atm nl] [-smooth sf] [-wavelength wl] [-incidence inc] [-range -rng] [-rms] [-dem]

input:
  intf.tab      -- list of unwrapped (filtered) interferograms:
                  format:  unwrap.grd corr.grd ref_id rep_id B_perp
  scene.tab     -- list of the SAR scenes in chronological order
                  format:  scene_id number_of_days
                  note:    the number_of_days is relative to a reference date
  N             -- number of the interferograms
  S             -- number of the SAR scenes
  xdim and ydim -- dimension of the interferograms
  -smooth sf    -- smoothing factors, default=0
  -atm nl       -- number of iterations for atmospheric correction, default=0(skip atm correction)
  -wavelength wl -- wavelength of the radar wave (m) default=0.236
  -incidence theta -- incidence angle of the radar wave (degree) default=37
  -range rng    -- range distance from the radar to the center of the interferogram (m) default=866000
  -rms          -- output RMS of the data misfit grids (mm): rms.grd
  -dem          -- output DEM error (m): dem.grd

output:
  disp_#.grd    -- cumulative displacement time series (mm) grids
  vel.grd       -- mean velocity (mm/yr) grids

example:
  sbas intf.tab scene.tab 88 28 700 1000

```

- N number of interferogram logs and S number of images

To get (xdim, xmin, xmax) and (ydim, ymin, ymax) run the following commands in one of the interferogram folders.

```

# tcsh
# gmt grdinfo unwrap.grd

```

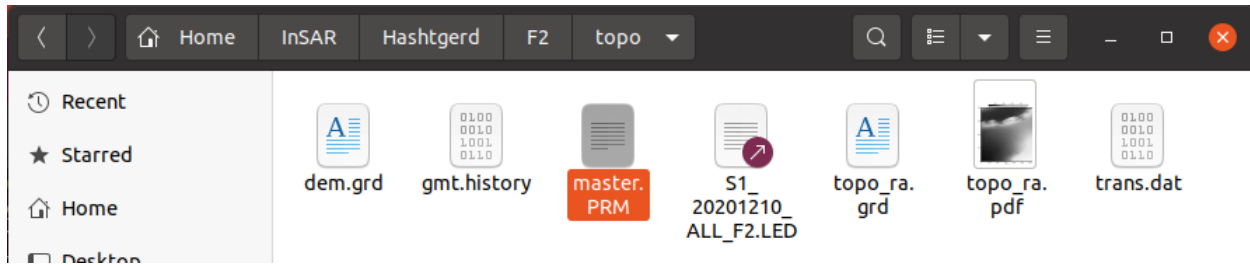
```

siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/intf_all/2020272_2020284$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/intf_all/2020272_2020284> gmt grdinfo unwrap.grd
unwrap.grd: Title: Produced by grdmath
unwrap.grd: Command: grdmath tmp.grd mask2_patch.grd MUL = tmp.grd
unwrap.grd: Remark:
unwrap.grd: Pixel node registration used [Cartesian grid]
unwrap.grd: Grid file format: nf = GMT netCDF format (32-bit float), COARDS, CF-1.5
unwrap.grd: x_min: 0 x_max: 6496 x_inc: 8 name: x n_columns: 812
unwrap.grd: y_min: 0 y_max: 2000 y_inc: 2 name: y n_rows: 1000
unwrap.grd: z_min: -7.03924226761 z_max: 6.28318548203 name: z
unwrap.grd: scale_factor: 1 add_offset: 0
unwrap.grd: format: classic
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/intf_all/2020272_2020284>

```

- n_columns = xdim
- n_rows = ydim

Wavelength: Radar wavelength in meters, look for "radar_wavelength" in the raw folder to find supermaster.PRM.



```

23 SC_identity           = 10
24 rng_samp_rate          = 64345238.125714
25 input_file             = S1_20200929_ALL_F2.raw
26 num_rng_bins           = 25652
27 bytes_per_line         = 102608
28 good_bytes_per_line    = 102608
29 PRF                     = 486.486310
30 pulse_dur              = 6.199593e-05
31 near_range              = 845481.851848
32 num_lines              = 12188
33 num_patches            = 1
34 SC_clock_start          = 2020272.6084004175
35 SC_clock_stop           = 2020272.6086903841
36 clock_start            = 272.608400417418
37 clock_stop              = 272.608690384087
38 led_file                = S1_20200929_ALL_F2.LED
39 orbdir = A
40 lookdir                = R
41 radar_wavelength        = 0.0554658
42 chirp_slope             = 7.79083e+11

```

Calculate the range for Sentinel-1 using the following formula:

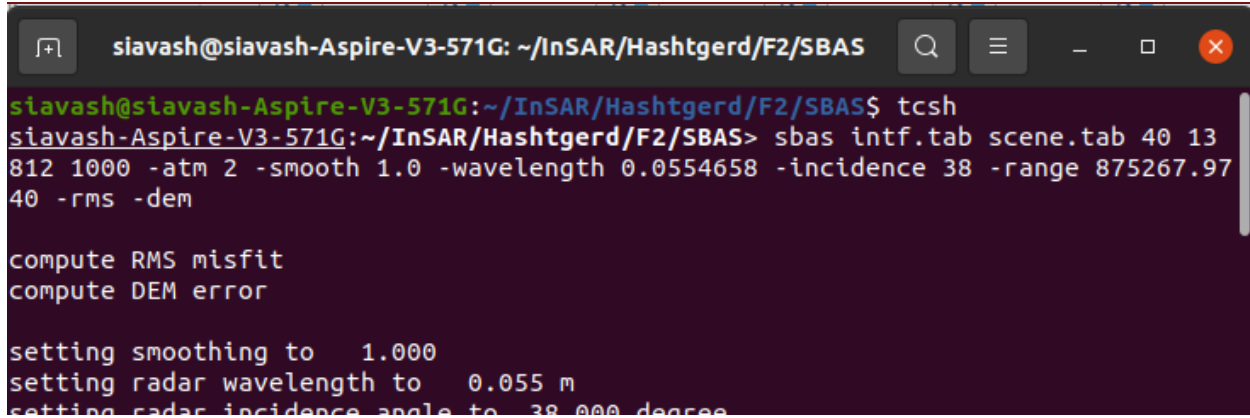
$$\text{rng_pixel_size} = (\text{speed of light}) / (\text{rng_samp_rate}) / 2$$

$$\text{rng} = \text{rng_pixel_size} \times (x_{\text{min}} + x_{\text{max}}) / 2 + 845000$$

- rng_samp_rate is available in supermaster.PRM.
- Speed of light The speed of light is $8 \times 10^8 \times 3$.
- Incidence To get the angle, you can use SNAP software or image information (Tie-Point Grids) .

Then open the Terminal in the SBAS folder and run the following commands.


```
# tcsh
# sbas intf.tab scene.tab 40 13 812 1000 -atm 2 -smooth 1.0 -wavelength 0.0554658
-incidence 38 -range 875267.97 -rms -dem
```

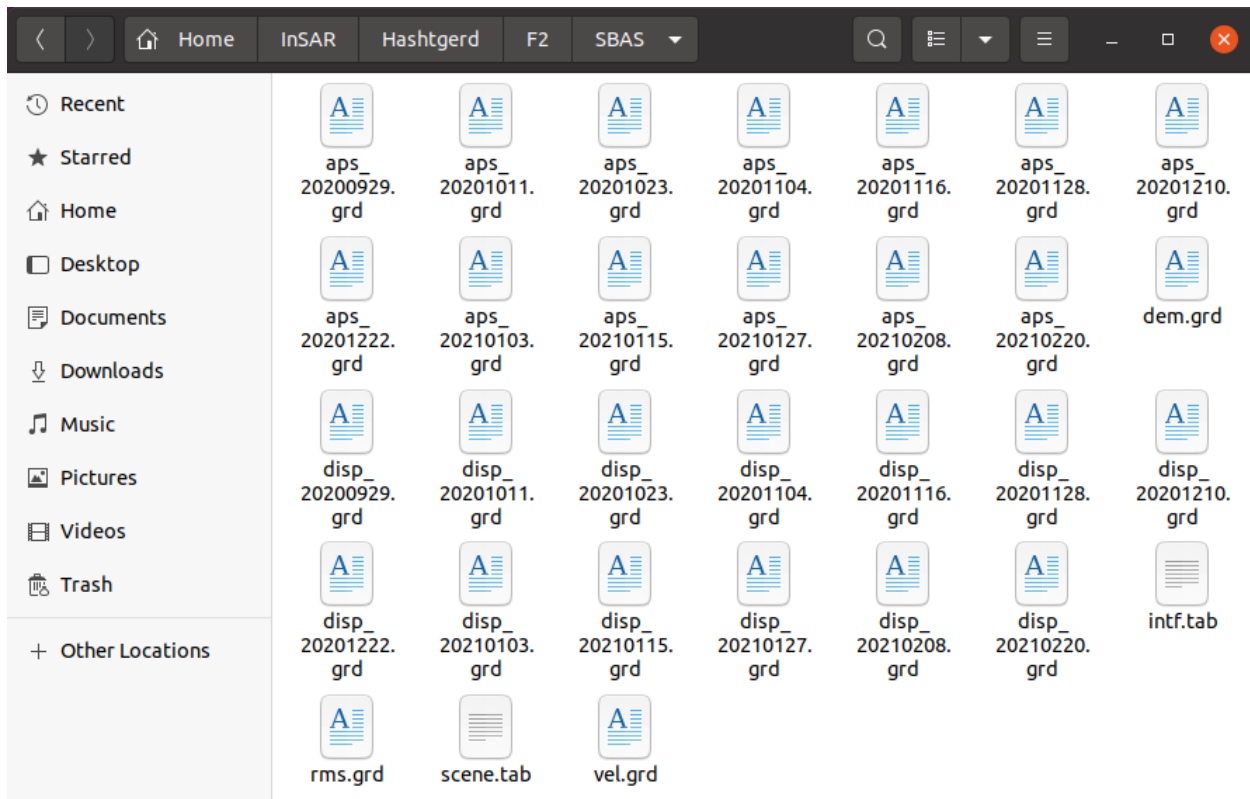


```
siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/SBAS
siavash@siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> sbas intf.tab scene.tab 40 13
812 1000 -atm 2 -smooth 1.0 -wavelength 0.0554658 -incidence 38 -range 875267.97
40 -rms -dem

compute RMS misfit
compute DEM error

setting smoothing to 1.000
setting radar wavelength to 0.055 m
setting radar incidence angle to 38.000 degree
```

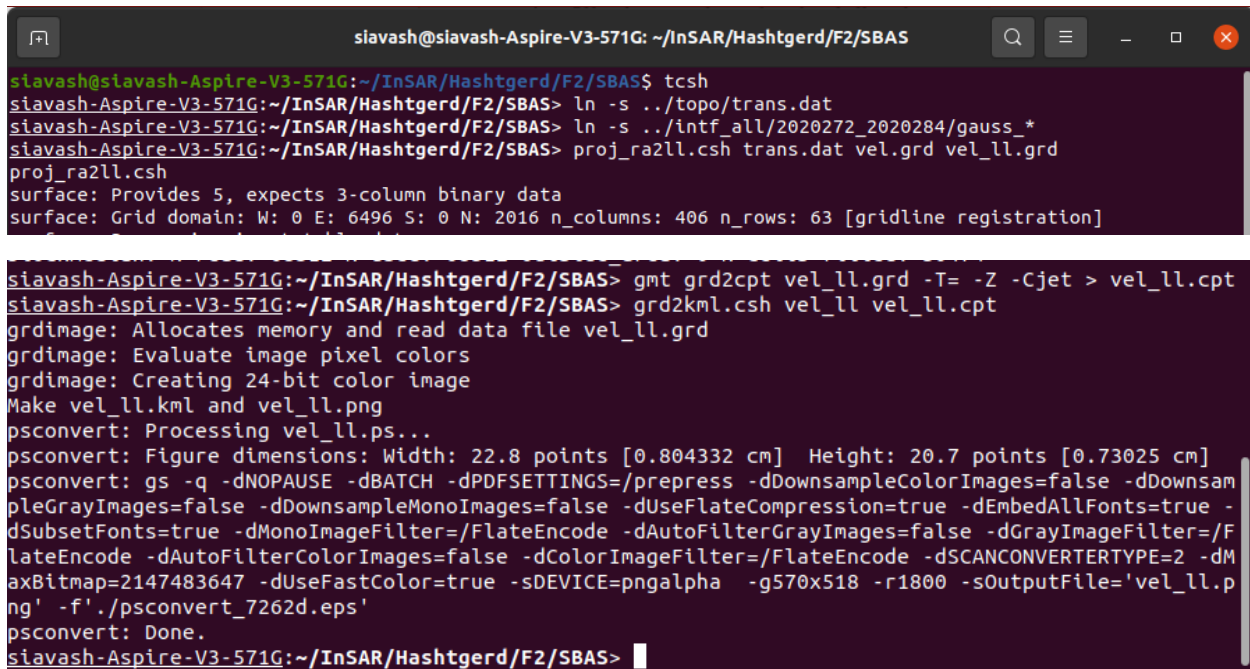
- In the above command, the number of interferograms and images and other numerical parameters depends on the user case and varies.



The output of the above command, the transfer time-series file named disp and .grd format and the average transfer speed on In mm/yr is called vel.grd

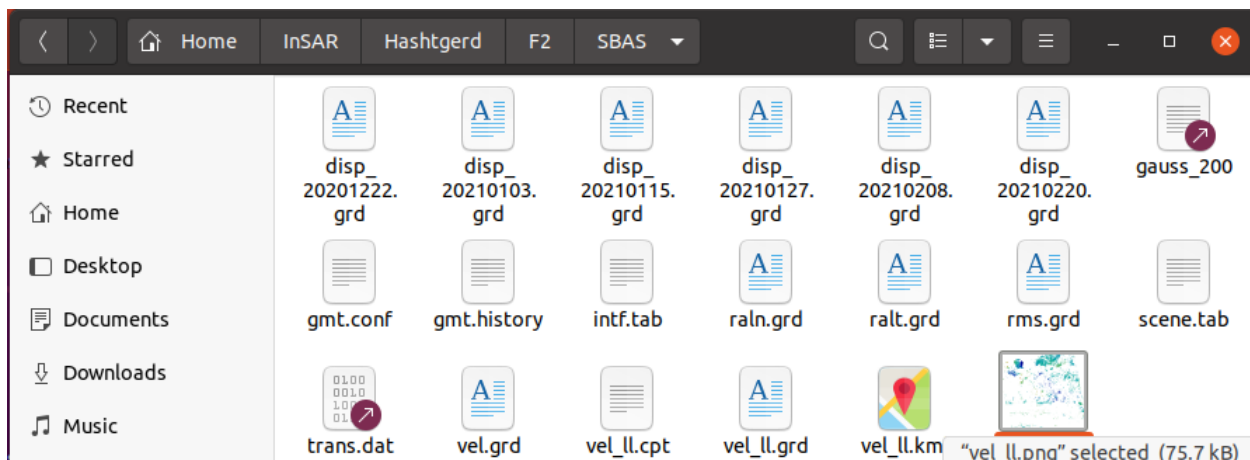
To convert vel.grd (Radar coordinates) file to lat / lon (latitude and longitude), open the Terminal in the SBAS folder and run the following commands.

```
# tcsh
# ln -s ../topo/trans.dat
# ln -s ../intf_all/2020272_2020284/gauss_*
# proj_ra2ll.csh trans.dat vel.grd vel_ll.grd
# gmt grd2cpt vel_ll.grd -T= -Z -Cjet > vel_ll.cpt
# grd2kml.csh vel_ll vel_ll.cpt
```

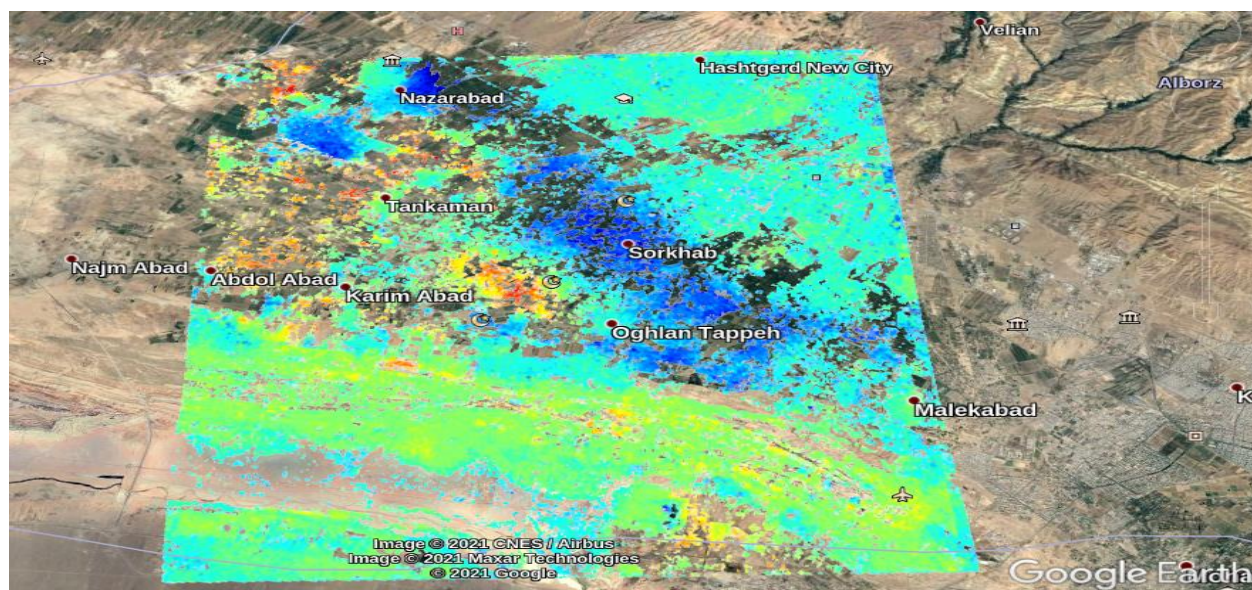


```
siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/SBAS
siavash@siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> ln -s ../topo/trans.dat
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> ln -s ../intf_all/2020272_2020284/gauss_*
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> proj_ra2ll.csh trans.dat vel.grd vel_ll.grd
proj_ra2ll.csh
surface: Provides 5, expects 3-column binary data
surface: Grid domain: W: 0 E: 6496 S: 0 N: 2016 n_columns: 406 n_rows: 63 [gridline registration]

siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> gmt grd2cpt vel_ll.grd -T= -Z -Cjet > vel_ll.cpt
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> grd2kml.csh vel_ll vel_ll.cpt
grdimage: Allocates memory and read data file vel_ll.grd
grdimage: Evaluate image pixel colors
grdimage: Creating 24-bit color image
Make vel_ll.kml and vel_ll.png
psconvert: Processing vel_ll.ps...
psconvert: Figure dimensions: Width: 22.8 points [0.804332 cm] Height: 20.7 points [0.73025 cm]
psconvert: gs -q -dNOPAUSE -dBATCH -dPDFSETTINGS=/prepress -dDownsampleColorImages=false -dDownsam
pleGrayImages=false -dDownsampleMonoImages=false -dUseFlateCompression=true -dEmbedAllFonts=true -d
SubsetFont=true -dMonoImageFilter=/FlateEncode -dAutoFilterGrayImages=false -dGrayImageFilter=/F
lateEncode -dAutoFilterColorImages=false -dColorImageFilter=/FlateEncode -dSCANCONVERTERTYPE=2 -dM
axBitmap=2147483647 -dUseFastColor=true -sDEVICE=pngalpha -g570x518 -r1800 -sOutputFile='vel_ll.p
ng' -f'./psconvert_7262d.eps'
psconvert: Done.
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS>
```



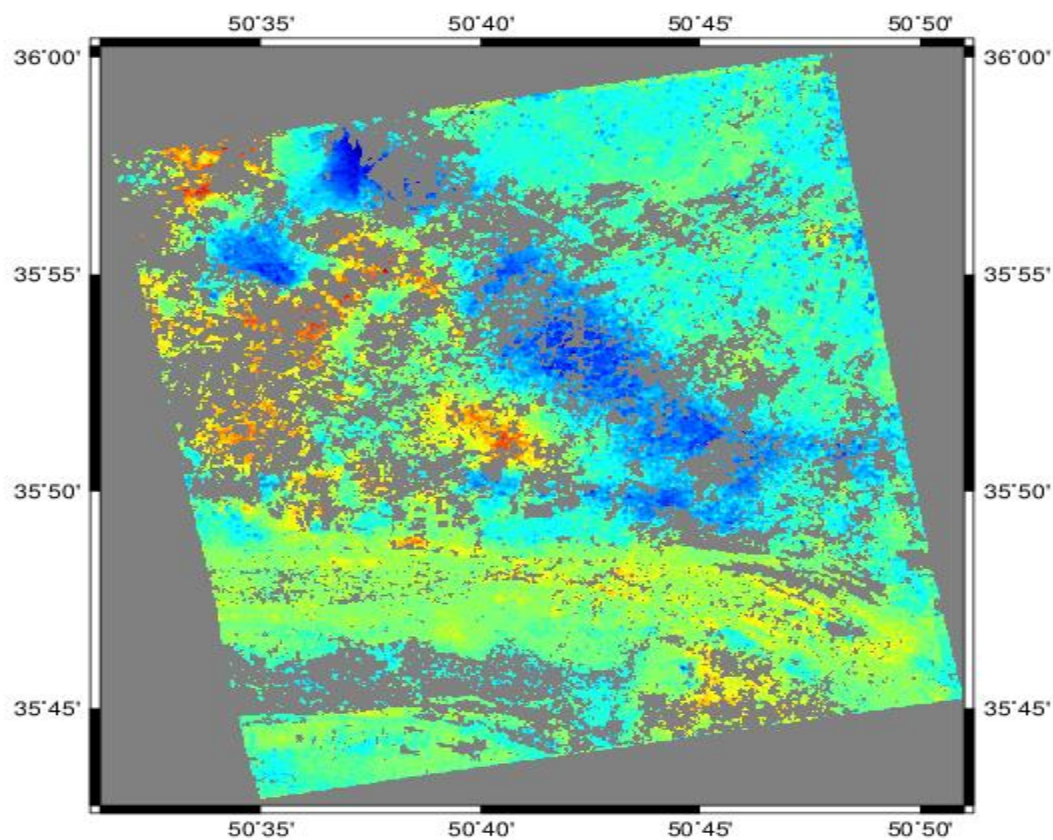
- vel_ll.png and vel_ll.kml files of the SBAS time-series map



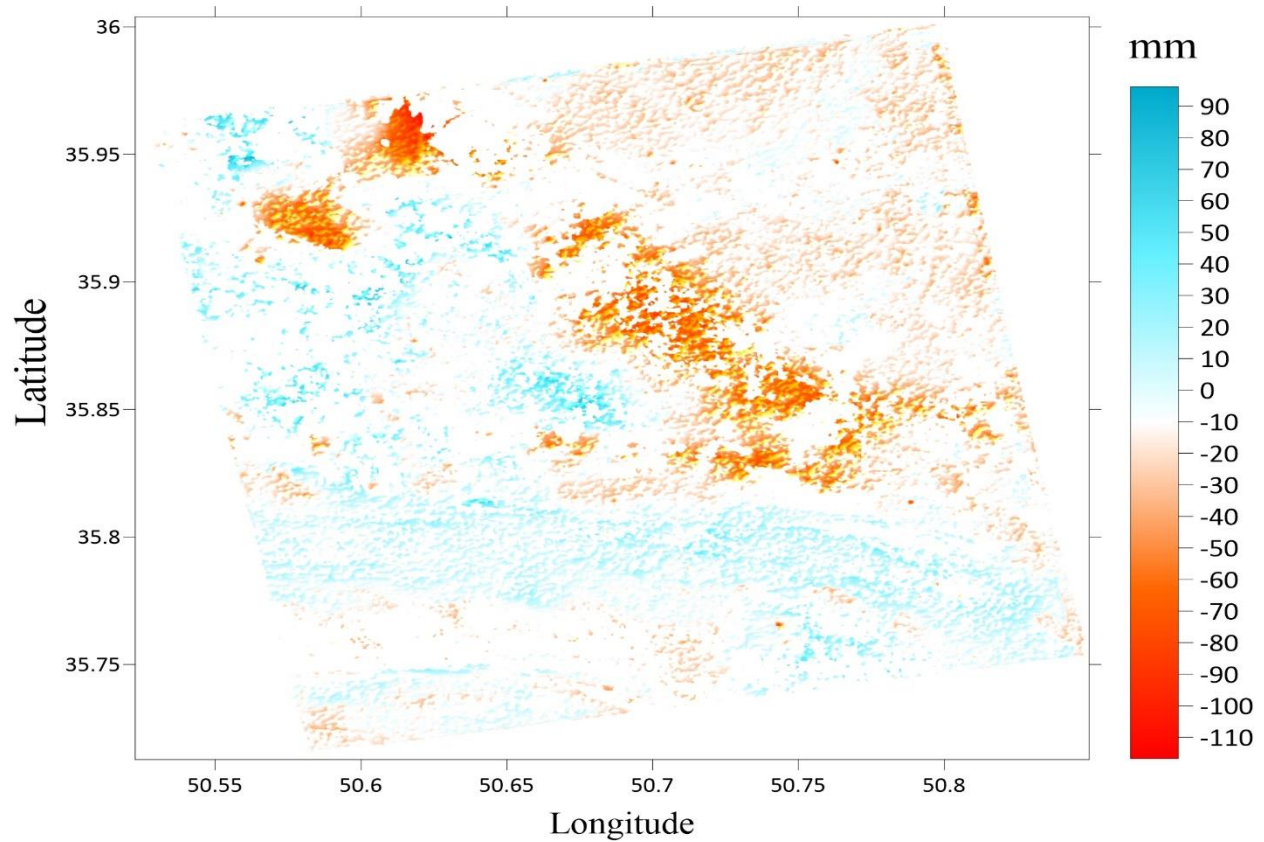
You can also get the Displacement map of the displacement by running the following command.

```
# gmt grdimage vel_ll.grd -JM6i -P -Bb -Cvel_ll.cpt -V -K > los_ll.ps
```

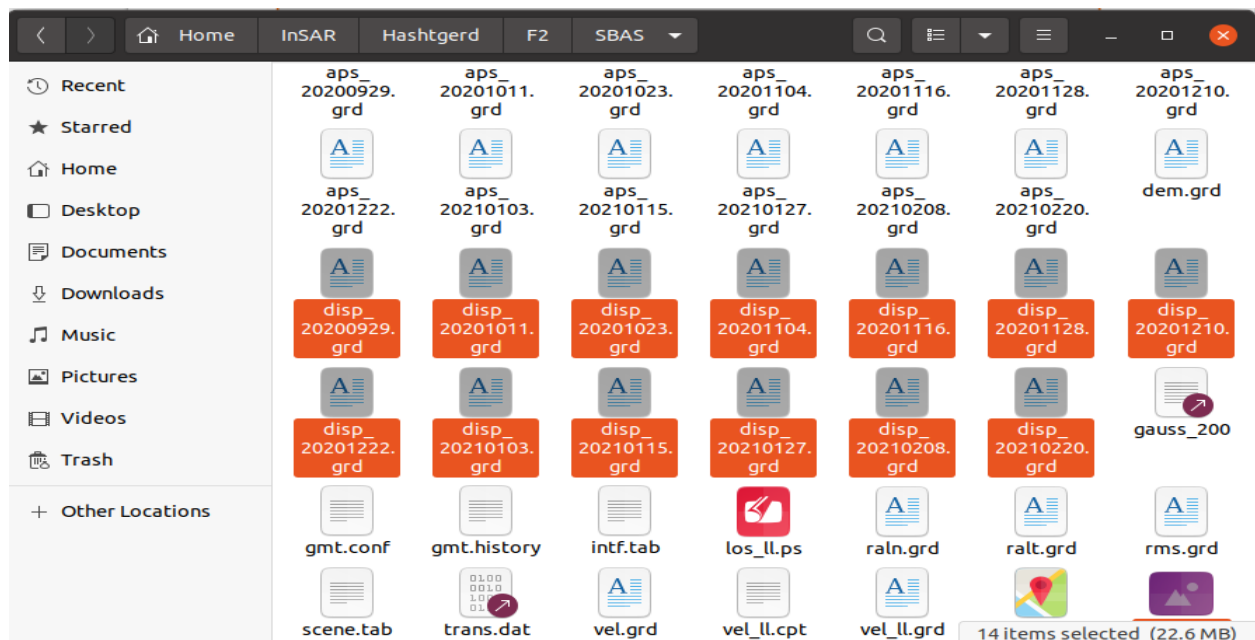
```
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> gmt grdimage vel_ll.grd -JM6i  
-P -Ba -Cvel_ll.cpt -V -K > los_ll.ps
```



You can also use vel_ll.grd file and software such as QGIS and Surfer to select Legend, for Displacement maps and other graphics.



The files of each image in format .grd show the amount of Displacement.

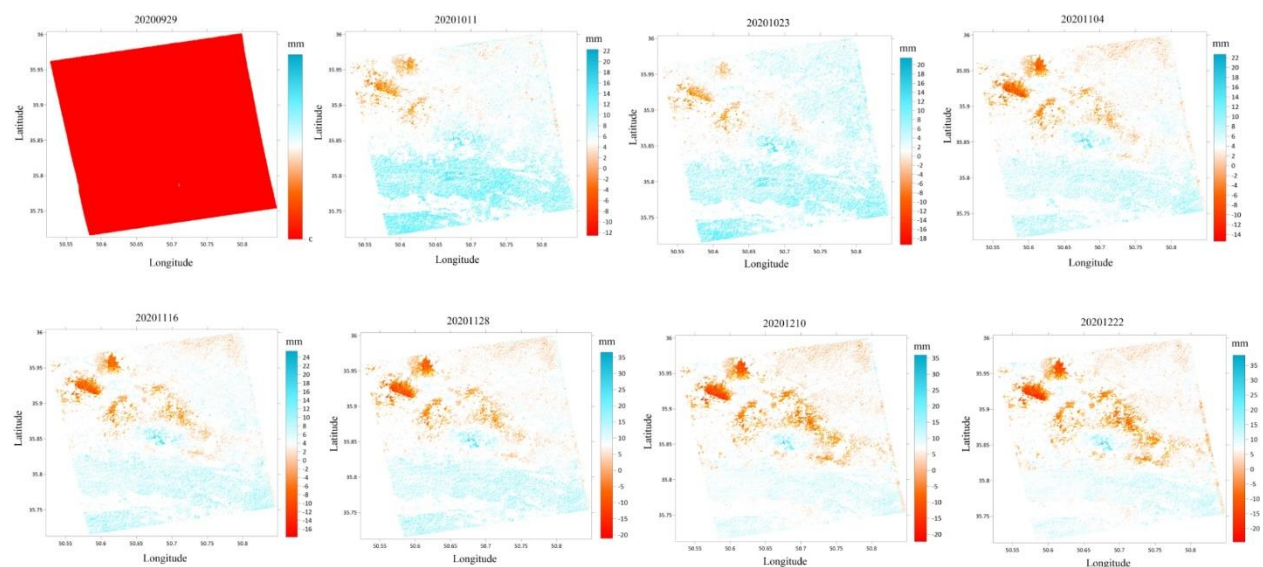


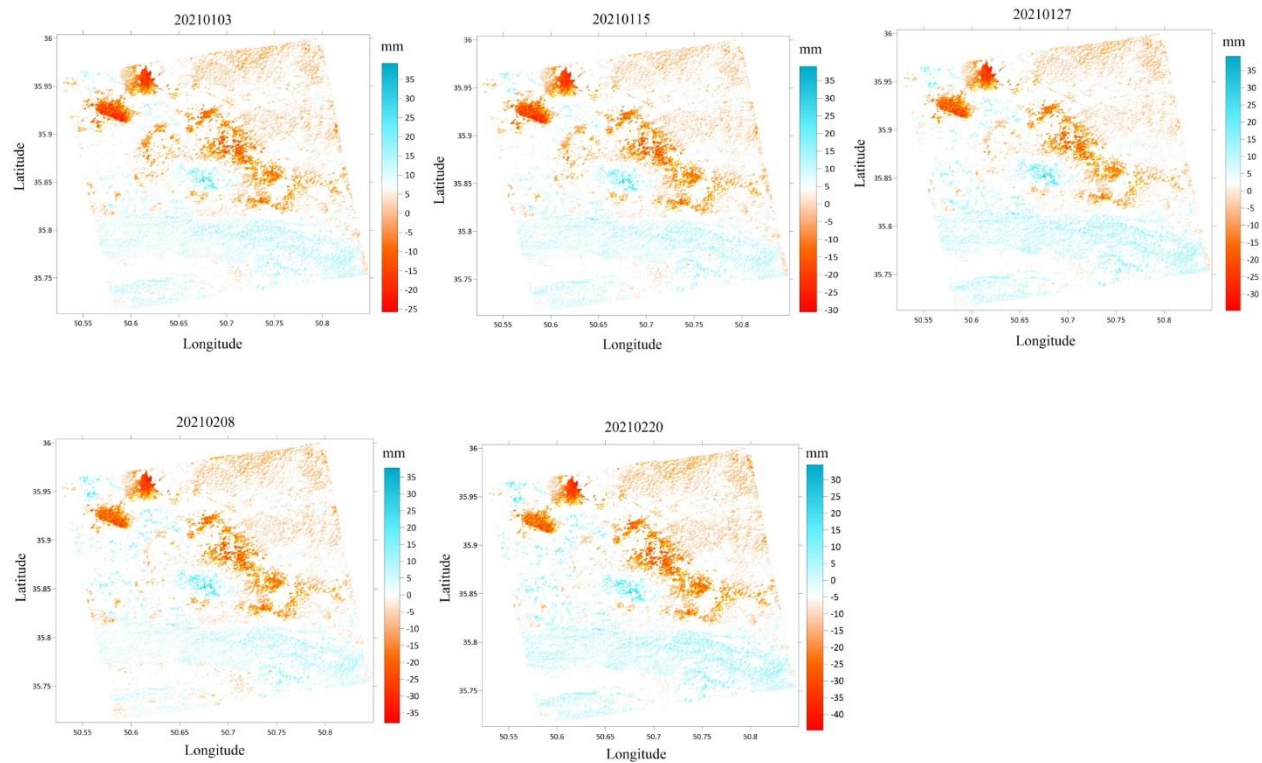
Then running the following commands to create the time-series of the Displacement value in the Terminal.

```
# proj_ra2ll.csh trans.dat disp_20200929.grd disp_20200929_ll.grd
# proj_ra2ll.csh trans.dat disp_20201011.grd disp_20201011_ll.grd
# proj_ra2ll.csh trans.dat disp_20201023.grd disp_20201023_ll.grd
# proj_ra2ll.csh trans.dat disp_20201104.grd disp_20201104_ll.grd
# proj_ra2ll.csh trans.dat disp_20201116.grd disp_20201116_ll.grd
# proj_ra2ll.csh trans.dat disp_20201128.grd disp_20201128_ll.grd
# proj_ra2ll.csh trans.dat disp_20201210.grd disp_20201210_ll.grd
# proj_ra2ll.csh trans.dat disp_20201222.grd disp_20201222_ll.grd
# proj_ra2ll.csh trans.dat disp_20210103.grd disp_20210103_ll.grd
# proj_ra2ll.csh trans.dat disp_20210115.grd disp_20210115_ll.grd
# proj_ra2ll.csh trans.dat disp_20210115.grd disp_20210115_ll.grd
# proj_ra2ll.csh trans.dat disp_20210127.grd disp_20210127_ll.grd
# proj_ra2ll.csh trans.dat disp_20210208.grd disp_20210208_ll.grd
# proj_ra2ll.csh trans.dat disp_20210220.grd disp_20210220_ll.grd
```

```
siavash@siavash-Aspire-V3-571G: ~/InSAR/Hashtgerd/F2/SBAS
siavash@siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS$ tcsh
siavash-Aspire-V3-571G:~/InSAR/Hashtgerd/F2/SBAS> proj_ra2ll.csh trans.dat disp_20200929.grd
disp_20200929_ll.grd
proj_ra2ll.csh
Sampling in geocoordinates with 50 meter pixels ...
blockmedian: Provides 3, expects 3-column binary data
blockmedian: Processing input table data
```

Then, using files disp_20200929_ll.grd, disp_20201011_ll.grd, etc., you can draw the Displacement map in software such as QGIS and Surfer.





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